

ACCELERATING CONVERGENCE TO EQUILIBRIUM IN GAUSSIAN DIFFUSIONS:
OPTIMAL RATES AND APPLICATIONS TO LANGEVIN DYNAMICS

MSc Thesis (*Afstudeerscriptie*)

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Abstract

Markov diffusions with the same invariant distribution can converge to equilibrium at significantly different rates. This thesis studies how the choice of linear drift and diffusion coefficients accelerates convergence to a given centred Gaussian distribution. We characterise the admissible invariant dynamics through a Lyapunov equation and identify the asymptotic Wasserstein contraction rate with the minimum real part of the drift spectrum. Under the total noise constraint $\text{tr}(\mathbf{D}) \leq \eta$, the optimal rate is $\frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1})$. We further apply this framework to Gaussian Langevin dynamics, extend the coupling analysis to strongly convex non-Gaussian potentials, and illustrate the acceleration in a two-dimensional Gaussian example.

Keywords: convergence to equilibrium; non-reversible diffusions; Ornstein–Uhlenbeck processes; Wasserstein distance; Langevin dynamics.

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Chapter 1

Introduction

Many problems in probability, statistics, and scientific computing require samples from a probability distribution that is known only through its density, often up to an unknown normalising constant. When direct sampling is difficult, Markov chain Monte Carlo methods offer an indirect route. They construct a Markov chain with the target as an invariant distribution and simulate it until the influence of its initial state has become small. Sampling may therefore be viewed as a convergence problem to equilibrium; see [LP17, Chapter 4] for the discrete-time setting and [BE26, Section 1.3] for the continuous-time framework used here.

The existence of the equilibrium is essential, but it does not determine how quickly that equilibrium is reached. A given probability measure can be invariant for many different processes, and their convergence rates may differ substantially. A process that preserves the target distribution but converges very slowly is of little practical use. Therefore, the dependence of the convergence rate on the choice of dynamics is a central object of study.

Roughly speaking, a process is *reversible* if its motion at equilibrium has no statistically preferred direction of time. Reversibility is often convenient, but it is not required for invariance. A non-reversible drift can change how the process moves through the state space while preserving the same target distribution, and may lead to faster convergence. The acceleration of Gaussian diffusions by antisymmetric perturbations is studied in [LNP13], while the optimisation problem allowing noise to act directly on only part of the state space is treated in [GM16].

The main research question of this thesis is: *Given a centred Gaussian target and a fixed noise budget, how should the noise and the non-reversible part of the drift be chosen so that the process forgets its initial state as rapidly as possible?*

Mathematical Framework and Main Results

Following the Gaussian case study in [BE26, Section 4.7], let the target distribution be $\mu = \mathcal{N}(0, \mathbf{C})$, where $\mathbf{C}$ is symmetric and positive definite. We consider linear diffusions of the form

$$dX_t = \sqrt{2}\sigma dB_t - \mathbf{A}X_t dt.$$

Here B_t is a standard d -dimensional Brownian motion, the matrix σ determines how noise enters the system, and $\mathbf{A}$ determines the linear drift. The Gaussian setting permits an explicit analysis.

Let $\mathbf{D} = \sigma\sigma^\top$. The target measure μ is *invariant* precisely when

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = 2\mathbf{D},$$

or, equivalently, when

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}$$

for an antisymmetric matrix $\mathbf{J}$. The positive semidefinite matrix $\mathbf{D}$ records where random forcing is applied, whereas $\mathbf{J}$ couples different directions without changing the invariant Gaussian measure. Thus, a coordinate need not be forced directly if the drift transmits noise to it from another part of the system.

Many choices of σ and $\mathbf{A}$ can preserve the same Gaussian distribution. We compare the resulting dynamics in *Wasserstein distance*. Start two processes from different points driven by the same Brownian motion. Then their common noise cancels in the difference, which satisfies a deterministic linear equation governed by $\mathbf{A}$. In the Gaussian case, this *synchronous coupling* is optimal, and the L^2 -Wasserstein contraction coefficient is exactly

$$\alpha_2(p_t) = \|e^{-t\mathbf{A}}\|.$$

Its asymptotic exponential rate is

$$\lambda^*(\mathbf{A}) := \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

The stochastic convergence problem is therefore reduced to a spectral optimisation problem.

A constraint on the coefficients is needed to make the optimisation meaningful. Without one, a rescaling of time would make the convergence rate arbitrarily large. In dimension two, suppose that $\mathbf{C} = \text{diag}(u, v)$. Under the unit noise budget $\text{tr}(\mathbf{D}) \leq 1$, the optimal non-reversible and reversible rates are

$$\lambda_{\text{nr}}^* = \frac{1}{2 \min\{u, v\}}, \quad \lambda_{\text{rev}}^* = \frac{1}{u + v}.$$

The acceleration factor is

$$\frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} = \frac{1 + \kappa(\mathbf{C})}{2},$$

where $\kappa(\mathbf{C})$ is the *condition number* of the covariance matrix, here the ratio of the larger variance to the smaller one. Thus there is no improvement when the two variances are equal, but the possible gain grows as the target becomes more anisotropic.

The same principle extends to arbitrary dimension. The main problem fixes the total noise through $\text{tr}(\mathbf{D}) \leq \eta$. If the state space has dimension d , the optimal rate is

$$\frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1}),$$

where $\lambda_{\max}(\mathbf{C}^{-1})$ is the largest eigenvalue of the inverse covariance matrix, corresponding to a direction of smallest variance. This is obtained by rescaling the normalisation used in [GM16, Theorem 2.1].

Langevin dynamics provides a natural application because its noise acts only on velocity and reaches position through deterministic coupling. We also compare one-, two- and three-level dynamics for a one-dimensional quadratic potential. Finally, we study which part of the Gaussian argument survives when the potential is no longer quadratic.

Literature and Thesis Outline

The main point of departure for this thesis is the Gaussian case study in [BE26, Section 4.7]. The thesis develops this material into a self-contained account of the optimisation problem. In particular, it supplies proofs for the two-dimensional statements in [BE26, Lemmas 4.33–4.34], and makes the optimal reversible comparison explicit. It also determines the equality cases in the fixed-drift problem, reformulates the arbitrary-dimensional theorem for a general noise budget, gives an explicit three-dimensional optimiser, and illustrates the two-dimensional acceleration through the corresponding drift fields. Finally, the Langevin discussion is extended by comparing overdamped, kinetic, and three-level dynamics, but the underlying optimal-rate results are attributed to their sources.

Chapter 2 develops the mathematical background needed in the later chapters. The linear-algebraic notation, spectral results and matrix norms mainly follow [SS90, Chapters I–II]. The semidefinite order and its behaviour under congruence transformations are taken from [Tro22, Proposition 12.2 and Definition 12.5]. The stochastic-calculus material is based on [Øks98, Chapters 2–5] and [Sep14, Chapters 1–4], while Gaussian random vectors are treated following [Bil95, Section 29]. The definitions of invariant measures, couplings, Wasserstein distances, and contraction coefficients follow [BE26, Sections 1.3, 3.3, and 4.1–4.3]. The matrix inequality used to obtain weighted contraction estimates is based on [AAS15, Lemma 2.11].

Chapter 3 contains the main Gaussian analysis. The characterisation of invariant Gaussian diffusions and the exact Wasserstein contraction formula follow [BE26, Theorems 4.31–4.32]. The two-dimensional problems are based on [BE26, Lemmas 4.33–4.34], for which complete proofs and further comparisons are provided here. The arbitrary-dimensional argument uses the trace bound and antisymmetric construction of [LNP13, Propositions 1–4 and Lemma 2], together with the optimisation result [GM16, Theorem 2.1]. The Gaussian kinetic Langevin calculation follows [BE26, Theorem 4.35]. The three-level process is the special case of the generalised Langevin equation described in [OP11, (7)], and its invariant law is obtained by specialising [OP11, (10)]. Part of the polynomial argument is adapted from [SB25, Lemma III.1].

Chapter 4 moves beyond Gaussian targets. It starts with the extension outlined in [BE26, page 133]. The use of synchronous coupling under uniform Hessian bounds is related to [DR20, Section 2, Theorem 1 and Proposition 1]. To explain what remains possible when these bounds are weakened, we refer to the contraction theorem and coupling construction in [EGZ19, Section 2.4, Theorem 2.3 and Sections 3–5].

Chapter 5 does not rely on an additional external theorem. It applies the two-dimensional results of *Chapter 3* to the target covariance $\text{diag}(9, 1)$. *Chapter 6* summarises the mathematical conclusions, their limitations, and directions for future work.

Chapter 2

Mathematical Background

This chapter collects the notation and mathematical results used throughout the thesis. We assume familiarity with elementary linear algebra, measure-theoretic probability, multivariable calculus, and differential equations.

2.1 Linear Algebra Preliminaries

We begin by fixing the matrix notation and recalling the linear-algebraic facts needed in the following chapters, with notation following [GM16, Notations] and [SS90, Chapter I, Section 1].

We write $\mathbf{I}_d$ for the $d \times d$ identity matrix, $\mathbf{M}^\top$ for the *transpose* of a real matrix $\mathbf{M}$, and $\mathbf{R}^{-\top} := (\mathbf{R}^{-1})^\top$ whenever $\mathbf{R}$ is invertible. For a square matrix $\mathbf{M}$, its *trace* and *determinant* are denoted by $\text{tr}(\mathbf{M})$ and $\det(\mathbf{M})$. In particular,

$$\text{tr}(\mathbf{M}\mathbf{N}) = \text{tr}(\mathbf{N}\mathbf{M}), \quad \text{tr}(\mathbf{M}^\top) = \text{tr}(\mathbf{M}),$$

whenever the expressions are defined.

The spaces of real *symmetric* and real *antisymmetric* matrices are denoted by

$$\mathcal{S}_d(\mathbb{R}) := \{\mathbf{S} \in \mathbb{R}^{d \times d} : \mathbf{S}^\top = \mathbf{S}\}, \quad \mathcal{A}_d(\mathbb{R}) := \{\mathbf{J} \in \mathbb{R}^{d \times d} : \mathbf{J}^\top = -\mathbf{J}\}.$$

Every matrix $\mathbf{M} \in \mathbb{R}^{d \times d}$ admits the unique decomposition

$$\mathbf{M} = \underbrace{\frac{\mathbf{M} + \mathbf{M}^\top}{2}}_{\mathbf{M}_s} + \underbrace{\frac{\mathbf{M} - \mathbf{M}^\top}{2}}_{\mathbf{M}_a},$$

where $\mathbf{M}_s$ is symmetric and $\mathbf{M}_a$ is antisymmetric. If $\mathbf{J} \in \mathcal{A}_d(\mathbb{R})$, then

$$x^\top \mathbf{J} x = \left(x^\top \mathbf{J} x\right)^\top = x^\top \mathbf{J}^\top x = -x^\top \mathbf{J} x,$$

and hence $x^\top \mathbf{J} x = 0$ for every $x \in \mathbb{R}^d$. Consequently, $x^\top \mathbf{M} x = x^\top \mathbf{M}_s x$, so a quadratic form only sees the symmetric part of a matrix. We shall also use $\text{tr}(\mathbf{J}) = 0$ and $\text{tr}(\mathbf{S}\mathbf{J}) = 0$ whenever $\mathbf{S}$ is symmetric and $\mathbf{J}$ is antisymmetric.

For $\mathbf{S} \in \mathcal{S}_d(\mathbb{R})$, the associated *quadratic* form is $q_{\mathbf{S}}(x) := x^\top \mathbf{S} x$. The matrix $\mathbf{S}$ is called *positive semidefinite*, written $\mathbf{S} \succeq 0$, if $q_{\mathbf{S}}(x) \geq 0$ for every $x \in \mathbb{R}^d$. It is called *positive definite*, written $\mathbf{S} \succ 0$,

if $q_{\mathbf{S}}(x) > 0$ for every non-zero $x \in \mathbb{R}^d$. We use the notation

$$\mathcal{S}_d^{\geq 0}(\mathbb{R}) := \{\mathbf{S} \in \mathcal{S}_d(\mathbb{R}) : \mathbf{S} \succeq 0\}, \quad \mathcal{S}_d^{> 0}(\mathbb{R}) := \{\mathbf{S} \in \mathcal{S}_d(\mathbb{R}) : \mathbf{S} \succ 0\}.$$

Definition 2.1 (Semidefinite Order, [Tro22, Definition 12.5]). For $\mathbf{S}, \mathbf{T} \in \mathcal{S}_d(\mathbb{R})$, we write $\mathbf{S} \preceq \mathbf{T}$ if $\mathbf{T} - \mathbf{S}$ is positive semidefinite. Equivalently, $x^\top \mathbf{S}x \leq x^\top \mathbf{T}x$ for every $x \in \mathbb{R}^d$. This partial order is called the *semidefinite order*, or the *Loewner order*. It should not be confused with entrywise comparison.

Proposition 2.2 ([Tro22, Proposition 12.2]). Let $\mathbf{S}, \mathbf{T} \in \mathcal{S}_d(\mathbb{R})$ and let $\mathbf{R} \in \mathbb{R}^{d \times m}$. If $\mathbf{S} \preceq \mathbf{T}$, then $\mathbf{R}^\top \mathbf{S} \mathbf{R} \preceq \mathbf{R}^\top \mathbf{T} \mathbf{R}$. If $\mathbf{R}$ is square and invertible, the converse also holds.

Proof. For every $z \in \mathbb{R}^m$,

$$z^\top (\mathbf{R}^\top \mathbf{T} \mathbf{R} - \mathbf{R}^\top \mathbf{S} \mathbf{R}) z = z^\top \mathbf{R}^\top (\mathbf{T} - \mathbf{S}) \mathbf{R} z = (\mathbf{R} z)^\top (\mathbf{T} - \mathbf{S}) (\mathbf{R} z) \geq 0.$$

This proves the first assertion. If $\mathbf{R}$ is invertible, the converse follows by applying the same argument to $\mathbf{R}^{-1}$. $\square$

Spectra and the Spectral Theorem

Next, we introduce the spectral quantities; see [SS90, Chapter I, Sections 3.1–3.3]. Let $\mathbf{A} \in \mathbb{R}^{d \times d}$. A number $\lambda \in \mathbb{C}$ is an *eigenvalue* of $\mathbf{A}$ if there exists a non-zero vector $x \in \mathbb{C}^d$ such that $\mathbf{A}x = \lambda x$. Such a vector is called a *right* eigenvector. A non-zero vector $y \in \mathbb{C}^d$ is a *left* eigenvector associated with λ if $y^\mathsf{H} \mathbf{A} = \lambda y^\mathsf{H}$.¹ The set of eigenvalues is the *spectrum* of $\mathbf{A}$, denoted by $\text{spec}(\mathbf{A})$. Equivalently, the eigenvalues are the roots of the *characteristic polynomial* $p_{\mathbf{A}}(z) := \det(z\mathbf{I}_d - \mathbf{A})$. The *spectral radius* of $\mathbf{A}$ and the minimum real part of its eigenvalues are defined by

$$\varrho(\mathbf{A}) := \max_{\lambda \in \text{spec}(\mathbf{A})} |\lambda|, \quad \lambda^*(\mathbf{A}) := \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

The *algebraic multiplicity* of an eigenvalue is its multiplicity as a root of $p_{\mathbf{A}}$, whereas its *geometric multiplicity* is $\dim \ker(\mathbf{A} - \lambda \mathbf{I}_d)$. An eigenvalue is *defective* if its geometric multiplicity is strictly smaller than its algebraic multiplicity, and *non-defective* if the two multiplicities coincide. A matrix is *diagonalisable* over $\mathbb{C}$ precisely when all its eigenvalues are non-defective.

Two matrices $\mathbf{A}$ and $\tilde{\mathbf{A}}$ are *similar* if $\tilde{\mathbf{A}} = \mathbf{R}^{-1} \mathbf{A} \mathbf{R}$ for some invertible matrix $\mathbf{R}$. Similar matrices have the same characteristic polynomial, and therefore the same spectrum. In particular, a change of coordinates does not change $\lambda^*(\mathbf{A})$.

Let $\lambda_1, \dots, \lambda_d$ be eigenvalues of $\mathbf{A}$, repeated according to their algebraic multiplicities. Then

$$p_{\mathbf{A}}(z) = \prod_{j=1}^d (z - \lambda_j), \quad \text{tr}(\mathbf{A}) = \sum_{j=1}^d \lambda_j, \quad \det(\mathbf{A}) = \prod_{j=1}^d \lambda_j.$$

If $\mathbf{A}$ is real, its non-real eigenvalues occur in complex-conjugate pairs. If $\mathbf{A}$ is block upper triangular with diagonal blocks $\mathbf{A}_{11}, \dots, \mathbf{A}_{kk}$, then

$$p_{\mathbf{A}}(z) = \prod_{j=1}^k p_{\mathbf{A}_{jj}}(z), \quad \text{spec}(\mathbf{A}) = \bigcup_{j=1}^k \text{spec}(\mathbf{A}_{jj}),$$

¹For a complex vector $z \in \mathbb{C}^d$, we write $z^\mathsf{H} := \bar{z}^\top$ for its *conjugate transpose*.

where algebraic multiplicities are retained.

Theorem 2.3 (Spectral Theorem, [SS90, Chapter I, Section 3.2]). *For every $\mathbf{S} \in \mathcal{S}_d(\mathbb{R})$, there exist an orthogonal matrix $\mathbf{Q}$ and real numbers $s_1, \dots, s_d$ such that*

$$\mathbf{S} = \mathbf{Q} \text{diag}(s_1, \dots, s_d) \mathbf{Q}^\top, \quad \mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_d.$$

The columns of $\mathbf{Q}$ form an orthonormal basis of eigenvectors of $\mathbf{S}$, and $s_1, \dots, s_d$ are its eigenvalues.

We use this standard theorem without proof. It immediately yields several consequences.

- Define

$$\lambda_{\min}(\mathbf{S}) := \min_{1 \leq i \leq d} s_i, \quad \lambda_{\max}(\mathbf{S}) := \max_{1 \leq i \leq d} s_i.$$

Then

$$\lambda_{\min}(\mathbf{S}) \|x\|^2 \leq x^\top \mathbf{S} x \leq \lambda_{\max}(\mathbf{S}) \|x\|^2$$

for every $x \in \mathbb{R}^d$. Thus, $\mathbf{S} \succeq 0$ if and only if all its eigenvalues are non-negative, and $\mathbf{S} \succ 0$ if and only if all its eigenvalues are strictly positive.

- If $\mathbf{S} \succeq 0$, its unique symmetric positive semidefinite square root is

$$\mathbf{S}^{\frac{1}{2}} = \mathbf{Q} \text{diag}(\sqrt{s_1}, \dots, \sqrt{s_d}) \mathbf{Q}^\top.$$

In particular, $\mathbf{S} = \mathbf{S}^{\frac{1}{2}} (\mathbf{S}^{\frac{1}{2}})^\top$, so every positive semidefinite matrix admits a factorisation of the form $\mathbf{S} = \mathbf{R} \mathbf{R}^\top$ for some $\mathbf{R} \in \mathbb{R}^{d \times d}$. When $\mathbf{S} \succ 0$, the matrices $\mathbf{S}^{-1}$ and $\mathbf{S}^{-\frac{1}{2}}$ are also symmetric and positive definite.

- Orthogonal changes of coordinates preserve symmetry, antisymmetry, positive definiteness and the semidefinite order. They also preserve the trace and spectrum:

$$\text{tr}(\mathbf{Q}^\top \mathbf{M} \mathbf{Q}) = \text{tr}(\mathbf{M}), \quad \text{spec}(\mathbf{Q}^\top \mathbf{M} \mathbf{Q}) = \text{spec}(\mathbf{M}).$$

- The eigenvalues of a real antisymmetric matrix are purely imaginary. Indeed, if $\mathbf{J}v = \lambda v$ for $v \in \mathbb{C}^d \setminus \{0\}$, then

$$\lambda \|v\|^2 = v^\mathbf{H} \mathbf{J} v = -\overline{v^\mathbf{H} \mathbf{J} v},$$

and therefore $\text{Re } \lambda = 0$.

Matrix Norms and Lyapunov Inequalities

Now we turn to the norms used in finite-time estimates; see [SS90, Chapter II, Sections 1–2]. For a matrix $\mathbf{M} \in \mathbb{R}^{r \times s}$, the *operator norm* (or *spectral norm*) induced by the Euclidean norms is

$$\|\mathbf{M}\| := \sup_{x \in \mathbb{R}^s \setminus \{0\}} \frac{\|\mathbf{M}x\|}{\|x\|} = \sup_{\substack{x \in \mathbb{R}^s \\ \|x\|=1}} \|\mathbf{M}x\|.$$

The *Frobenius norm* is

$$\|\mathbf{M}\|_F := \left(\sum_{i=1}^r \sum_{j=1}^s |M_{ij}|^2 \right)^{\frac{1}{2}} = \sqrt{\text{tr}(\mathbf{M}^\top \mathbf{M})}.$$

The operator norm satisfies

$$\|\mathbf{M}x\| \leq \|\mathbf{M}\|\|x\|, \quad \|\mathbf{M}\mathbf{N}\| \leq \|\mathbf{M}\|\|\mathbf{N}\|,$$

whenever the products are defined. For a square matrix,

$$\varrho(\mathbf{M}) \leq \|\mathbf{M}\|, \quad \|\mathbf{M}\|^2 = \lambda_{\max}(\mathbf{M}^\top \mathbf{M}).$$

Moreover,

$$\|\mathbf{M}\| \leq \|\mathbf{M}\|_F \leq \sqrt{\text{rank}(\mathbf{M})} \|\mathbf{M}\| \leq \sqrt{\min\{r, s\}} \|\mathbf{M}\|.$$

Definition 2.4 ([SS90, Chapter II, Corollary 1.5]). Let $\mathbf{G} \in \mathcal{S}_d^{>0}(\mathbb{R})$. For $x, y \in \mathbb{R}^d$, define

$$\langle x, y \rangle_{\mathbf{G}} := x^\top \mathbf{G} y, \quad \|x\|_{\mathbf{G}} := \sqrt{\langle x, x \rangle_{\mathbf{G}}} = \sqrt{x^\top \mathbf{G} x}.$$

Then $\langle \cdot, \cdot \rangle_{\mathbf{G}}$ is an inner product on $\mathbb{R}^d$, and $\|\cdot\|_{\mathbf{G}}$ is its induced norm, called the $\mathbf{G}$ -weighted Euclidean norm.

The *spectral theorem* gives the quantitative equivalence

$$\sqrt{\lambda_{\min}(\mathbf{G})} \|x\| \leq \|x\|_{\mathbf{G}} \leq \sqrt{\lambda_{\max}(\mathbf{G})} \|x\|.$$

It is convenient to denote the spectral condition number of $\mathbf{G}$ by

$$\bar{\kappa}(\mathbf{G}) := \frac{\lambda_{\max}(\mathbf{G})}{\lambda_{\min}(\mathbf{G})} = \|\mathbf{G}\| \|\mathbf{G}^{-1}\|.$$

The factor $\sqrt{\bar{\kappa}(\mathbf{G})}$ measures the cost of passing between the weighted and Euclidean geometries.

Definition 2.5 (Matrix Exponential, [SS90, Chapter II, Section 2, Exercise 2.22]). For $\mathbf{A} \in \mathbb{R}^{d \times d}$ and $t \in \mathbb{R}$, the matrix exponential of $t\mathbf{A}$ is defined by

$$e^{t\mathbf{A}} := \sum_{k=0}^{\infty} \frac{t^k \mathbf{A}^k}{k!}.$$

Let $\mathbf{A} \in \mathbb{R}^{d \times d}$ and $s, t \in \mathbb{R}$. Then

$$e^{0\mathbf{A}} = \mathbf{I}_d, \quad e^{s\mathbf{A}} e^{t\mathbf{A}} = e^{(s+t)\mathbf{A}}, \quad (e^{t\mathbf{A}})^{-1} = e^{-t\mathbf{A}}, \quad (e^{t\mathbf{A}})^\top = e^{t\mathbf{A}^\top}.$$

If $\mathbf{R}$ is invertible, then

$$e^{t\mathbf{R}^{-1}\mathbf{A}\mathbf{R}} = \mathbf{R}^{-1} e^{t\mathbf{A}} \mathbf{R}.$$

Moreover,

$$\frac{d}{dt} e^{t\mathbf{A}} = \mathbf{A} e^{t\mathbf{A}} = e^{t\mathbf{A}} \mathbf{A}.$$

Consequently, the solution of $\dot{x}_t = -\mathbf{A}x_t$ with $x_0 = x$ is $x_t = e^{-t\mathbf{A}}x$.

We shall also use the spectral mapping identity for matrix exponentials. It follows directly from the Jordan canonical form together with the similarity rule for matrix exponentials; see [SS90, Chapter I, Sections 3.3–3.4, and Chapter II, Exercise 2.22]. Since this is a standard consequence of these results, we state it without proof.

Proposition 2.6. *Let $\mathbf{A} \in \mathbb{R}^{d \times d}$. Then, for every $t \in \mathbb{R}$,*

$$\text{spec}(e^{t\mathbf{A}}) = \left\{ e^{t\lambda} : \lambda \in \text{spec}(\mathbf{A}) \right\}.$$

In particular, for $t \geq 0$,

$$\varrho(e^{-t\mathbf{A}}) = e^{-t\lambda^*(\mathbf{A})}, \quad \lambda^*(\mathbf{A}) = \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

The following formulation combines the two cases of [AAS15, Lemma 2.11] and states the ε -estimate for every matrix. Here we give a direct proof using the real invariant-subspace decomposition associated with the Jordan canonical form; see [SS90, Chapter I, Sections 3.3–3.4].

Lemma 2.7. *Let $\mathbf{A} \in \mathbb{R}^{d \times d}$ and set*

$$\lambda^* := \lambda^*(\mathbf{A}) = \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

If every eigenvalue satisfying $\text{Re } \lambda = \lambda^$ is non-defective, then there exists $\mathbf{G} \in \mathcal{S}_d^{>0}(\mathbb{R})$ such that*

$$\mathbf{A}^\top \mathbf{G} + \mathbf{G} \mathbf{A} \succeq 2\lambda^* \mathbf{G}.$$

For an arbitrary matrix $\mathbf{A}$ and every $\varepsilon > 0$, there exists $\mathbf{G}_\varepsilon \in \mathcal{S}_d^{>0}(\mathbb{R})$ such that

$$\mathbf{A}^\top \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon \mathbf{A} \succeq 2(\lambda^* - \varepsilon) \mathbf{G}_\varepsilon.$$

Proof. We first prove the assertion with the loss ε . Set

$$\mathbf{B}_\varepsilon := \mathbf{A} - (\lambda^* - \varepsilon) \mathbf{I}_d.$$

Every eigenvalue of $\mathbf{B}_\varepsilon$ has real part at least ε . By the Jordan canonical form, each entry of $e^{-t\mathbf{B}_\varepsilon}$ is a finite sum of terms of the form $p(t)e^{-t\lambda}$, where p is a polynomial and $\text{Re } \lambda \geq \varepsilon$. Since $p(t)e^{-\frac{\varepsilon t}{2}}$ is bounded on $[0, \infty)$, there exists a constant $C_\varepsilon > 0$ such that

$$\|e^{-t\mathbf{B}_\varepsilon}\| \leq C_\varepsilon e^{-\frac{\varepsilon t}{2}}, \quad t \geq 0.$$

Consequently,

$$\int_0^\infty \|e^{-t\mathbf{B}_\varepsilon}\|^2 dt < \infty.$$

Define

$$\mathbf{G}_\varepsilon := \int_0^\infty e^{-t\mathbf{B}_\varepsilon^\top} e^{-t\mathbf{B}_\varepsilon} dt.$$

The matrix $\mathbf{G}_\varepsilon$ is symmetric. For every $v \neq 0$, the function $t \mapsto \|e^{-t\mathbf{B}_\varepsilon} v\|^2$ is continuous and has value $\|v\|^2 > 0$ at $t = 0$. Therefore,

$$v^\top \mathbf{G}_\varepsilon v = \int_0^\infty v^\top e^{-t\mathbf{B}_\varepsilon^\top} e^{-t\mathbf{B}_\varepsilon} v dt = \int_0^\infty \|e^{-t\mathbf{B}_\varepsilon} v\|^2 dt > 0.$$

Thus $\mathbf{G}_\varepsilon$ is positive definite. Furthermore,

$$\frac{d}{dt} \left(e^{-t\mathbf{B}_\varepsilon^\top} e^{-t\mathbf{B}_\varepsilon} \right) = -\mathbf{B}_\varepsilon^\top e^{-t\mathbf{B}_\varepsilon^\top} e^{-t\mathbf{B}_\varepsilon} - e^{-t\mathbf{B}_\varepsilon^\top} e^{-t\mathbf{B}_\varepsilon} \mathbf{B}_\varepsilon.$$

The estimate above implies that $e^{-t\mathbf{B}_\varepsilon} \rightarrow 0$ as $t \rightarrow \infty$. Therefore, integrating from 0 to ∞ gives the *Lyapunov identity*

$$\mathbf{B}_\varepsilon^\top \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon \mathbf{B}_\varepsilon = \mathbf{I}_d.$$

Since $\mathbf{A} = \mathbf{B}_\varepsilon + (\lambda^* - \varepsilon)\mathbf{I}_d$, it follows that

$$\begin{aligned} \mathbf{A}^\top \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon \mathbf{A} &= \left(\mathbf{B}_\varepsilon^\top + (\lambda^* - \varepsilon)\mathbf{I}_d \right) \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon (\mathbf{B}_\varepsilon + (\lambda^* - \varepsilon)\mathbf{I}_d) \\ &= \mathbf{B}_\varepsilon^\top \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon \mathbf{B}_\varepsilon + 2(\lambda^* - \varepsilon)\mathbf{G}_\varepsilon \\ &= \mathbf{I}_d + 2(\lambda^* - \varepsilon)\mathbf{G}_\varepsilon \succeq 2(\lambda^* - \varepsilon)\mathbf{G}_\varepsilon. \end{aligned}$$

We now suppose that every eigenvalue satisfying $\operatorname{Re} \lambda = \lambda^*$ is non-defective. Let E_* be the real invariant subspace associated with these boundary eigenvalues, and set $r := \dim_{\mathbb{R}} E_*$. By the real form of the Jordan decomposition, there exists a real invertible matrix $\mathbf{S}$ such that

$$\hat{\mathbf{A}} := \mathbf{S}^{-1} \mathbf{A} \mathbf{S} = \begin{pmatrix} \lambda^* \mathbf{I} + \mathbf{K} & 0 \\ 0 & \mathbf{A}_+ \end{pmatrix},$$

where $\mathbf{K}^\top = -\mathbf{K}$ and every eigenvalue of $\mathbf{A}_+$ has real part strictly greater than λ^* . When $r = d$, the lower block and all subsequent terms involving $\mathbf{A}_+$, $\mathbf{B}_+$, and $\mathbf{G}_+$ are omitted, and we take $\mathbf{H} = \mathbf{I}_d$. Define

$$\mathbf{B}_+ := \mathbf{A}_+ - \lambda^* \mathbf{I}_{d-r}, \quad \mathbf{G}_+ := \int_0^\infty e^{-t\mathbf{B}_+^\top} e^{-t\mathbf{B}_+} dt.$$

By the calculation in the first part, $\mathbf{B}_+^\top \mathbf{G}_+ + \mathbf{G}_+ \mathbf{B}_+ = \mathbf{I}_{d-r}$. Using $\mathbf{A}_+ = \mathbf{B}_+ + \lambda^* \mathbf{I}_{d-r}$, we obtain

$$\mathbf{A}_+^\top \mathbf{G}_+ + \mathbf{G}_+ \mathbf{A}_+ = 2\lambda^* \mathbf{G}_+ + \mathbf{I}_{d-r}.$$

Now let

$$\mathbf{H} := \begin{pmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{G}_+ \end{pmatrix}, \quad \mathbf{G} := \mathbf{S}^{-\top} \mathbf{H} \mathbf{S}^{-1}.$$

Then $\mathbf{G}$ is real, symmetric and positive definite. Consequently,

$$\begin{aligned} \mathbf{A}^\top \mathbf{G} + \mathbf{G} \mathbf{A} &= \mathbf{S}^{-\top} \left(\hat{\mathbf{A}}^\top \mathbf{H} + \mathbf{H} \hat{\mathbf{A}} \right) \mathbf{S}^{-1} \\ &= \mathbf{S}^{-\top} \begin{pmatrix} 2\lambda^* \mathbf{I}_r & 0 \\ 0 & 2\lambda^* \mathbf{G}_+ + \mathbf{I}_{d-r} \end{pmatrix} \mathbf{S}^{-1} \\ &\succeq 2\lambda^* \mathbf{S}^{-\top} \mathbf{H} \mathbf{S}^{-1} = 2\lambda^* \mathbf{G}, \end{aligned}$$

which completes the proof. □

2.2 Gaussian Processes and Stochastic Differential Equations

Throughout this section, d denotes the dimension of the state space and m the dimension of the driving Brownian motion (BM). For the standard background, see [Øks98, Chapters 2–5] and [Sep14, Chapters 1–4].

Gaussian Random Vectors

The characteristic-function formulation below follows [Sep14, Example 1.18(iv)], while its equivalent characterisation by one-dimensional projections follows [Bil95, Section 29].

Definition 2.8 (Gaussian Random Vectors). *An $\mathbb{R}^d$ -valued random vector X is called Gaussian if $u^\top X$ is a one-dimensional Gaussian random variable for every $u \in \mathbb{R}^d$. Its mean and covariance matrix are denoted by*

$$m_X := \mathbb{E}[X], \quad \mathbf{C}_X := \text{Cov}(X) = \mathbb{E}\left[(X - m_X)(X - m_X)^\top\right],$$

and we write $X \sim \mathcal{N}(m_X, \mathbf{C}_X)$. Equivalently, the characteristic function of X is

$$\mathbb{E}\left[e^{iu^\top X}\right] = \exp\left(iu^\top m_X - \frac{1}{2}u^\top \mathbf{C}_X u\right), \quad u \in \mathbb{R}^d.$$

The covariance matrix $\mathbf{C}_X$ is symmetric and positive semidefinite, but it need not be invertible. Thus degenerate Gaussian distributions are included, and $\mathcal{N}(m, 0) = \delta_m$ is the point mass at m . If $\mathbf{C}_X \succ 0$, then $\mathcal{N}(m_X, \mathbf{C}_X)$ has density

$$x \mapsto \frac{1}{(2\pi)^{\frac{d}{2}} \det(\mathbf{C}_X)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - m_X)^\top \mathbf{C}_X^{-1}(x - m_X)\right)$$

with respect to Lebesgue measure. For any square-integrable random vector X ,

$$\mathbb{E}\|X\|^2 = \|m_X\|^2 + \text{tr}(\mathbf{C}_X).$$

Proposition 2.9 ([Bil95, Section 29]). *Let $X \sim \mathcal{N}(m_X, \mathbf{C}_X)$ in $\mathbb{R}^d$.*

(i) *If $\mathbf{R} \in \mathbb{R}^{k \times d}$ and $a \in \mathbb{R}^k$, then*

$$\mathbf{R}X + a \sim \mathcal{N}\left(\mathbf{R}m_X + a, \mathbf{R}\mathbf{C}_X\mathbf{R}^\top\right).$$

(ii) *Let $Y \sim \mathcal{N}(m_Y, \mathbf{C}_Y)$ in $\mathbb{R}^r$, independently of X . For $\mathbf{R} \in \mathbb{R}^{k \times d}$, $\mathbf{S} \in \mathbb{R}^{k \times r}$, and $a \in \mathbb{R}^k$,*

$$\mathbf{R}X + \mathbf{S}Y + a \sim \mathcal{N}\left(\mathbf{R}m_X + \mathbf{S}m_Y + a, \mathbf{R}\mathbf{C}_X\mathbf{R}^\top + \mathbf{S}\mathbf{C}_Y\mathbf{S}^\top\right).$$

In particular, independent Gaussian random vectors are jointly Gaussian.

Gaussian Processes and Filtrations

Let $\mathcal{T} \subseteq [0, \infty)$. An $\mathbb{R}^d$ -valued *stochastic process* indexed by $\mathcal{T}$ is a family $X = (X_t)_{t \in \mathcal{T}}$ of $\mathbb{R}^d$ -valued random vectors defined on a common probability space. For a fixed $\omega \in \Omega$, the map $t \mapsto X_t(\omega)$ is called a *sample path* of X . The following vector-valued definition is adapted from [Sep14, Exercise 1.19].

Definition 2.10 (Gaussian Processes). An $\mathbb{R}^d$ -valued stochastic process $(X_t)_{t \in \mathcal{T}}$ is called a *Gaussian process* if, for every $q \in \mathbb{N}$ and $t_1, \dots, t_q \in \mathcal{T}$, the random vector

$$\begin{pmatrix} X_{t_1} \\ \vdots \\ X_{t_q} \end{pmatrix} \in \mathbb{R}^{qd}$$

is Gaussian.

Thus Gaussianity is a statement about all *finite-dimensional distributions* of the process. It is not enough that each X_t be Gaussian: the joint law at every finite collection of times must also be Gaussian.

A *filtration* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a family of sub- σ -algebras $(\mathcal{F}_t)_{t \geq 0}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ whenever $0 \leq s \leq t$. Then $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ is called a *filtered probability space*. A process $(X_t)_{t \geq 0}$ is adapted to the filtration if X_t is $\mathcal{F}_t$ -measurable for every $t \geq 0$. Its natural filtration is $\mathcal{F}_t^X := \sigma(X_s : 0 \leq s \leq t)$.

From now on, the filtration is assumed to satisfy the usual conditions.² Unless stated otherwise, all processes are adapted to the filtration under consideration.

Brownian Motion

The following is the standard m -dimensional extension of [Sep14, Definition 2.26].

Definition 2.11 (Standard BM). An adapted $\mathbb{R}^m$ -valued process $B = (B_t)_{t \geq 0}$ is called a *standard m -dimensional BM with respect to $(\mathcal{F}_t)_{t \geq 0}$* if:

- (i) $B_0 = 0$ almost surely (a.s.).
- (ii) Almost every path $t \mapsto B_t(\omega)$ is continuous.
- (iii) For every $0 \leq s < t$, the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ and $B_t - B_s \sim \mathcal{N}(0, (t - s)\mathbf{I}_m)$.

The distribution of an increment depends only on its length $t - s$, so BM has *stationary* increments. Its increments over disjoint time intervals are independent. Moreover, for every fixed $r \geq 0$, the shifted process

$$B_u^{(r)} := B_{r+u} - B_r, \quad u \geq 0,$$

is again a standard BM and is independent of $\mathcal{F}_r$; see [Sep14, Proposition 2.29(b)].

The Itô Integral

Brownian paths are almost surely of unbounded variation on every non-trivial compact interval, so the classical Riemann–Stieltjes construction does not provide a general pathwise integral [Sep14, Corollary 2.43]. The Itô integral is instead obtained as an L^2 -limit of integrals of adapted step processes [Øks98, Section 3.1]; below we use its standard componentwise extension to $\mathbb{R}^{d \times m}$ -valued integrands.

²The filtration $(\mathcal{F}_t)_{t \geq 0}$ is complete ($\mathcal{F}_0$ contains all subsets of $\mathbb{P}$ -null sets in $\mathcal{F}$) and right-continuous.

Definition 2.12 (Itô Integral, [Øks98, Definitions 3.1.6 and 3.3.1]). *Fix $T > 0$. An elementary adapted process with values in $\mathbb{R}^{d \times m}$ is a process of the form*

$$H_t = \sum_{k=0}^{N-1} H_k \mathbb{1}_{(t_k, t_{k+1}]}(t), \quad 0 = t_0 < t_1 < \dots < t_N = T,$$

where each H_k is $\mathcal{F}_{t_k}$ -measurable and square-integrable. Its Itô integral is defined by

$$\int_0^T H_s dB_s := \sum_{k=0}^{N-1} H_k (B_{t_{k+1}} - B_{t_k}).$$

The coefficient H_k is chosen using the information available at time t_k , before the next Brownian increment is observed. This is the essential adaptedness requirement.

Now let H be a jointly measurable, adapted $\mathbb{R}^{d \times m}$ -valued process satisfying

$$\mathbb{E} \left[\int_0^T \|H_t\|_{\mathbb{F}}^2 dt \right] < \infty.$$

The approximation construction in [Øks98, Section 3.1] provides elementary adapted processes $H^{(q)}$ such that

$$\mathbb{E} \left[\int_0^T \|H_t^{(q)} - H_t\|_{\mathbb{F}}^2 dt \right] \rightarrow 0.$$

The Itô integral of H is defined by the $L^2(\Omega; \mathbb{R}^d)$ -limit³

$$\int_0^T H_t dB_t := \lim_{q \rightarrow \infty} \int_0^T H_t^{(q)} dB_t.$$

Corollary 2.13 (Itô Isometry, [Øks98, Corollary 3.1.7 and Theorem 3.2.1(iii)]). *If H satisfies the assumptions in the preceding definition, then*

$$\mathbb{E} \left[\int_0^T H_t dB_t \right] = 0, \quad \mathbb{E} \left[\left\| \int_0^T H_t dB_t \right\|^2 \right] = \mathbb{E} \left[\int_0^T \|H_t\|_{\mathbb{F}}^2 dt \right].$$

The following result is the finite-dimensional matrix-valued version of the isonormal-process construction.

Proposition 2.14 ([Sep14, Section 9.1.1 and Exercise 9.7(c)]). *For $j = 1, \dots, q$, let $F_j : [0, T] \rightarrow \mathbb{R}^{r_j \times m}$ be deterministic measurable functions satisfying*

$$\int_0^T \|F_j(s)\|_{\mathbb{F}}^2 ds < \infty,$$

and define

$$Z_j := \int_0^T F_j(s) dB_s \in \mathbb{R}^{r_j}.$$

³Here $L^2(\Omega; \mathbb{R}^d)$ is shorthand for $L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^d)$, the space of square-integrable $\mathbb{R}^d$ -valued random variables. Thus $Y_q \rightarrow Y$ in $L^2(\Omega; \mathbb{R}^d)$ means that $\mathbb{E}[\|Y_q - Y\|^2] \rightarrow 0$.

Then

$$\begin{pmatrix} Z_1 \\ \vdots \\ Z_q \end{pmatrix} \in \mathbb{R}^{r_1 + \dots + r_q}$$

is a centred Gaussian random vector, with cross-covariance blocks

$$\text{Cov}(Z_i, Z_j) = \mathbb{E}[Z_i Z_j^\top] = \int_0^T F_i(s) F_j(s)^\top ds.$$

Stochastic Differential Equations

Let B be a standard m -dimensional BM and let

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \Sigma : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$$

be measurable functions. We consider the stochastic differential equation (SDE)

$$dX_t = b(t, X_t)dt + \Sigma(t, X_t)dB_t, \quad X_0 = \xi.$$

This differential notation is shorthand for an integral equation.

The following definition is adapted from [Øks98, Sections 5.2–5.3]. To remain within the L^2 -construction of the Itô integral given above, we impose the slightly stronger condition of square-integrability in expectation on the diffusion coefficient.

Definition 2.15 (Strong Solutions). *Suppose that the filtered probability space and the BM B have already been fixed, and let ξ be a $\mathcal{F}_0$ -measurable initial random vector. A continuous adapted process $X = (X_t)_{0 \leq t \leq T}$ is called a strong solution if*

$$\int_0^T \|b(s, X_s)\| ds < \infty \quad a.s.,$$

and

$$\mathbb{E} \left[\int_0^T \|\Sigma(s, X_s)\|_F^2 ds \right] < \infty,$$

and if

$$X_t = \xi + \int_0^t b(s, X_s) ds + \int_0^t \Sigma(s, X_s) dB_s$$

holds almost surely for all $t \in [0, T]$.

Theorem 2.16 (Existence and Uniqueness for SDEs, [Øks98, Theorem 5.2.1]). *Let $T > 0$, and let*

$$b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \Sigma : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$$

be measurable functions. Suppose that there exist constants $C, L > 0$ such that

$$\|b(t, x)\| + \|\Sigma(t, x)\|_F \leq C(1 + \|x\|), \quad \|b(t, x) - b(t, y)\| + \|\Sigma(t, x) - \Sigma(t, y)\|_F \leq L\|x - y\|$$

for every $t \in [0, T]$ and $x, y \in \mathbb{R}^d$. Let ξ be an $\mathcal{F}_0$ -measurable $\mathbb{R}^d$ -valued random variable, independent

of $\sigma(B_s : s \geq 0)$, and satisfying $\mathbb{E}[\|\xi\|^2] < \infty$. Then the SDE

$$dX_t = b(t, X_t)dt + \Sigma(t, X_t)dB_t, \quad X_0 = \xi, \quad 0 \leq t \leq T$$

has a unique continuous strong solution on $[0, T]$.

Itô's Formula

Theorem 2.17 (The General Itô Formula, [Øks98, Theorem 4.2.1]). *Let X satisfy*

$$dX_t = b(t, X_t)dt + \Sigma(t, X_t)dB_t$$

in $\mathbb{R}^d$, and let $f \in C^{1,2}([0, T] \times \mathbb{R}^d; \mathbb{R})$.⁴ Then

$$\begin{aligned} df(t, X_t) = & \left[\partial_t f(t, X_t) + \nabla f(t, X_t)^\top b(t, X_t) + \frac{1}{2} \text{tr} \left(\Sigma(t, X_t) \Sigma(t, X_t)^\top \nabla^2 f(t, X_t) \right) \right] dt \\ & + \nabla f(t, X_t)^\top \Sigma(t, X_t) dB_t. \end{aligned}$$

Here ∇f and $\nabla^2 f$ denote the gradient and Hessian with respect to the spatial variable.

Let $\mathbf{M} : [0, T] \rightarrow \mathbb{R}^{r \times d}$ be continuously differentiable and deterministic, and let X satisfy an SDE of the preceding form. Applying *Itô's formula* componentwise gives

$$d(\mathbf{M}(t)X_t) = \dot{\mathbf{M}}(t)X_t dt + \mathbf{M}(t)dX_t. \quad (2.1)$$

2.3 Markov Semigroups, Generators, and Invariant Measures

Let $X = (X_t)_{t \geq 0}$ be an $\mathbb{R}^d$ -valued stochastic process adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$. For each $x \in \mathbb{R}^d$, let $\mathbb{P}_x$ denote the *law* governing X when the process is started from x , so that $\mathbb{P}_x(X_0 = x) = 1$. We denote *expectation* with respect to $\mathbb{P}_x$ by $\mathbb{E}_x$. We also write $\mathcal{B}(\mathbb{R}^d)$ for the Borel σ -algebra on $\mathbb{R}^d$, and $B_b(\mathbb{R}^d)$ for the space of bounded Borel-measurable functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

Transition Kernels and Transition Semigroups

The notation and basic facts in this subsection are adapted from [Sep14, Definition 2.22 and (2.26)-(2.28)]. A map

$$p : \mathbb{R}^d \times \mathcal{B}(\mathbb{R}^d) \longrightarrow [0, 1]$$

is called a *Markov kernel* if, for every $x \in \mathbb{R}^d$, the map $A \mapsto p(x, A)$ is a probability measure and, for every $A \in \mathcal{B}(\mathbb{R}^d)$, the map $x \mapsto p(x, A)$ is Borel measurable.

Definition 2.18 (Time-Homogeneous Markov Processes). *For $t \geq 0$, define*

$$p_t(x, A) := \mathbb{P}_x(X_t \in A), \quad x \in \mathbb{R}^d, \quad A \in \mathcal{B}(\mathbb{R}^d).$$

⁴Here $C^{1,2}([0, T] \times \mathbb{R}^d; \mathbb{R})$ denotes the space of real-valued functions that are continuously differentiable once in time and twice in space.

Suppose that p_t is a Markov kernel for every $t \geq 0$. For $f \in B_b(\mathbb{R}^d)$, set

$$P_t f(x) := \int_{\mathbb{R}^d} f(y) p_t(x, dy) = \mathbb{E}_x[f(X_t)].$$

The family $(X, \mathbb{P}_x)_{x \in \mathbb{R}^d}$ is called a *time-homogeneous Markov process* if, for every $f \in B_b(\mathbb{R}^d)$ and all $s, t \geq 0$,

$$\mathbb{E}_x[f(X_{s+t}) \mid \mathcal{F}_s] = P_t f(X_s) \quad \mathbb{P}_x\text{-a.s.}$$

Since $\mathbb{P}_x(X_0 = x) = 1$, we have $P_0 = \text{Id}$, where Id denotes the identity operator on $B_b(\mathbb{R}^d)$. Moreover, the *Markov property* implies that for $s, t \geq 0$,

$$P_{s+t} f(x) = \mathbb{E}_x[f(X_{s+t})] = \mathbb{E}_x[\mathbb{E}_x[f(X_{s+t}) \mid \mathcal{F}_s]] = \mathbb{E}_x[P_t f(X_s)] = P_s(P_t f)(x).$$

The family $(P_t)_{t \geq 0}$ is therefore called the *transition semigroup* of the process.

A transition kernel also acts on probability measures. If ν is a probability measure on $\mathbb{R}^d$, define

$$(\nu p_t)(A) := \int_{\mathbb{R}^d} p_t(x, A) \nu(dx), \quad A \in \mathcal{B}(\mathbb{R}^d).$$

If the initial state has distribution ν , then $\text{Law}(X_t) = \nu p_t$. Moreover, for every $f \in B_b(\mathbb{R}^d)$, *Fubini's theorem* gives

$$\int_{\mathbb{R}^d} f(y) (\nu p_t)(dy) = \int_{\mathbb{R}^d} P_t f(x) \nu(dx).$$

Invariant Measures and Reversibility

Definition 2.19 (Invariant Measures, [BE26, Section 1.3]). A probability measure μ on $\mathbb{R}^d$ is called *invariant* for $(p_t)_{t \geq 0}$ if $\mu p_t = \mu$ for every $t \geq 0$. Equivalently,

$$\int_{\mathbb{R}^d} P_t f d\mu = \int_{\mathbb{R}^d} f d\mu$$

for every bounded Borel function f .

If X_0 has distribution μ , invariance implies that X_t also has distribution μ for every $t \geq 0$. In the thesis, such an invariant probability measure is also called an *equilibrium* distribution. Convergence to equilibrium means that νp_t converges to μ , in a specified metric or topology, as $t \rightarrow \infty$. Invariance alone does not guarantee such convergence, nor does it guarantee uniqueness.

Following [BE26, (1.6)–(1.7)], a transition semigroup $(P_t)_{t \geq 0}$ is called *reversible* with respect to a probability measure μ if

$$\mu(dx) p_t(x, dy) = \mu(dy) p_t(y, dx)$$

as measures on $\mathbb{R}^d \times \mathbb{R}^d$, for every $t \geq 0$. Equivalently,

$$\int_{\mathbb{R}^d} f P_t g d\mu = \int_{\mathbb{R}^d} g P_t f d\mu$$

for all $f, g \in B_b(\mathbb{R}^d)$.

Reversibility implies invariance. Taking $f \equiv 1$ and using $P_t 1 = 1$ gives

$$\int P_t g d\mu = \int g d\mu.$$

The converse does not hold in general: a process may preserve μ without being reversible. Such non-reversible dynamics are central to this thesis.

Generators and Infinitesimal Invariance

Definition 2.20 (Infinitesimal Generator, [Øks98, Definition 7.3.1]). *Let $(P_t)_{t \geq 0}$ be a transition semigroup. Its infinitesimal generator L is defined by*

$$Lf(x) := \lim_{t \downarrow 0} \frac{P_t f(x) - f(x)}{t},$$

whenever the limit exists. The collection of functions $f \in B_b(\mathbb{R}^d)$ for which this limit exists for every $x \in \mathbb{R}^d$ is called the domain of the generator and is denoted by $\text{Dom}(L)$.

Following [Øks98, Theorem 7.3.3], let $X = (X_t)_{t \geq 0}$ be the time-homogeneous Markov process determined by the SDE

$$dX_t = b(X_t)dt + \Sigma(X_t)dB_t$$

in $\mathbb{R}^d$, where B is an m -dimensional BM, and b, Σ satisfy the linear-growth and global Lipschitz conditions in *Theorem 2.16*. Set $a(x) := \Sigma(x)\Sigma(x)^\top$. For $f \in C_c^2(\mathbb{R}^d)$,⁵ define the differential operator

$$\mathcal{L}f(x) := b(x)^\top \nabla f(x) + \frac{1}{2} \text{tr}(a(x) \nabla^2 f(x)).$$

Then $C_c^2(\mathbb{R}^d) \subseteq \text{Dom}(L)$. For every $f \in C_c^2(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$,

$$Lf(x) = \lim_{t \downarrow 0} \frac{P_t f(x) - f(x)}{t} = \mathcal{L}f(x).$$

For standard BM, $b = 0$ and $a = \mathbf{I}_d$, so $\mathcal{L}f = \frac{1}{2} \Delta f$, where

$$\Delta f := \text{tr}(\nabla^2 f) = \sum_{i=1}^d \partial_{ii} f$$

is the Laplace operator.

Definition 2.21 (Formal Adjoint, [BE26, Section 7.1]). *In addition to the linear-growth and global Lipschitz conditions imposed above, assume that $b \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ and $a \in C^2(\mathbb{R}^d; \mathbb{R}^{d \times d})$. The formal adjoint $\mathcal{L}^*$ of $\mathcal{L}$, with respect to Lebesgue measure, is the differential operator determined by*

$$\int_{\mathbb{R}^d} (\mathcal{L}f)(x)g(x)dx = \int_{\mathbb{R}^d} f(x)(\mathcal{L}^*g)(x)dx$$

for all $f, g \in C_c^\infty(\mathbb{R}^d)$.

⁵Here $C_c^2(\mathbb{R}^d)$ denotes the space of twice continuously differentiable functions with compact support. The support of f is defined by $\text{supp}(f) := \overline{\{x \in \mathbb{R}^d : f(x) \neq 0\}}$.

Under these assumptions, integration by parts gives

$$\mathcal{L}^* g = - \sum_{i=1}^d \partial_i (b_i g) + \frac{1}{2} \sum_{i,j=1}^d \partial_{ij} (a_{ij} g).$$

Definition 2.22 (Infinitesimal Invariance, [BE26, Section 7.1]). *A probability measure μ on $\mathbb{R}^d$ is called infinitesimally invariant for $\mathcal{L}$ if*

$$\int_{\mathbb{R}^d} \mathcal{L}f(x) \mu(dx) = 0, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

Equivalently, $\mathcal{L}^ \mu = 0$ in the distributional sense. If μ has density ρ with respect to Lebesgue measure, then infinitesimal invariance is equivalent to $\mathcal{L}^* \rho = 0$ as a distribution.*

We will only use the implication from invariance to infinitesimal invariance. Under the assumptions above, [BE26, Theorem 7.1 and Corollary 7.2] show that if μ is invariant, then

$$\int_{\mathbb{R}^d} \mathcal{L}f d\mu = 0, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

Equivalently, $\mathcal{L}^* \mu = 0$ in the distributional sense.

2.4 Couplings and Wasserstein Contraction

Throughout this section, $E = \mathbb{R}^d$ and d_E denotes a continuous metric on E .

Definition 2.23 (Couplings, [BE26, Definitions 3.3, 4.1(i)]). *We write $\mathcal{P}(E)$ for the collection of probability measures on E .*

- (i) *Let $\mu, \nu \in \mathcal{P}(E)$. A coupling of μ and ν is a probability measure $\gamma \in \mathcal{P}(E \times E)$ with marginals μ and ν . The collection of all such couplings is denoted by $\Pi(\mu, \nu)$. A coupling γ can be realised by random variables X and Y on a common probability space such that $(X, Y) \sim \gamma$.*
- (ii) *A coupling of two E -valued stochastic processes $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ is a process $(\tilde{X}_t, \tilde{Y}_t)_{t \geq 0}$ whose coordinate processes have the same laws as $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$, respectively. In particular, for every $t \geq 0$, $\text{Law}(\tilde{X}_t, \tilde{Y}_t) \in \Pi(\text{Law}(X_t), \text{Law}(Y_t))$.*

Definition 2.24 (Wasserstein Distance, [BE26, Definition 4.5]). *Let $q \in [1, \infty)$. Define*

$$\mathcal{P}_q(E) := \left\{ \mu \in \mathcal{P}(E) : \int_E d_E(x, x_0)^q \mu(dx) < \infty \right\},$$

where $x_0 \in E$ is arbitrary. The definition is independent of x_0 by the triangle inequality. For $\mu, \nu \in \mathcal{P}_q(E)$, their L^q -Wasserstein distance is

$$\mathcal{W}_{q,d_E}(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \left(\int_{E \times E} d_E(x, y)^q \pi(dx dy) \right)^{\frac{1}{q}} = \inf_{\substack{X \sim \mu \\ Y \sim \nu}} (\mathbb{E}[d_E(X, Y)^q])^{\frac{1}{q}},$$

where the second infimum is taken over all couplings of μ and ν . When the underlying metric is clear, we simply write $\mathcal{W}_q$ for $\mathcal{W}_{q,d_E}$.

A function $c : E \times E \rightarrow [0, \infty]$ is called *lower semicontinuous* if, whenever $(x_n, y_n) \rightarrow (x, y)$, one has $c(x, y) \leq \liminf_{n \rightarrow \infty} c(x_n, y_n)$.

Theorem 2.25 (Existence of Optimal Couplings, [BE26, Theorem 4.6]). *Suppose that the underlying distance function $d_E : E \times E \rightarrow [0, \infty)$ is lower semicontinuous. Then, for every $q \in [1, \infty)$ and every $\mu, \nu \in \mathcal{P}_q(E)$, there exists a coupling $\gamma \in \Pi(\mu, \nu)$ such that*

$$\mathcal{W}_{q,d_E}(\mu, \nu)^q = \int_{E \times E} d_E(x, y)^q \gamma(dx dy).$$

Such a coupling γ is called an *optimal coupling* of μ and ν with respect to $\mathcal{W}_{q,d_E}$.

In the following two examples, $d_E(x, y) := \|x - y\|$ is the Euclidean metric.

Example 2.26 (Synchronous Coupling, [BE26, Section 4.2]). *Let W be an $\mathbb{R}^d$ -valued random variable with finite second moment, and let $p(a, \cdot) := \text{Law}(a + W)$. For $a, b \in \mathbb{R}^d$, consider $X = a + W$, and $Y = b + W$. The pair (X, Y) is called the *synchronous coupling* because both components use the same realisation of the noise.*

Let (U, V) be any coupling of $p(a, \cdot)$ and $p(b, \cdot)$. Since $U - a$ and $V - b$ have the same law as W , $\mathbb{E}[(U - a) - (V - b)] = 0$. Consequently,

$$\begin{aligned} \mathbb{E}[\|U - V\|^2] &= \mathbb{E}[\|(a - b) + (U - a) - (V - b)\|^2] \\ &= \|a - b\|^2 + \mathbb{E}[\|(U - a) - (V - b)\|^2] + 2(a - b)^\top \mathbb{E}[(U - a) - (V - b)] \\ &= \|a - b\|^2 + \mathbb{E}[\|(U - a) - (V - b)\|^2] \geq \|a - b\|^2. \end{aligned}$$

On the other hand, the synchronous coupling satisfies $X - Y = a - b$ almost surely and therefore attains this lower bound. Hence $\mathcal{W}_2(p(a, \cdot), p(b, \cdot)) = \|a - b\|$. Thus the synchronous coupling is optimal.

Example 2.27 (Wasserstein Distance between One-Dimensional Gaussians, [BE26, Section 4.2]). *Let $X \sim \mathcal{N}(m, \sigma^2)$ and $Y \sim \mathcal{N}(\tilde{m}, \tilde{\sigma}^2)$ for $\sigma, \tilde{\sigma} \geq 0$. Then*

$$\mathcal{W}_2(\mathcal{N}(m, \sigma^2), \mathcal{N}(\tilde{m}, \tilde{\sigma}^2)) = \sqrt{|m - \tilde{m}|^2 + |\sigma - \tilde{\sigma}|^2}.$$

Indeed, for an arbitrary coupling, write

$$X = m + \sigma Z_1, \quad Y = \tilde{m} + \tilde{\sigma} Z_2, \quad \sigma, \tilde{\sigma} > 0,$$

where Z_1 and Z_2 are standard normal random variables. Then

$$\mathbb{E}[|X - Y|^2] = (m - \tilde{m})^2 + \sigma^2 + \tilde{\sigma}^2 - 2\sigma\tilde{\sigma}\mathbb{E}[Z_1 Z_2].$$

By the Cauchy–Schwarz inequality,

$$\mathbb{E}[Z_1 Z_2] \leq \sqrt{\mathbb{E}[Z_1^2] \mathbb{E}[Z_2^2]} = 1.$$

Therefore,

$$\mathbb{E}[|X - Y|^2] \geq (m - \tilde{m})^2 + (\sigma - \tilde{\sigma})^2.$$

Equality is attained by choosing $Z_1 = Z_2$. Taking the infimum over all couplings proves the formula. The degenerate cases $\sigma = 0$ or $\tilde{\sigma} = 0$ follow in the same way.

Definition 2.28 (Wasserstein Contraction Coefficient, [BE26, Definition 4.12]). *Let $q \in [1, \infty)$, and let p be a Markov kernel on E that maps $\mathcal{P}_q(E)$ into itself. Its global Wasserstein contraction coefficient is*

$$\alpha_q(p) := \sup_{\substack{\mu, \nu \in \mathcal{P}_q(E) \\ \mu \neq \nu}} \frac{\mathcal{W}_{q,d_E}(\mu p, \nu p)}{\mathcal{W}_{q,d_E}(\mu, \nu)}.$$

Thus $\alpha_q(p)$ is the Lipschitz constant of the map $\mu \mapsto \mu p$ on the space of probability measures equipped with the Wasserstein distance. The terminology *contraction* is commonly used even though $\alpha_q(p)$ may be larger than 1.

Theorem 2.29 ([BE26, Theorem 4.14(1)]). *Let $q \in [1, \infty)$, and let p be a Markov kernel on E that maps $\mathcal{P}_q(E)$ into itself. Then*

$$\alpha_q(p) = \sup_{\substack{x, y \in E \\ x \neq y}} \frac{\mathcal{W}_{q,d_E}(p(x, \cdot), p(y, \cdot))}{d_E(x, y)}.$$

Assume that p_t maps $\mathcal{P}_q(E)$ into itself for every $t \geq 0$. Then the definition of α_q and the semigroup property imply that

$$\alpha_q(p_{s+t}) \leq \alpha_q(p_s) \alpha_q(p_t), \quad s, t \geq 0.$$

Indeed, for $\mu, \nu \in \mathcal{P}_q(E)$,

$$\mathcal{W}_q(\mu p_{s+t}, \nu p_{s+t}) = \mathcal{W}_q((\mu p_s) p_t, (\nu p_s) p_t) \leq \alpha_q(p_t) \mathcal{W}_q(\mu p_s, \nu p_s) \leq \alpha_q(p_t) \alpha_q(p_s) \mathcal{W}_q(\mu, \nu).$$

If $\mu \in \mathcal{P}_q(E)$ is invariant, then for every $\nu \in \mathcal{P}_q(E)$,

$$\mathcal{W}_q(\nu p_t, \mu) = \mathcal{W}_q(\nu p_t, \mu p_t) \leq \alpha_q(p_t) \mathcal{W}_q(\nu, \mu).$$

Consequently, if $M, c > 0$ and $\alpha_q(p_t) \leq M e^{-ct}$ for all $t \geq 0$, then

$$\mathcal{W}_q(\nu p_t, \mu) \leq M e^{-ct} \mathcal{W}_q(\nu, \mu).$$

This is exponential convergence to equilibrium in the Wasserstein distance.

Chapter 3

Accelerating Convergence to Equilibrium for the Gaussian Case

This chapter investigates how a linear diffusion can be designed to preserve a prescribed centred Gaussian distribution and to converge towards it as rapidly as possible. In the Gaussian setting, both invariance and convergence reduce to explicit matrix calculations. Following [BE26, Section 4.7], consider the linear diffusion

$$dX_t = \sqrt{2}\sigma dB_t - \mathbf{A}X_t dt, \quad (3.1)$$

where $B = (B_t)_{t \geq 0}$ is a standard d -dimensional BM, $\sigma \in \mathbb{R}^{d \times d}$ determines how the noise enters the system, and $\mathbf{A} \in \mathbb{R}^{d \times d}$ is the drift matrix. At this stage, σ and $\mathbf{A}$ are arbitrary. The resulting process is called a d -dimensional *Ornstein–Uhlenbeck process*. We first derive its explicit representation.

Lemma 3.1 ([BE26, (4.57)]). *Let X_0 be an $\mathbb{R}^d$ -valued random variable independent of B , and suppose that $\mathbb{E}[\|X_0\|^2] < \infty$. Then the linear SDE (3.1) with initial condition X_0 has a unique global strong solution, given by*

$$X_t = e^{-t\mathbf{A}}X_0 + \sqrt{2} \int_0^t e^{(s-t)\mathbf{A}}\sigma dB_s. \quad (3.2)$$

Proof. Fix $T > 0$. Set $b(t, x) := -\mathbf{A}x$ and $\Sigma(t, x) := \sqrt{2}\sigma$. For every $t \in [0, T]$ and $x, y \in \mathbb{R}^d$,

$$\|b(t, x) - b(t, y)\| + \|\Sigma(t, x) - \Sigma(t, y)\|_F = \|\mathbf{A}(x - y)\| \leq \|\mathbf{A}\|\|x - y\|,$$

and, for a suitable constant $C > 0$,

$$\|b(t, x)\| + \|\Sigma(t, x)\|_F \leq \|\mathbf{A}\|\|x\| + \sqrt{2}\|\sigma\|_F \leq C(1 + \|x\|).$$

Thus the hypotheses of *Theorem 2.16* are satisfied, and (3.1) has a unique strong solution on $[0, T]$. Since $T > 0$ was arbitrary, uniqueness on each finite interval makes these solutions agree on overlaps, so they form a unique global strong solution.

We now identify this solution explicitly. Set $Y_t := e^{t\mathbf{A}}X_t$. By the *product rule* (2.1),

$$\begin{aligned} dY_t &= d(e^{t\mathbf{A}}X_t) \\ &= d(e^{t\mathbf{A}})X_t + e^{t\mathbf{A}}dX_t \\ &= \mathbf{A}e^{t\mathbf{A}}X_t dt + e^{t\mathbf{A}}\left(\sqrt{2}\sigma dB_t - \mathbf{A}X_t dt\right) \\ &= \mathbf{A}e^{t\mathbf{A}}X_t dt - e^{t\mathbf{A}}\mathbf{A}X_t dt + e^{t\mathbf{A}}\sqrt{2}\sigma dB_t = e^{t\mathbf{A}}\sqrt{2}\sigma dB_t, \end{aligned}$$

where $\mathbf{A}$ commutes with $e^{t\mathbf{A}}$. Therefore,

$$e^{t\mathbf{A}}X_t - X_0 = Y_t - Y_0 = \sqrt{2} \int_0^t e^{s\mathbf{A}}\sigma dB_s.$$

Multiplying both sides by $e^{-t\mathbf{A}}$, we obtain

$$X_t = e^{-t\mathbf{A}}X_0 + \sqrt{2}e^{-t\mathbf{A}} \int_0^t e^{s\mathbf{A}}\sigma dB_s = e^{-t\mathbf{A}}X_0 + \sqrt{2} \int_0^t e^{(s-t)\mathbf{A}}\sigma dB_s,$$

which proves (3.2). $\square$

The Markov property is included in [BE26, Theorem 4.31], where the coefficients have already been restricted to preserve the target Gaussian measure. The argument itself uses only the explicit representation (3.2), so we prove it here for arbitrary σ and $\mathbf{A}$.

Proposition 3.2 (Markov Property). *Under the assumptions of the preceding lemma, let $(X_t)_{t \geq 0}$ be the solution of (3.1), and set $\mathcal{F}_t := \sigma(X_0, B_s : 0 \leq s \leq t)$. Let $\tilde{B}$ be a standard d -dimensional BM and, for $h \geq 0$, define*

$$\tilde{Z}_h := \sqrt{2} \int_0^h e^{(s-h)\mathbf{A}}\sigma d\tilde{B}_s.$$

For $x \in \mathbb{R}^d$ and $E \in \mathcal{B}(\mathbb{R}^d)$, set

$$p_h(x, E) := \mathbb{P}\left(e^{-h\mathbf{A}}x + \tilde{Z}_h \in E\right),$$

and, for $f \in B_b(\mathbb{R}^d)$,

$$P_h f(x) := \int_{\mathbb{R}^d} f(y) p_h(x, dy) = \mathbb{E}\left[f\left(e^{-h\mathbf{A}}x + \tilde{Z}_h\right)\right].$$

Then p_h is a Markov kernel for every $h \geq 0$, and, for $0 \leq r \leq t$,

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_r] = P_{t-r}f(X_r) \quad a.s.$$

Consequently, $(X_t)_{t \geq 0}$ is a time-homogeneous Markov process with transition semigroup $(P_t)_{t \geq 0}$.

Proof. Let ν_h denote the law of $\tilde{Z}_h$. For each fixed $x \in \mathbb{R}^d$, the map $E \mapsto p_h(x, E)$ is the law of $e^{-h\mathbf{A}}x + \tilde{Z}_h$, and is therefore a probability measure. On the other hand, for every $E \in \mathcal{B}(\mathbb{R}^d)$,

$$p_h(x, E) = \int_{\mathbb{R}^d} \mathbb{1}_E\left(e^{-h\mathbf{A}}x + z\right) \nu_h(dz).$$

Because $(x, z) \mapsto \mathbb{1}_E(e^{-h\mathbf{A}}x + z)$ is Borel measurable, the map $x \mapsto p_h(x, E)$ is also Borel measurable.

Thus p_h is a Markov kernel. For $0 \leq r \leq t$, (3.2) implies

$$X_r = e^{-r\mathbf{A}}X_0 + \sqrt{2} \int_0^r e^{(s-r)\mathbf{A}} \sigma dB_s.$$

Multiplying both sides by $e^{-(t-r)\mathbf{A}}$ gives

$$e^{-(t-r)\mathbf{A}}X_r = e^{-t\mathbf{A}}X_0 + \sqrt{2} \int_0^r e^{-(t-r)\mathbf{A}} e^{(s-r)\mathbf{A}} \sigma dB_s = e^{-t\mathbf{A}}X_0 + \sqrt{2} \int_0^r e^{(s-t)\mathbf{A}} \sigma dB_s.$$

Splitting the stochastic integral in (3.2) at time r yields

$$X_t = e^{-t\mathbf{A}}X_0 + \sqrt{2} \int_0^r e^{(s-t)\mathbf{A}} \sigma dB_s + \sqrt{2} \int_r^t e^{(s-t)\mathbf{A}} \sigma dB_s = e^{-(t-r)\mathbf{A}}X_r + \sqrt{2} \int_r^t e^{(s-t)\mathbf{A}} \sigma dB_s.$$

For $u \geq 0$, set $W_u := B_{r+u} - B_r$. By the stationary and independent increments of BM, $W = (W_u)_{u \geq 0}$ is a standard d -dimensional BM, and it is independent of $\sigma(B_s : 0 \leq s \leq r)$. Since X_0 is independent of the entire BM B by assumption, W is independent of $\mathcal{F}_r = \sigma(X_0, B_s : 0 \leq s \leq r)$. Writing $h := t - r$, the change of variable $s = r + u$ yields

$$\int_r^t e^{(s-t)\mathbf{A}} \sigma dB_s = \int_0^h e^{(u-h)\mathbf{A}} \sigma dW_u.$$

Thus

$$X_t = e^{-h\mathbf{A}}X_r + Z_{r,h},$$

where

$$Z_{r,h} := \sqrt{2} \int_0^h e^{(u-h)\mathbf{A}} \sigma dW_u.$$

The random vector X_r is $\mathcal{F}_r$ -measurable, whereas $Z_{r,h}$ is independent of $\mathcal{F}_r$. Since W and $\tilde{B}$ have the same law, $Z_{r,h} \sim \tilde{Z}_h$. For $x, z \in \mathbb{R}^d$, define

$$\Phi_h(x, z) := f\left(e^{-h\mathbf{A}}x + z\right).$$

Because f is bounded and Borel measurable, Φ_h is Borel measurable, and $\Phi_h(X_r, Z_{r,h})$ is integrable. Define

$$g_h(x) := \mathbb{E}\left[\Phi_h\left(x, \tilde{Z}_h\right)\right] = \mathbb{E}\left[f\left(e^{-h\mathbf{A}}x + \tilde{Z}_h\right)\right] = P_h f(x).$$

Then $\mathbb{E}[\Phi_h(x, Z_{r,h})] = g_h(x)$. In particular, although $Z_{r,h}$ is indexed by r , the function g_h does not depend on r . Since X_r is $\mathcal{F}_r$ -measurable and $Z_{r,h}$ is independent of $\mathcal{F}_r$, [Sep14, Theorem 1.26(x)] gives

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_r] = \mathbb{E}[\Phi_h(X_r, Z_{r,h}) \mid \mathcal{F}_r] = g_h(X_r) = P_h f(X_r) = P_{t-r} f(X_r) \quad a.s.$$

This is precisely the time-homogeneous Markov property. As shown in *Chapter 2*, it also implies $P_0 = \text{Id}$ and $P_{s+t} = P_s P_t$ for $s, t \geq 0$. Therefore, $(P_t)_{t \geq 0}$ is the transition semigroup of $(X_t)_{t \geq 0}$. $\square$

3.1 Invariant Gaussian Measures

Fix a centred Gaussian measure $\mu = \mathcal{N}(0, \mathbf{C})$ on $\mathbb{R}^d$, where the covariance matrix $\mathbf{C} \in \mathbb{R}^{d \times d}$ is symmetric and positive definite. We want to determine which coefficients in (3.1) preserve this measure. As in *Section 2.3*, invariance implies infinitesimal invariance.

Set $\mathbf{D} := \sigma\sigma^\top$. By *Section 2.1*, it is symmetric and positive semidefinite. The next lemma is a sign-adjusted version of [GM16, Lemma 3.1]. We remove the noise-budget constraint, which is irrelevant to invariance, add an equivalent formulation, and give a direct proof using the formal adjoint.

Lemma 3.3. *Let $\mu = \mathcal{N}(0, \mathbf{C})$, where $\mathbf{C} \in \mathbb{R}^{d \times d}$ is symmetric and positive definite. Consider the differential operator associated with (3.1):*

$$\mathcal{L}f(x) = -(\mathbf{A}x)^\top \nabla f(x) + \text{tr}(\mathbf{D}\nabla^2 f(x)), \quad f \in C_c^\infty(\mathbb{R}^d). \quad (3.3)$$

Then the following statements are equivalent:

- (i) μ is infinitesimally invariant for $\mathcal{L}$.
- (ii) $\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = 2\mathbf{D}$.
- (iii) There exists an antisymmetric matrix $\mathbf{J} \in \mathbb{R}^{d \times d}$ such that $\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}$.

Whenever these conditions hold, $\mathbf{J}$ is uniquely determined by $\mathbf{J} = \mathbf{A}\mathbf{C} - \mathbf{D}$.

Proof. Set $\mathbf{S} := \mathbf{C}^{-1}$. Since $\mathbf{C}$ is symmetric and positive definite, the same is true for $\mathbf{S}$. The density of μ with respect to Lebesgue measure is

$$\psi(x) = Z^{-1} \exp\left(-\frac{1}{2}x^\top \mathbf{S}x\right), \quad Z = (2\pi)^{\frac{d}{2}}(\det \mathbf{C})^{\frac{1}{2}}.$$

Since $\mathbf{S}^\top = \mathbf{S}$, for every $k \in \{1, \dots, d\}$,

$$\partial_k(x^\top \mathbf{S}x) = \partial_k\left(\sum_{i,j=1}^d x_i S_{ij} x_j\right) = \sum_{i=1}^d x_i S_{ik} + \sum_{j=1}^d S_{kj} x_j = 2(\mathbf{S}x)_k.$$

Consequently, $\nabla(x^\top \mathbf{S}x) = 2\mathbf{S}x$. By the chain rule, differentiating the density $\psi(x)$ gives

$$\nabla\psi(x) = \psi(x)\nabla\left(-\frac{1}{2}x^\top \mathbf{S}x\right) = -\mathbf{S}x\psi(x). \quad (3.4)$$

Let $f \in C_c^\infty(\mathbb{R}^d)$. Because $\mathbf{D}$ is constant,

$$\text{tr}(\mathbf{D}\nabla^2 f(x)) = \sum_{i,j=1}^d D_{ij} \partial_{ij} f(x) = \sum_{i=1}^d \partial_i \left(\sum_{j=1}^d D_{ij} \partial_j f(x) \right) = \nabla \cdot (\mathbf{D}\nabla f)(x).$$

So (3.3) can be rewritten as

$$\mathcal{L}f(x) = -(\mathbf{A}x)^\top \nabla f(x) + \nabla \cdot (\mathbf{D}\nabla f)(x).$$

Hence

$$\int_{\mathbb{R}^d} \mathcal{L}f(x)\mu(dx) = \int_{\mathbb{R}^d} \left(-(\mathbf{A}x)^\top \nabla f(x) + \nabla \cdot (\mathbf{D}\nabla f)(x) \right) \psi(x) dx.$$

For the drift part, integration by parts gives

$$\begin{aligned} - \int_{\mathbb{R}^d} (\mathbf{A}x)^\top \nabla f(x) \psi(x) dx &= - \sum_{i=1}^d \int_{\mathbb{R}^d} (\mathbf{A}x)_i \partial_i f(x) \psi(x) dx \\ &= - \sum_{i=1}^d \left(\int_{\mathbb{R}^d} \partial_i (f(x) (\mathbf{A}x)_i \psi(x)) dx - \int_{\mathbb{R}^d} f(x) \partial_i ((\mathbf{A}x)_i \psi(x)) dx \right) \\ &= \sum_{i=1}^d \int_{\mathbb{R}^d} f(x) \partial_i ((\mathbf{A}x)_i \psi(x)) dx = \int_{\mathbb{R}^d} f(x) \nabla \cdot (\mathbf{A}x \psi(x)) dx. \end{aligned} \quad (3.5)$$

The boundary terms vanish because f and its derivatives are compactly supported. For the diffusion part, since $\mathbf{D}^\top = \mathbf{D}$,

$$\int_{\mathbb{R}^d} \nabla \cdot (\mathbf{D}\nabla f)(x) \psi(x) dx = - \int_{\mathbb{R}^d} (\mathbf{D}\nabla f)(x) \cdot \nabla \psi(x) dx = \int_{\mathbb{R}^d} f(x) \nabla \cdot (\mathbf{D}\nabla \psi(x)) dx. \quad (3.6)$$

Combining (3.5) and (3.6), we obtain

$$\begin{aligned} \int_{\mathbb{R}^d} \mathcal{L}f(x)\mu(dx) &= \int_{\mathbb{R}^d} \left(-(\mathbf{A}x)^\top \nabla f(x) + \nabla \cdot (\mathbf{D}\nabla f)(x) \right) \psi(x) dx \\ &= - \int_{\mathbb{R}^d} (\mathbf{A}x)^\top \nabla f(x) \psi(x) dx + \int_{\mathbb{R}^d} \nabla \cdot (\mathbf{D}\nabla f)(x) \psi(x) dx \\ &= \int_{\mathbb{R}^d} f(x) \nabla \cdot (\mathbf{A}x \psi(x)) dx + \int_{\mathbb{R}^d} f(x) \nabla \cdot (\mathbf{D}\nabla \psi(x)) dx \\ &= \int_{\mathbb{R}^d} f(x) (\nabla \cdot (\mathbf{A}x \psi(x)) + \nabla \cdot (\mathbf{D}\nabla \psi(x))) dx = \int_{\mathbb{R}^d} f(x) \mathcal{L}^* \psi(x) dx, \end{aligned} \quad (3.7)$$

where the formal adjoint of $\mathcal{L}$ with respect to Lebesgue measure is

$$\mathcal{L}^*g(x) = \nabla \cdot (\mathbf{A}xg(x)) + \nabla \cdot (\mathbf{D}\nabla g(x)), \quad g \in C_c^\infty(\mathbb{R}^d).$$

Although ψ is not compactly supported, the integration-by-parts identity (3.7) remains valid because $f \in C_c^\infty(\mathbb{R}^d)$. By (3.4),

$$\mathcal{L}^* \psi(x) = \nabla \cdot (\mathbf{A}x \psi(x)) + \nabla \cdot (\mathbf{D}\nabla \psi(x)) = \nabla \cdot (\mathbf{A}x \psi(x)) - \nabla \cdot (\mathbf{D}\mathbf{S}x \psi(x)) = \nabla \cdot ((\mathbf{A} - \mathbf{D}\mathbf{S})x \psi(x)).$$

Set $\mathbf{M} := \mathbf{A} - \mathbf{D}\mathbf{S}$. For any constant matrix $\mathbf{M}$, $\nabla \cdot (\mathbf{M}x) = \text{tr}(\mathbf{M})$. Then by the product rule,

$$\mathcal{L}^* \psi(x) = \nabla \cdot (\mathbf{M}x \psi(x)) = \text{tr}(\mathbf{M}) \psi(x) + (\mathbf{M}x)^\top \nabla \psi(x) = \left(\text{tr}(\mathbf{M}) - x^\top \mathbf{M}^\top \mathbf{S}x \right) \psi(x). \quad (3.8)$$

We first prove that (i) implies (iii). Suppose that μ is infinitesimally invariant for $\mathcal{L}$. Then, by (3.7),

$$\int_{\mathbb{R}^d} f(x) \mathcal{L}^* \psi(x) dx = 0, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

Thus $\mathcal{L}^* \psi(x) = 0$ Lebesgue-*a.e.* Since the function $\mathcal{L}^* \psi(x)$ in (3.8) is smooth, the equality holds

pointwise¹:

$$\mathcal{L}^*\psi(x) = 0, \quad \forall x \in \mathbb{R}^d.$$

Since $\psi(x) > 0$, (3.8) gives

$$\text{tr}(\mathbf{M}) - x^\top \mathbf{M}^\top \mathbf{S} x = 0, \quad \forall x \in \mathbb{R}^d.$$

Taking $x = 0$ yields $\text{tr}(\mathbf{M}) = 0$. It follows that $x^\top \mathbf{M}^\top \mathbf{S} x = 0$ for $x \in \mathbb{R}^d$. Then for arbitrary $x, y \in \mathbb{R}^d$,

$$\begin{aligned} 0 &= (x + y)^\top (\mathbf{M}^\top \mathbf{S}) (x + y) - x^\top (\mathbf{M}^\top \mathbf{S}) x - y^\top (\mathbf{M}^\top \mathbf{S}) y \\ &= x^\top (\mathbf{M}^\top \mathbf{S}) y + y^\top (\mathbf{M}^\top \mathbf{S}) x = x^\top (\mathbf{M}^\top \mathbf{S} + \mathbf{S} \mathbf{M}) y. \end{aligned}$$

Hence $\mathbf{M}^\top \mathbf{S} + \mathbf{S} \mathbf{M} = 0$. Multiplying from the left and the right by $\mathbf{C} = \mathbf{S}^{-1}$ gives

$$\mathbf{S}^{-1} (\mathbf{M}^\top \mathbf{S} + \mathbf{S} \mathbf{M}) \mathbf{S}^{-1} = \mathbf{C} \mathbf{M}^\top + \mathbf{M} \mathbf{C} = 0,$$

so that $\mathbf{M} \mathbf{C}$ is antisymmetric. Set $\mathbf{J} := \mathbf{M} \mathbf{C}$. Then

$$\mathbf{J}^\top = (\mathbf{M} \mathbf{C})^\top = \mathbf{C} \mathbf{M}^\top = -\mathbf{M} \mathbf{C} = -\mathbf{J}.$$

Thus $\mathbf{J}$ is antisymmetric. Moreover,

$$\mathbf{J} = \mathbf{M} \mathbf{C} = (\mathbf{A} - \mathbf{D} \mathbf{S}) \mathbf{C} = \mathbf{A} \mathbf{C} - \mathbf{D} \mathbf{S} \mathbf{C} = \mathbf{A} \mathbf{C} - \mathbf{D},$$

and therefore $\mathbf{A} \mathbf{C} = \mathbf{D} + \mathbf{J}$, which implies $\mathbf{A} = (\mathbf{D} + \mathbf{J}) \mathbf{C}^{-1}$. This proves (iii).

Conversely, suppose that (iii) holds. Since $\mathbf{S} = \mathbf{C}^{-1}$, we obtain $\mathbf{M} = \mathbf{A} - \mathbf{D} \mathbf{S} = \mathbf{J} \mathbf{S}$. Then (3.8) becomes

$$\mathcal{L}^*\psi(x) = (\text{tr}(\mathbf{J} \mathbf{S}) + x^\top \mathbf{S} \mathbf{J} \mathbf{S} x) \psi(x). \quad (3.9)$$

Because $\mathbf{J}^\top = -\mathbf{J}$ and $\mathbf{S}^\top = \mathbf{S}$,

$$\text{tr}(\mathbf{J} \mathbf{S}) = \text{tr}((\mathbf{J} \mathbf{S})^\top) = \text{tr}(\mathbf{S} \mathbf{J}^\top) = -\text{tr}(\mathbf{S} \mathbf{J}) = -\text{tr}(\mathbf{J} \mathbf{S}),$$

which implies $\text{tr}(\mathbf{J} \mathbf{S}) = 0$. Similarly, $\mathbf{S} \mathbf{J} \mathbf{S}$ is antisymmetric:

$$(\mathbf{S} \mathbf{J} \mathbf{S})^\top = \mathbf{S} \mathbf{J}^\top \mathbf{S} = -\mathbf{S} \mathbf{J} \mathbf{S}.$$

Consequently, $x^\top \mathbf{S} \mathbf{J} \mathbf{S} x = 0$ for $x \in \mathbb{R}^d$. Therefore, (3.9) gives $\mathcal{L}^*\psi(x) = 0$. By (3.7), we conclude that

$$\int_{\mathbb{R}^d} \mathcal{L} f(x) \mu(dx) = 0, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

Thus μ is infinitesimally invariant for $\mathcal{L}$, proving (i).

It remains to relate (ii) and (iii). If (iii) holds, then $\mathbf{A} \mathbf{C} = \mathbf{D} + \mathbf{J}$. Taking transposes and using

$$\mathbf{C}^\top = \mathbf{C}, \quad \mathbf{D}^\top = \mathbf{D}, \quad \mathbf{J}^\top = -\mathbf{J}$$

¹Two locally integrable functions define the same distribution if and only if they are equal almost everywhere [Fol99, Section 9.1]. Since $\mathcal{L}^*\psi$ is continuous, the equality holds everywhere.

gives $\mathbf{CA}^\top = \mathbf{D} - \mathbf{J}$. Adding these two identities yields

$$\mathbf{AC} + \mathbf{CA}^\top = 2\mathbf{D},$$

which is (ii). Conversely, suppose that (ii) holds and set $\mathbf{J} := \mathbf{AC} - \mathbf{D}$. Then

$$\mathbf{J}^\top = \mathbf{CA}^\top - \mathbf{D} = 2\mathbf{D} - \mathbf{AC} - \mathbf{D} = -(\mathbf{AC} - \mathbf{D}) = -\mathbf{J},$$

which means $\mathbf{J}$ is antisymmetric and $\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}$. Finally, if any matrix $\tilde{\mathbf{J}}$ satisfies $\mathbf{A} = (\mathbf{D} + \tilde{\mathbf{J}})\mathbf{C}^{-1}$, we also have $\tilde{\mathbf{J}} = \mathbf{AC} - \mathbf{D}$. Hence $\mathbf{J}$ is unique, and this completes the proof. $\square$

For the present linear Gaussian diffusion, the corresponding reversibility criterion is $\mathbf{A} = \mathbf{DC}^{-1}$; see [GM16, page 4], after translating its opposite sign convention for the drift matrix. By the uniqueness of the decomposition in the preceding lemma, this condition is equivalent to $\mathbf{J} = 0$. Thus, $\mathbf{DC}^{-1}$ is the reversible part of the drift matrix, whereas $\mathbf{JC}^{-1}$ is its non-reversible part.

We next analyse the Gaussian structure of the explicit solution (3.2). We first verify that X is a Gaussian process and then derive its mean and covariance equations, which correspond to [BE26, (4.58)–(4.59)]. These quantities will identify its transition kernel and prove invariance.

Let $X_0 \sim \mathcal{N}(m_0, \Sigma_0)$ be independent of the BM $(B_t)_{t \geq 0}$, where $m_0 \in \mathbb{R}^d$ and $\Sigma_0 \in \mathbb{R}^{d \times d}$ is symmetric and positive semidefinite. By the explicit solution (3.2),

$$X_t = e^{-t\mathbf{A}}X_0 + Z_t, \quad Z_t := \sqrt{2} \int_0^t e^{(s-t)\mathbf{A}} \sigma dB_s.$$

We first identify the finite-dimensional distributions of $(Z_t)_{t \geq 0}$. Fix $n \in \mathbb{N}$ and $t_1, \dots, t_n \geq 0$, and set $T := \max_{1 \leq i \leq n} t_i$. For each $i = 1, \dots, n$, define the deterministic matrix-valued function

$$F_i(s) := \sqrt{2} \mathbf{1}_{[0, t_i]}(s) e^{(s-t_i)\mathbf{A}} \sigma, \quad 0 \leq s \leq T.$$

Since $s \mapsto e^{(s-t_i)\mathbf{A}} \sigma$ is continuous on the compact interval $[0, t_i]$, each F_i belongs to $L^2([0, T]; \mathbb{R}^{d \times d})$. Moreover,

$$Z_{t_i} = \int_0^T F_i(s) dB_s.$$

Therefore, it follows from *Proposition 2.14* that

$$\begin{pmatrix} Z_{t_1} \\ \vdots \\ Z_{t_n} \end{pmatrix}$$

is a centred Gaussian random vector with

$$\text{Cov}(Z_{t_i}, Z_{t_j}) = 2 \int_0^{t_i \wedge t_j} e^{(s-t_i)\mathbf{A}} \mathbf{D} e^{(s-t_j)\mathbf{A}^\top} ds.$$

In particular, $Z_t \sim \mathcal{N}(0, \Gamma_t)$, where

$$\Gamma_t := 2 \int_0^t e^{(s-t)\mathbf{A}} \mathbf{D} e^{(s-t)\mathbf{A}^\top} ds.$$

Because the times $t_1, \dots, t_n$ were arbitrary, $(Z_t)_{t \geq 0}$ is a Gaussian process.

It remains to verify the same conclusion for $(X_t)_{t \geq 0}$. The random vector

$$\begin{pmatrix} e^{-t_1 \mathbf{A}} X_0 \\ \vdots \\ e^{-t_n \mathbf{A}} X_0 \end{pmatrix}$$

is a linear transformation of the Gaussian vector X_0 , and is therefore Gaussian. Furthermore, it is independent of

$$\begin{pmatrix} Z_{t_1} \\ \vdots \\ Z_{t_n} \end{pmatrix},$$

because X_0 is independent of the BM. Hence

$$\begin{pmatrix} X_{t_1} \\ \vdots \\ X_{t_n} \end{pmatrix} = \begin{pmatrix} e^{-t_1 \mathbf{A}} X_0 \\ \vdots \\ e^{-t_n \mathbf{A}} X_0 \end{pmatrix} + \begin{pmatrix} Z_{t_1} \\ \vdots \\ Z_{t_n} \end{pmatrix}$$

is Gaussian. Thus $(X_t)_{t \geq 0}$ is a Gaussian process.

The mean of X_t is

$$m_t := \mathbb{E}[X_t] = e^{-t \mathbf{A}} m_0,$$

while its covariance matrix is

$$\Sigma_t := \text{Cov}(X_t) = \text{Cov}(e^{-t \mathbf{A}} X_0) + \text{Cov}(Z_t) = \underbrace{e^{-t \mathbf{A}} \Sigma_0 e^{-t \mathbf{A}^\top}}_{:= \mathbf{H}_t} + \Gamma_t = \mathbf{H}_t + \Gamma_t.$$

Differentiating the expression for m_t gives the ordinary differential equation (ODE):

$$\frac{d}{dt} m_t = -\mathbf{A} m_t, \quad m_{t=0} = m_0. \quad (3.10)$$

Differentiating Γ_t by the Leibniz rule gives

$$\begin{aligned} \frac{d}{dt} \Gamma_t &= 2\mathbf{D} - 2 \int_0^t \mathbf{A} e^{(s-t) \mathbf{A}} \mathbf{D} e^{(s-t) \mathbf{A}^\top} ds - 2 \int_0^t e^{(s-t) \mathbf{A}} \mathbf{D} e^{(s-t) \mathbf{A}^\top} \mathbf{A}^\top ds \\ &= -\mathbf{A} \Gamma_t - \Gamma_t \mathbf{A}^\top + 2\mathbf{D}, \quad \Gamma_0 = 0. \end{aligned}$$

Since

$$\frac{d}{dt} \mathbf{H}_t = -\mathbf{A} \mathbf{H}_t - \mathbf{H}_t \mathbf{A}^\top,$$

we conclude that

$$\begin{aligned} \frac{d}{dt} \Sigma_t &= -\mathbf{A}(\mathbf{H}_t + \Gamma_t) - (\mathbf{H}_t + \Gamma_t) \mathbf{A}^\top + 2\mathbf{D} \\ &= -\mathbf{A} \Sigma_t - \Sigma_t \mathbf{A}^\top + 2\mathbf{D}, \quad \Sigma_{t=0} = \Sigma_0. \end{aligned} \quad (3.11)$$

Theorem 3.4 ([BE26, Theorem 4.31]). *Assume that the equivalent conditions in Lemma 3.3 hold.*

Then the SDE (3.1) defines a time-homogeneous Markov process with invariant measure $\mathcal{N}(0, \mathbf{C})$ and transition kernel

$$p_t(x, \cdot) = \mathcal{N}(e^{-t\mathbf{A}}x, \mathbf{\Gamma}_t), \quad x \in \mathbb{R}^d, \quad (3.12)$$

where $\mathbf{\Gamma}_t$ is the unique solution of the covariance equation in (3.11) with initial condition $\mathbf{\Gamma}_0 = 0$.

Proof. The time-homogeneous Markov property is established by *Proposition 3.2*. We therefore begin by identifying the transition kernel. Fix $x \in \mathbb{R}^d$, and consider the deterministic initial condition $X_0 = x$. The explicit solution (3.2) becomes

$$X_t = e^{-t\mathbf{A}}x + Z_t,$$

where $Z_t \sim \mathcal{N}(0, \mathbf{\Gamma}_t)$. Consequently, $X_t \sim \mathcal{N}(e^{-t\mathbf{A}}x, \mathbf{\Gamma}_t)$. Since $\mathbf{\Gamma}_t$ is the unique solution of the covariance equation in (3.11) with $\mathbf{\Gamma}_0 = 0$, this proves (3.12).

It remains to prove the invariance. Suppose $X_0 \sim \mathcal{N}(0, \mathbf{C})$ and X_0 is independent of the BM. Then the mean equation (3.10) with $m_0 = 0$ admits a unique solution $m_t \equiv 0$ for $t \geq 0$. By *Lemma 3.3*, $-\mathbf{A}\mathbf{C} - \mathbf{C}\mathbf{A}^\top + 2\mathbf{D} = 0$, so the constant function $\mathbf{\Sigma}_t \equiv \mathbf{C}$ is the unique solution of the covariance equation (3.11) with $\mathbf{\Sigma}_0 = \mathbf{C}$ for $t \geq 0$. Consequently, $X_t \sim \mathcal{N}(0, \mathbf{C})$ for $t \geq 0$, and hence $\mathcal{N}(0, \mathbf{C})$ is an invariant measure. $\square$

Combining this theorem with the implication from invariance to infinitesimal invariance, we conclude that $\mathcal{N}(0, \mathbf{C})$ is invariant for (3.1) if and only if

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = 2\mathbf{D},$$

or equivalently,

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}$$

for a unique antisymmetric matrix $\mathbf{J}$.

3.2 Optimal Convergence Rates

We first verify that p_t preserves finite q -th moments. Fix $t \geq 0$ and $q \in [1, \infty)$, and let $X_0 \sim \nu \in \mathcal{P}_q(\mathbb{R}^d)$ be independent of the driving BM. By (3.2) and *Minkowski's inequality*,

$$(\mathbb{E}[\|X_t\|^q])^{\frac{1}{q}} \leq \|e^{-t\mathbf{A}}\|(\mathbb{E}[\|X_0\|^q])^{\frac{1}{q}} + (\mathbb{E}[\|Z_t\|^q])^{\frac{1}{q}} < \infty,$$

because Z_t is Gaussian. Since X_t has distribution νp_t , it follows that $\nu p_t \in \mathcal{P}_q(\mathbb{R}^d)$. Thus p_t maps $\mathcal{P}_q(\mathbb{R}^d)$ into itself. By (3.12), $p_t(x, \cdot)$ and $p_t(y, \cdot)$ are translations of the same centred Gaussian measure, so their L^2 -Wasserstein distance is determined entirely by the separation of their means.

The following result is adapted from [BE26, Theorem 4.32]. We retain its exact finite-time identity. For the large-time behaviour, we express the exponential decay rate through a logarithmic-slope limit.

Theorem 3.5 (Wasserstein Contraction Rates of Gaussian Diffusions). *For every $t \in [0, \infty)$,*

$$\alpha_2(p_t) = \|e^{-t\mathbf{A}}\| = \varrho\left(e^{-t\mathbf{A}^\top}e^{-t\mathbf{A}}\right)^{\frac{1}{2}}.$$

Moreover,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \alpha_2(p_t) = -\lambda^*, \quad \lambda^* := \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

In particular, if $\lambda^* > 0$, then $\log \alpha_2(p_t) \sim -\lambda^* t$ as $t \rightarrow \infty$.

Proof. Let $t \geq 0$ and $x, y \in \mathbb{R}^d$. By (3.12),

$$p_t(x, \cdot) = \mathcal{N}(e^{-t\mathbf{A}}x, \mathbf{\Gamma}_t), \quad p_t(y, \cdot) = \mathcal{N}(e^{-t\mathbf{A}}y, \mathbf{\Gamma}_t).$$

These two measures are translations of the same centred Gaussian distribution. Therefore, applying the synchronous-coupling calculation in *Example 2.26* with translation vectors $e^{-t\mathbf{A}}x$ and $e^{-t\mathbf{A}}y$, we obtain

$$\mathcal{W}_2(p_t(x, \cdot), p_t(y, \cdot)) = \|e^{-t\mathbf{A}}(x - y)\|.$$

By *Theorem 2.29*,

$$\alpha_2(p_t) = \sup_{x \neq y} \frac{\|e^{-t\mathbf{A}}(x - y)\|}{\|x - y\|} = \|e^{-t\mathbf{A}}\| = \varrho(e^{-t\mathbf{A}^\top} e^{-t\mathbf{A}})^{\frac{1}{2}}.$$

It remains to determine the asymptotic rate. By *Proposition 2.6*,

$$\alpha_2(p_t) = \|e^{-t\mathbf{A}}\| \geq \varrho(e^{-t\mathbf{A}}) = e^{-t\lambda^*}.$$

Hence,

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \alpha_2(p_t) \geq -\lambda^*.$$

Conversely, fix $\varepsilon > 0$. By *Lemma 2.7*, there exists $\mathbf{G}_\varepsilon \in \mathcal{S}_d^{>0}(\mathbb{R})$ such that

$$\mathbf{A}^\top \mathbf{G}_\varepsilon + \mathbf{G}_\varepsilon \mathbf{A} \succeq 2(\lambda^* - \varepsilon) \mathbf{G}_\varepsilon.$$

For $z \in \mathbb{R}^d$, set $z_t = e^{-t\mathbf{A}}z$. Differentiation gives

$$\frac{d}{dt} \|z_t\|_{\mathbf{G}_\varepsilon}^2 \leq -2(\lambda^* - \varepsilon) \|z_t\|_{\mathbf{G}_\varepsilon}^2.$$

Therefore,

$$\|e^{-t\mathbf{A}}z\| \leq \sqrt{\kappa(\mathbf{G}_\varepsilon)} e^{-(\lambda^* - \varepsilon)t} \|z\|,$$

and hence

$$\alpha_2(p_t) \leq \sqrt{\kappa(\mathbf{G}_\varepsilon)} e^{-(\lambda^* - \varepsilon)t}.$$

It follows that

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \alpha_2(p_t) \leq -\lambda^* + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \alpha_2(p_t) = -\lambda^*.$$

If $\lambda^* > 0$, dividing by $-\lambda^* t$ also gives $\log \alpha_2(p_t) \sim -\lambda^* t$. □

Twisted Metrics

The asymptotic Wasserstein contraction rate describes the exponential slope observed at large times, but one can also derive corresponding non-asymptotic bounds. We can show this by considering *twisted metrics*. Let X_t^x and X_t^y be two solutions of the Ornstein–Uhlenbeck equation with initial conditions $x, y \in \mathbb{R}^d$, driven by the same BM:

$$\begin{aligned} dX_t^x &= -\mathbf{A}X_t^x dt + \sqrt{2}\sigma dB_t, & X_0^x &= x, \\ dX_t^y &= -\mathbf{A}X_t^y dt + \sqrt{2}\sigma dB_t, & X_0^y &= y. \end{aligned}$$

Define their difference by $\Delta_t := X_t^x - X_t^y$. The stochastic terms cancel under the synchronous coupling, so Δ_t satisfies the deterministic equation

$$\frac{d}{dt}\Delta_t = -\mathbf{A}\Delta_t, \quad \Delta_t = e^{-t\mathbf{A}}(x - y).$$

In the Euclidean norm,

$$\frac{d}{dt}\|\Delta_t\|^2 = -\Delta_t^\top (\mathbf{A}^\top + \mathbf{A})\Delta_t.$$

Even when $\lambda^*(\mathbf{A}) > 0$, the symmetric part $\frac{1}{2}(\mathbf{A}^\top + \mathbf{A})$ need not be positive definite. Consequently, $t \mapsto \|\Delta_t\|$ need not be decreasing. To recover a monotone distance, let $\mathbf{G} \in \mathcal{S}_d^{>0}(\mathbb{R})$ and use the $\mathbf{G}$ -weighted norm introduced in *Definition 2.4*. The corresponding distance

$$d_{\mathbf{G}}(x, y) := \|x - y\|_{\mathbf{G}}$$

is called a *twisted metric*. Along the difference process,

$$\begin{aligned} \frac{d}{dt}\|\Delta_t\|_{\mathbf{G}}^2 &= \frac{d}{dt}\left(\Delta_t^\top \mathbf{G} \Delta_t\right) \\ &= \left(\frac{d}{dt}\Delta_t\right)^\top \mathbf{G} \Delta_t + \Delta_t^\top \mathbf{G} \left(\frac{d}{dt}\Delta_t\right) \\ &= (-\mathbf{A}\Delta_t)^\top \mathbf{G} \Delta_t + \Delta_t^\top \mathbf{G} (-\mathbf{A}\Delta_t) = -\Delta_t^\top (\mathbf{A}^\top \mathbf{G} + \mathbf{G} \mathbf{A}) \Delta_t. \end{aligned} \tag{3.13}$$

By *Lemma 2.7*, we may take $c = \lambda^*$ if the eigenvalues on the spectral boundary are non-defective, and $c = \lambda^* - \varepsilon > 0$ in the general case. For the corresponding matrix $\mathbf{G}$, (3.13) gives

$$\frac{d}{dt}\|\Delta_t\|_{\mathbf{G}}^2 \leq -2c\|\Delta_t\|_{\mathbf{G}}^2.$$

Multiplying by e^{2ct} , we obtain

$$\frac{d}{dt}\left(e^{2ct}\|\Delta_t\|_{\mathbf{G}}^2\right) \leq 0.$$

Hence

$$\|\Delta_t\|_{\mathbf{G}}^2 \leq e^{-2ct}\|\Delta_0\|_{\mathbf{G}}^2,$$

and therefore

$$\|\Delta_t\|_{\mathbf{G}} \leq e^{-ct}\|\Delta_0\|_{\mathbf{G}}.$$

Since $\lambda_{\min}(\mathbf{G})\|z\|^2 \leq \|z\|_{\mathbf{G}}^2 \leq \lambda_{\max}(\mathbf{G})\|z\|^2$, we obtain

$$\|\Delta_t\| \leq \lambda_{\min}(\mathbf{G})^{-\frac{1}{2}} \|\Delta_t\|_{\mathbf{G}} \leq \lambda_{\min}(\mathbf{G})^{-\frac{1}{2}} e^{-ct} \|\Delta_0\|_{\mathbf{G}} \leq \sqrt{\frac{\lambda_{\max}(\mathbf{G})}{\lambda_{\min}(\mathbf{G})}} e^{-ct} \|\Delta_0\| \leq \sqrt{\kappa(\mathbf{G})} e^{-ct} \|\Delta_0\|.$$

Thus,

$$\|X_t^x - X_t^y\| \leq \sqrt{\kappa(\mathbf{G})} e^{-ct} \|x - y\|.$$

The synchronous construction provides a coupling of $p_t(x, \cdot)$ and $p_t(y, \cdot)$. Consequently,

$$\mathcal{W}_q(p_t(x, \cdot), p_t(y, \cdot)) \leq (\mathbb{E}\|X_t^x - X_t^y\|^q)^{\frac{1}{q}} = \|X_t^x - X_t^y\| \leq \sqrt{\kappa(\mathbf{G})} e^{-ct} \|x - y\|.$$

Taking the supremum over $x \neq y$ and applying *Theorem 2.29* yields

$$\alpha_q(p_t) \leq \sqrt{\kappa(\mathbf{G})} e^{-ct}, \quad q \in [1, \infty), \quad t \geq 0.$$

Having related the spectral rate to a finite-time contraction estimate, we now turn to its optimisation. By *Theorem 3.5*, for a fixed invariant covariance matrix $\mathbf{C}$, the asymptotic Wasserstein contraction rate is determined by

$$\lambda^*(\mathbf{A}) := \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda, \quad \mathbf{A} = (\sigma\sigma^\top + \mathbf{J})\mathbf{C}^{-1}, \quad \mathbf{D} = \sigma\sigma^\top.$$

Thus, we seek to maximise the smallest real part of the eigenvalues over the admissible matrices $\mathbf{D}$ and $\mathbf{J}$. The *spectral theorem* yields an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{\Lambda} := \mathbf{Q}^\top \mathbf{C} \mathbf{Q} = \text{diag}(\sigma_1^2, \dots, \sigma_d^2),$$

where $\sigma_1^2, \dots, \sigma_d^2$ are the eigenvalues of $\mathbf{C}$. Under the orthogonal change of coordinates, set $\mathbf{Y} := \mathbf{Q}^\top \mathbf{X}$. Then the Gaussian transformation rule from *Section 2.2* gives $\mathbf{Y} \sim \mathcal{N}(0, \mathbf{\Lambda})$ whenever $\mathbf{X} \sim \mathcal{N}(0, \mathbf{C})$. The transformed coefficient matrices are

$$\tilde{\mathbf{D}} = \mathbf{Q}^\top \mathbf{D} \mathbf{Q}, \quad \tilde{\mathbf{J}} = \mathbf{Q}^\top \mathbf{J} \mathbf{Q}, \quad \tilde{\mathbf{A}} = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = (\tilde{\mathbf{D}} + \tilde{\mathbf{J}}) \mathbf{\Lambda}^{-1}.$$

The orthogonal-conjugation properties established in *Section 2.1* imply that

$$\tilde{\mathbf{D}} \succeq 0, \quad \tilde{\mathbf{J}}^\top = -\tilde{\mathbf{J}}, \quad \text{tr}(\tilde{\mathbf{D}}) = \text{tr}(\mathbf{D}).$$

Moreover, $\tilde{\mathbf{A}}$ is similar to $\mathbf{A}$, and hence $\lambda^*(\tilde{\mathbf{A}}) = \lambda^*(\mathbf{A})$. Therefore, the orthogonal transformation preserves both the admissible class and the asymptotic convergence rate. We may therefore assume, without loss of generality, that $\mathbf{C}$ is diagonal.

3.2.1 Two-Dimensional Cases

We begin with dimension two, that is, the smallest dimension in which a non-zero antisymmetric component of the drift can occur. In dimension one, an antisymmetric matrix $\mathbf{J} = (j)$ must satisfy $j = -j$, and therefore $j = 0$. Hence, a one-dimensional drift contains no non-reversible antisymmetric

component. In dimension two, set

$$\mathbf{C} = \begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix},$$

where $u, v > 0$ are the variances along the two principal coordinate directions. Consider the symmetric diffusion matrix and the antisymmetric perturbation:

$$\mathbf{D} := \sigma\sigma^\top = \begin{pmatrix} \varepsilon & \delta \\ \delta & \gamma \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix},$$

where $\varepsilon, \gamma \geq 0$ denote the noise intensities in the two coordinate directions, $\delta \in \mathbb{R}$ is their cross-covariance, and $\beta \in \mathbb{R}$ determines the strength and orientation of the non-reversible drift. Since $\mathbf{D}$ must be positive semidefinite, its parameters satisfy $\delta^2 \leq \varepsilon\gamma$. The corresponding drift matrix is

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1} = \begin{pmatrix} \frac{\varepsilon}{u} & \frac{\beta+\delta}{v} \\ -\frac{\beta+\delta}{u} & \frac{\gamma}{v} \end{pmatrix}, \quad (3.14)$$

where the contribution of the antisymmetric perturbation to the drift matrix is

$$\mathbf{J}\mathbf{C}^{-1} = \begin{pmatrix} 0 & \frac{\beta}{v} \\ -\frac{\beta}{u} & 0 \end{pmatrix}.$$

Consequently, for $x = (x_1, x_2)^\top$, the corresponding non-reversible drift vector field is

$$-\mathbf{J}\mathbf{C}^{-1}x = \begin{pmatrix} -\frac{\beta x_2}{v} \\ \frac{\beta x_1}{u} \end{pmatrix}.$$

When $\beta \neq 0$, the velocity in the first coordinate depends on the position in the second coordinate, and vice versa. This means that the antisymmetric drift couples the two principal directions, and a displacement in either direction is continuously transferred into the other.

The characteristic equation for (3.14) is

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \left(\frac{\varepsilon}{u} - \lambda\right)\left(\frac{\gamma}{v} - \lambda\right) - \frac{\delta^2 - \beta^2}{uv} = 0,$$

or equivalently,

$$\lambda^2 - \left(\frac{\varepsilon}{u} + \frac{\gamma}{v}\right)\lambda + \frac{\varepsilon\gamma + \beta^2 - \delta^2}{uv} = 0.$$

It follows that the eigenvalues of $\mathbf{A}$ are

$$\lambda_{\pm} = \frac{1}{2}\left(\frac{\varepsilon}{u} + \frac{\gamma}{v}\right) \pm \sqrt{\frac{1}{4}\left(\frac{\varepsilon}{u} - \frac{\gamma}{v}\right)^2 + \frac{\delta^2 - \beta^2}{uv}}.$$

If the expression under the square root is non-negative, the two eigenvalues are real. If it is negative, they form a complex-conjugate pair. Writing $a_+ := \max\{a, 0\}$, the minimum real part of the eigenvalues is

$$\lambda^* = \frac{1}{2}\left(\frac{\varepsilon}{u} + \frac{\gamma}{v}\right) - \sqrt{\left[\frac{1}{4}\left(\frac{\varepsilon}{u} - \frac{\gamma}{v}\right)^2 + \frac{\delta^2 - \beta^2}{uv}\right]_+}.$$

For fixed ε , γ , and β , this expression is non-increasing as δ^2 increases. Moreover, whenever $\mathbf{D} \succeq 0$,

replacing δ by zero preserves the diagonal noise intensities and the trace, while it can only leave the convergence rate unchanged or increase it. So it is sufficient to let $\delta = 0$, and we are therefore led to maximise

$$\lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) - \sqrt{\left[\frac{1}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{\beta^2}{uv} \right]_+}. \quad (3.15)$$

The next *Lemma 3.6* optimises the antisymmetric drift β , while the noise intensities ε and γ are fixed. The lemma is adapted from [BE26, Lemma 4.33(1)]. Since the original reference does not provide a proof, we include one here for completeness.

Lemma 3.6. *For every fixed $\varepsilon, \gamma \in [0, \infty)$,*

$$\max_{\beta \in \mathbb{R}} \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right).$$

The maximum is attained for every $\beta \in \mathbb{R}$ satisfying

$$|\beta| \geq \frac{1}{2} \left| \sqrt{\frac{v}{u}} \varepsilon - \sqrt{\frac{u}{v}} \gamma \right|.$$

Proof. The sum of the two eigenvalues is

$$\lambda_+ + \lambda_- = \text{tr}(\mathbf{A}) = \frac{\varepsilon}{u} + \frac{\gamma}{v}.$$

So the smaller of their real parts cannot exceed their average:

$$\lambda^*(\varepsilon, \gamma, \beta) \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right)$$

for any $\beta \in \mathbb{R}$. The same upper bound also follows directly from (3.15), since the square-root term is non-negative. The equality holds when the positive part under the square root vanishes:

$$\frac{1}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{\beta^2}{uv} \leq 0,$$

which is equivalent to

$$\beta^2 \geq \frac{uv}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 = \frac{1}{4} \left(\frac{v\varepsilon - u\gamma}{\sqrt{uv}} \right)^2 = \frac{1}{4} \left(\sqrt{\frac{v}{u}} \varepsilon - \sqrt{\frac{u}{v}} \gamma \right)^2.$$

Taking square roots yields the claimed condition for $|\beta|$. □

The lemma shows that, for fixed noise intensities, the antisymmetric drift can equalise the real parts of the two eigenvalues. However, it cannot increase their sum, which is fixed by $\text{tr}(\mathbf{A})$. So we aim to optimise the common real part by constraining the total amount of noise.

Total Noise Constraint

Without a constraint on the coefficients, the convergence rate can be made arbitrarily large simply by rescaling the diffusion and drift matrices. We therefore fix the scale by imposing the noise constraint

$$\varepsilon + \gamma = \text{tr}(\sigma \sigma^\top) \leq 1. \quad (3.16)$$

Over an Euler step of length h , the stochastic increment is

$$\Delta_h X := \sqrt{2}\sigma(B_{t+h} - B_t).$$

Since $B_{t+h} - B_t \sim \mathcal{N}(0, h\mathbf{I}_2)$, its covariance is $\text{Cov}(\Delta_h X) = 2h\sigma\sigma^\top$. Consequently,

$$\mathbb{E}[\|\Delta_h X\|^2] = 2\mathbb{E}[\|\sigma(B_{t+h} - B_t)\|^2] = 2h \text{tr}(\sigma\sigma^\top) \leq 2h.$$

Thus, the constraint normalises the total diffusion budget, enabling comparison of different noise allocations. However, it does not restrict the antisymmetric drift parameter β , which will motivate the drift and noise constraint in the next subsection. We now consider the following optimisation problem:

$$\begin{aligned} \text{maximise} \quad & \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) - \sqrt{\left[\frac{1}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{\beta^2}{uv} \right]_+}, \\ \text{subject to} \quad & \varepsilon, \gamma \geq 0, \quad \varepsilon + \gamma = \text{tr}(\sigma\sigma^\top) \leq 1, \quad \beta \in \mathbb{R}. \end{aligned}$$

Theorem 3.7 ([BE26, Lemma 4.33(2)]). *Under the noise constraint (3.16),*

$$\max_{\substack{\varepsilon, \gamma \geq 0, \varepsilon + \gamma \leq 1 \\ \beta \in \mathbb{R}}} \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \max \left\{ \frac{1}{u}, \frac{1}{v} \right\}. \quad (3.17)$$

If $u \geq v$, one optimal choice is

$$\varepsilon = 0, \quad \gamma = 1, \quad |\beta| \geq \frac{1}{2} \sqrt{\frac{u}{v}}.$$

Proof. By Lemma 3.6, for any fixed $\varepsilon, \gamma \geq 0$,

$$\lambda^*(\varepsilon, \gamma, \beta) \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right), \quad \beta \in \mathbb{R}.$$

Using the noise constraint (3.16), we obtain

$$\lambda^*(\varepsilon, \gamma, \beta) \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) \leq \frac{1}{2} \max \left\{ \frac{1}{u}, \frac{1}{v} \right\} (\varepsilon + \gamma) \leq \frac{1}{2} \max \left\{ \frac{1}{u}, \frac{1}{v} \right\}.$$

It remains to show that this upper bound can be attained. Suppose that $u \geq v$. Then

$$\max \left\{ \frac{1}{u}, \frac{1}{v} \right\} = \frac{1}{v}.$$

With $\varepsilon = 0$ and $\gamma = 1$, the upper bound becomes

$$\frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) = \frac{1}{2v}.$$

The equality is attained whenever

$$|\beta| \geq \frac{1}{2} \left| \sqrt{\frac{v}{u}} \varepsilon - \sqrt{\frac{u}{v}} \gamma \right| = \frac{1}{2} \sqrt{\frac{u}{v}}.$$

□

Remark. If $v \geq u$, the analogous optimal choice is obtained by interchanging the two coordinates:

$$\varepsilon = 1, \quad \gamma = 0, \quad |\beta| \geq \frac{1}{2} \sqrt{\frac{v}{u}}.$$

The proof follows by symmetry.

To see the improvement produced by the antisymmetric drift, we compare the optimal non-reversible rate with the optimal reversible rate under the same noise constraint. In the reversible case, $\beta = 0$, and hence

$$\lambda^*(\varepsilon, \gamma, 0) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) - \frac{1}{2} \left| \frac{\varepsilon}{u} - \frac{\gamma}{v} \right| = \min \left\{ \frac{\varepsilon}{u}, \frac{\gamma}{v} \right\} =: r. \quad (3.18)$$

Let the optimal reversible rate be

$$\lambda_{\text{rev}}^* := \max_{\substack{\varepsilon, \gamma \geq 0 \\ \varepsilon + \gamma \leq 1}} \lambda^*(\varepsilon, \gamma, 0).$$

Then $\varepsilon \geq ur$ and $\gamma \geq vr$. Consequently, the noise constraint implies $1 \geq \varepsilon + \gamma \geq (u + v)r$, and therefore

$$r = \min \left\{ \frac{\varepsilon}{u}, \frac{\gamma}{v} \right\} \leq \frac{1}{u + v}.$$

This upper bound is attained when the two coordinate rates are balanced:

$$\frac{\varepsilon}{u} = \frac{\gamma}{v}, \quad \varepsilon + \gamma = 1.$$

Solving these equations yields

$$\varepsilon_{\text{rev}} = \frac{u}{u + v}, \quad \gamma_{\text{rev}} = \frac{v}{u + v}, \quad \lambda_{\text{rev}}^* = \frac{1}{u + v}.$$

By contrast, *Theorem 3.7* gives the optimal non-reversible rate:

$$\lambda_{\text{nr}}^* := \max_{\substack{\varepsilon, \gamma \geq 0, \varepsilon + \gamma \leq 1 \\ \beta \in \mathbb{R}}} \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \max \left\{ \frac{1}{u}, \frac{1}{v} \right\} = \frac{1}{2 \min \{u, v\}}.$$

Then the improvement is

$$\frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} = \frac{u + v}{2 \min \{u, v\}} = \frac{1 + \kappa(\mathbf{C})}{2},$$

where $\kappa(\mathbf{C}) := \frac{\max \{u, v\}}{\min \{u, v\}}$ denotes the condition number of the covariance matrix. Thus, there is no improvement in the isotropic case $u = v$, for which $\kappa(\mathbf{C}) = 1$. However, the improvement becomes arbitrarily large as the covariance becomes increasingly anisotropic:

$$\frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} \sim \frac{\kappa(\mathbf{C})}{2}, \quad \kappa(\mathbf{C}) \rightarrow \infty.$$

Suppose that $u \geq v$. Then

$$\lambda_{\text{nr}}^* = \frac{1}{2v}, \quad \frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} = \frac{1}{2} \left(1 + \frac{u}{v} \right) \sim \frac{u}{2v}, \quad \frac{u}{v} \rightarrow \infty.$$

In the reversible optimum, when $u \gg v$, $\varepsilon_{\text{rev}} \approx 1$ and $\gamma_{\text{rev}} \approx 0$. Therefore, nearly all the noise budget is spent directly on the slow direction associated with the larger variance u . Because the two coordinates

are uncoupled, the overall convergence rate is determined by the slower of the two.

The optimal non-reversible construction takes a strikingly different approach. It places all the noise in the coordinate corresponding to the smaller variance v , by choosing

$$\mathbf{D} = \sigma\sigma^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The drift matrix becomes

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1} = \begin{pmatrix} 0 & \frac{\beta}{v} \\ -\frac{\beta}{u} & \frac{1}{v} \end{pmatrix},$$

whose eigenvalues are

$$\lambda_{\pm} = \frac{1}{2v} \pm \sqrt{\frac{1}{4v^2} - \frac{\beta^2}{uv}}.$$

If $|\beta| \geq \frac{1}{2}\sqrt{\frac{u}{v}}$, the expression under the square root is non-positive, and hence

$$\operatorname{Re} \lambda_+ = \operatorname{Re} \lambda_- = \frac{1}{2v}, \quad \lambda_+ + \lambda_- = \operatorname{tr}(\mathbf{A}) = \frac{1}{v},$$

where the antisymmetric drift couples the two coordinate directions. Indeed, let $e_2 = (0, 1)^\top$ denote the direction in which the noise is injected; then $\mathbf{A}e_2 = \left(\frac{\beta}{v}, \frac{1}{v}\right)^\top$. For $\beta \neq 0$, $\operatorname{span}\{e_2, \mathbf{A}e_2\} = \mathbb{R}^2$, allowing degenerate noise to produce convergence in both directions.

However, this acceleration comes at the cost of a large drift. Attaining the optimal non-reversible rate requires

$$|\beta| \geq \frac{1}{2}\sqrt{\frac{u}{v}} = \frac{1}{2}\sqrt{\kappa(\mathbf{C})}.$$

Hence, when the covariance is strongly anisotropic, the required drift grows with $\sqrt{\kappa(\mathbf{C})}$. Such a large drift may cause numerical instability and require smaller time-discretisation step sizes. This motivates fixing the strength of the antisymmetric drift and studying the remaining noise intensity.

Fixed Drift and Noise Constraint

Throughout this subsection, assume that $u \geq v > 0$. We fix the antisymmetric drift parameter $\beta \in \mathbb{R}$, and the noise intensity $\varepsilon \geq 0$ in the direction corresponding to the larger variance u . Then we optimise over the remaining noise intensity $\gamma \geq 0$. The total noise constraint (3.16) is *not* imposed in this optimisation problem.

The following theorem is adapted from [BE26, Lemma 4.34]. Since the original reference does not provide a proof, we include one here for completeness.

Theorem 3.8. *For every fixed $\beta \in \mathbb{R}$ and $\varepsilon \geq 0$,*

$$\max_{\gamma \geq 0} \lambda^*(\varepsilon, \gamma, \beta) = \frac{\varepsilon}{u} + \frac{|\beta|}{\sqrt{uv}}. \quad (3.19)$$

One maximiser is

$$\gamma^* = \varepsilon \frac{v}{u} + 2|\beta| \sqrt{\frac{v}{u}}. \quad (3.20)$$

If $\beta \neq 0$, this maximiser is unique. If $\beta = 0$, every $\gamma \geq \varepsilon \frac{v}{u}$ is a maximiser.

Proof. We rewrite (3.15) as

$$\lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) - \frac{1}{2} \sqrt{\left[\left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{4\beta^2}{uv} \right]_+}.$$

We first suppose that

$$\frac{\gamma}{v} - \frac{\varepsilon}{u} \leq \frac{2|\beta|}{\sqrt{uv}}.$$

Since the square-root term is non-negative,

$$\lambda^*(\varepsilon, \gamma, \beta) \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\varepsilon}{u} + \frac{2|\beta|}{\sqrt{uv}} \right) = \frac{\varepsilon}{u} + \frac{|\beta|}{\sqrt{uv}}.$$

Now suppose that

$$\frac{\gamma}{v} - \frac{\varepsilon}{u} > \frac{2|\beta|}{\sqrt{uv}}.$$

In this case, the expression inside the positive part is strictly positive. Moreover,

$$\left(\frac{\gamma}{v} - \frac{\varepsilon}{u} \right)^2 - \frac{4\beta^2}{uv} = \left(\frac{\gamma}{v} - \frac{\varepsilon}{u} - \frac{2|\beta|}{\sqrt{uv}} \right) \left(\frac{\gamma}{v} - \frac{\varepsilon}{u} + \frac{2|\beta|}{\sqrt{uv}} \right) \geq \left(\frac{\gamma}{v} - \frac{\varepsilon}{u} - \frac{2|\beta|}{\sqrt{uv}} \right)^2.$$

Taking square roots gives

$$\sqrt{\left(\frac{\gamma}{v} - \frac{\varepsilon}{u} \right)^2 - \frac{4\beta^2}{uv}} \geq \frac{\gamma}{v} - \frac{\varepsilon}{u} - \frac{2|\beta|}{\sqrt{uv}}. \quad (3.21)$$

Consequently,

$$\lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) - \frac{1}{2} \sqrt{\left(\frac{\gamma}{v} - \frac{\varepsilon}{u} \right)^2 - \frac{4\beta^2}{uv}} \leq \frac{1}{2} \left[\frac{\varepsilon}{u} + \frac{\gamma}{v} - \left(\frac{\gamma}{v} - \frac{\varepsilon}{u} - \frac{2|\beta|}{\sqrt{uv}} \right) \right] = \frac{\varepsilon}{u} + \frac{|\beta|}{\sqrt{uv}}.$$

Combining the two cases, we obtain the upper bound in (3.19) for every $\gamma \geq 0$.

To show that this upper bound is attained, choose γ^* as in (3.20). Then

$$\frac{\gamma^*}{v} - \frac{\varepsilon}{u} = \frac{2|\beta|}{\sqrt{uv}}, \quad (3.22)$$

and hence

$$\left(\frac{\varepsilon}{u} - \frac{\gamma^*}{v} \right)^2 - \frac{4\beta^2}{uv} = 0.$$

Therefore, the square-root term vanishes, and

$$\lambda^*(\varepsilon, \gamma^*, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma^*}{v} \right) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\varepsilon}{u} + \frac{2|\beta|}{\sqrt{uv}} \right) = \frac{\varepsilon}{u} + \frac{|\beta|}{\sqrt{uv}},$$

which proves (3.19).

It remains to determine which equality cases apply. When $\beta \neq 0$, the inequality (3.21) is strict, so equality cannot occur in the second case. In the first case, equality requires both the square-root term to vanish and (3.22) to hold, so γ^* is the unique maximiser. When $\beta = 0$, $\lambda^*(\varepsilon, \gamma, 0) = \min \left\{ \frac{\varepsilon}{u}, \frac{\gamma}{v} \right\}$ and therefore every $\gamma \geq \varepsilon \frac{v}{u}$ is a maximiser, completing the proof. $\square$

Next, we assess the acceleration. Let

$$\kappa := \kappa(\mathbf{C}) = \frac{u}{v} \geq 1, \quad B := |\beta|,$$

and denote the optimal rate for fixed ε and B by

$$\lambda_{\text{dn}}^*(\varepsilon, B) := \max_{\gamma \geq 0} \lambda^*(\varepsilon, \gamma, \beta).$$

Theorem 3.8 yields

$$\gamma^*(\varepsilon, B) = \frac{\varepsilon}{\kappa} + \frac{2B}{\sqrt{\kappa}}, \quad \lambda_{\text{dn}}^*(\varepsilon, B) = \frac{\varepsilon}{u} + \frac{B}{\sqrt{uv}} = \frac{\varepsilon + B\sqrt{\kappa}}{u}.$$

We first compare this rate with that of the reversible dynamics under the same fixed value of ε . When $\beta = 0$, equation (3.18) gives

$$\lambda^*(\varepsilon, \gamma, 0) = \min \left\{ \frac{\varepsilon}{u}, \frac{\gamma}{v} \right\}.$$

Therefore,

$$\bar{\lambda}_{\text{rev}}^*(\varepsilon) := \max_{\gamma \geq 0} \lambda^*(\varepsilon, \gamma, 0) = \frac{\varepsilon}{u}.$$

The maximum is attained for every $\gamma \geq \varepsilon \frac{v}{u}$. It follows that the antisymmetric drift produces the additive improvement

$$\lambda_{\text{dn}}^*(\varepsilon, B) - \bar{\lambda}_{\text{rev}}^*(\varepsilon) = \frac{B}{\sqrt{uv}}.$$

If $\varepsilon > 0$, the corresponding acceleration factor is

$$\frac{\lambda_{\text{dn}}^*(\varepsilon, B)}{\bar{\lambda}_{\text{rev}}^*(\varepsilon)} = 1 + \frac{B\sqrt{\kappa}}{\varepsilon}.$$

Thus, for fixed $B, \varepsilon > 0$, the relative improvement grows like $\sqrt{\kappa}$ as the covariance becomes increasingly anisotropic. Suppose now $0 \leq \varepsilon \leq 1$ and $B > 0$ is fixed. Since $u \geq v$,

$$\frac{\varepsilon}{u} \leq \frac{1}{u} \leq \frac{1}{\sqrt{uv}} \leq \frac{1}{v}.$$

Consequently,

$$\frac{B}{\sqrt{uv}} \leq \lambda_{\text{dn}}^*(\varepsilon, B) = \frac{\varepsilon}{u} + \frac{B}{\sqrt{uv}} \leq \frac{B+1}{\sqrt{uv}}.$$

Hence, the optimal rate in (3.19) is of order $\frac{1}{\sqrt{uv}}$. The limiting case $\varepsilon = 0$ is particularly revealing:

$$\bar{\lambda}_{\text{rev}}^*(0) = 0, \quad \lambda_{\text{dn}}^*(0, B) = \frac{B}{\sqrt{uv}} > 0, \quad B > 0.$$

Without the antisymmetric coupling, the reversible system does not converge in the first coordinate, whereas the non-reversible system has a strictly positive convergence rate.

When we examine the optimal noise intensity, the absence of an explicit constraint on γ does not lead to an arbitrarily large noise intensity. For $0 \leq \varepsilon \leq 1$,

$$0 \leq \gamma^*(\varepsilon, B) = \frac{\varepsilon}{\kappa} + \frac{2B}{\sqrt{\kappa}} \leq \varepsilon + 2B \leq 1 + 2B.$$

Moreover, for fixed B and ε , $\gamma^*(\varepsilon, B) \rightarrow 0$ as $\kappa \rightarrow \infty$. For fixed u, v , and B , the optimal rate is strictly increasing in ε . Therefore, if the only additional restriction is $0 \leq \varepsilon \leq 1$, the optimum is attained at

$$\varepsilon_{\text{opt}} = 1, \quad \gamma_{\text{opt}} = \frac{1}{\kappa} + \frac{2B}{\sqrt{\kappa}},$$

and the corresponding rate is

$$\lambda_B^* := \max_{\substack{0 \leq \varepsilon \leq 1 \\ \gamma \geq 0}} \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{u} + \frac{B}{\sqrt{uv}}.$$

Nevertheless, when $\kappa \gg 1$, taking $\varepsilon = 0$ gives a suboptimal alternative:

$$\gamma_0 = \frac{2B}{\sqrt{\kappa}} = 2B\sqrt{\frac{v}{u}}, \quad \lambda_0 := \lambda_{\text{dn}}^*(0, B) = \frac{B}{\sqrt{uv}}.$$

Since $\gamma_0 \rightarrow 0$ as $\kappa \rightarrow \infty$,

$$\text{tr}(\sigma\sigma^\top) = \varepsilon + \gamma_0 = \frac{2B}{\sqrt{\kappa}} \rightarrow 0.$$

Therefore, in the strongly anisotropic regime, a positive convergence rate can be obtained while injecting only a small amount of noise into a single coordinate. The relative loss compared with the optimum is

$$\frac{\lambda_B^* - \lambda_0}{\lambda_B^*} = \frac{\frac{1}{u}}{\frac{1}{u} + \frac{B}{\sqrt{uv}}} = \frac{1}{1 + B\sqrt{\kappa}} \sim \frac{1}{B\sqrt{\kappa}}.$$

Finally, we compare the fixed-drift rate with the optimal rate under the total noise constraint. When $u \geq v$, *Theorem 3.7* gives $\lambda_{\text{noise}}^* = \frac{1}{2v}$. For fixed B and $0 \leq \varepsilon \leq 1$,

$$\frac{\lambda_{\text{dn}}^*(\varepsilon, B)}{\lambda_{\text{noise}}^*} = 2v \left(\frac{\varepsilon}{u} + \frac{B}{\sqrt{uv}} \right) = \frac{2\varepsilon}{\kappa} + \frac{2B}{\sqrt{\kappa}} \rightarrow 0$$

as $\kappa \rightarrow \infty$. Thus, although a fixed antisymmetric drift gives a substantial acceleration relative to the reversible dynamics, it cannot achieve a rate of the same order as the unrestricted optimum under the total noise constraint when the covariance becomes strongly anisotropic. To determine how the drift must scale, let $B = B_\kappa$ depend on the condition number. The preceding ratio shows that B_κ must grow on the scale $\sqrt{\kappa}$. For example, when $\varepsilon = 0$, choosing $B_\kappa = \frac{\sqrt{\kappa}}{2}$ gives

$$\gamma^* = \frac{2B_\kappa}{\sqrt{\kappa}} = 1, \quad \lambda_{\text{dn}}^*(0, B_\kappa) = \frac{B_\kappa}{\sqrt{uv}} = \frac{1}{2v} = \lambda_{\text{noise}}^*,$$

which is exactly the optimal construction under the total noise constraint. Therefore, the acceleration is a trade-off between noise allocation and drift strength.

3.2.2 Higher-Dimensional Cases

In dimension two, the characteristic equation can be solved explicitly. However, in higher dimensions, this approach quickly becomes impractical. Fortunately, explicit formulas for the individual eigenvalues are unnecessary. The trace controls their real parts, while an appropriate antisymmetric perturbation can make all these real parts equal, leading to an optimal convergence rate in every dimension.

The following *Proposition 3.9* provides the linear-algebraic ingredient needed for *Theorem 3.11*. Its

upper bound is [LNP13, Proposition 1(iii)], while attainment is stated in [LNP13, Proposition 4]. The construction below follows [LNP13, Propositions 2–3 and Lemma 2], rewritten in our notation.

Proposition 3.9. *Let $\mathbf{S} \in \mathcal{S}_d^{>0}(\mathbb{R})$ be real, symmetric and positive definite. Then, for every $\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})$,*

$$\lambda^*(\mathbf{S} + \tilde{\mathbf{J}}) \leq \frac{\text{tr}(\mathbf{S})}{d}.$$

Moreover, there exists $\tilde{\mathbf{J}}^* \in \mathcal{A}_d(\mathbb{R})$ such that

$$\lambda^*(\mathbf{S} + \tilde{\mathbf{J}}^*) = \frac{\text{tr}(\mathbf{S})}{d}.$$

Consequently,

$$\max_{\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})} \lambda^*(\mathbf{S} + \tilde{\mathbf{J}}) = \frac{\text{tr}(\mathbf{S})}{d}.$$

Proof. We first establish the upper bound. Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\mathbf{S} + \tilde{\mathbf{J}}$, counted with algebraic multiplicity. Since every real antisymmetric matrix has zero trace,

$$\sum_{k=1}^d \text{Re } \lambda_k = \text{Re } \text{tr}(\mathbf{S} + \tilde{\mathbf{J}}) = \text{tr}(\mathbf{S}).$$

Therefore,

$$\lambda^*(\mathbf{S} + \tilde{\mathbf{J}}) = \min_{1 \leq k \leq d} \text{Re } \lambda_k \leq \frac{1}{d} \sum_{k=1}^d \text{Re } \lambda_k = \frac{\text{tr}(\mathbf{S})}{d}.$$

We now construct an antisymmetric matrix for which equality holds. We begin by proving that there exists an orthonormal basis $\{\psi_1, \dots, \psi_d\}$ of $\mathbb{R}^d$ satisfying

$$\psi_k^\top \mathbf{S} \psi_k = \frac{\text{tr}(\mathbf{S})}{d}, \quad k = 1, \dots, d.$$

We prove this assertion by induction on d . The case $d = 1$ is immediate. Suppose that $d \geq 2$ and that the assertion holds in dimension $d - 1$. If

$$\mathbf{S} = \frac{\text{tr}(\mathbf{S})}{d} \mathbf{I}_d,$$

then every orthonormal basis has the required property. Otherwise,

$$\lambda_{\min}(\mathbf{S}) < \frac{\text{tr}(\mathbf{S})}{d} < \lambda_{\max}(\mathbf{S}).$$

Let $u_{\min}$ and $u_{\max}$ be unit eigenvectors associated with $\lambda_{\min}(\mathbf{S})$ and $\lambda_{\max}(\mathbf{S})$, respectively. Define

$$\psi(t) = \cos(t)u_{\min} + \sin(t)u_{\max}, \quad 0 \leq t \leq \frac{\pi}{2},$$

and consider the continuous function

$$f(t) := \psi(t)^\top \mathbf{S} \psi(t).$$

We have

$$f(0) = \lambda_{\min}(\mathbf{S}), \quad f\left(\frac{\pi}{2}\right) = \lambda_{\max}(\mathbf{S}).$$

By the *intermediate value theorem*, there exists $t_0 \in (0, \frac{\pi}{2})$ such that

$$f(t_0) = \frac{\text{tr}(\mathbf{S})}{d}.$$

Set $\psi_1 := \psi(t_0)$, and let $V := \psi_1^\perp = \{x \in \mathbb{R}^d : \psi_1^\top x = 0\}$. Denote the orthogonal projection onto V by $\mathbf{P}_V = \mathbf{I}_d - \psi_1 \psi_1^\top$. Then

$$\mathbf{P}_V x = x - \psi_1 (\psi_1^\top x) = x, \quad x \in V.$$

Define the compression of $\mathbf{S}$ to V by

$$\mathbf{S}_V : V \rightarrow V, \quad \mathbf{S}_V x := \mathbf{P}_V \mathbf{S} x.$$

For $x, y \in V$,

$$x^\top \mathbf{S}_V x = x^\top \mathbf{P}_V \mathbf{S} x = x^\top \mathbf{S} x > 0$$

whenever $x \neq 0$, and

$$x^\top \mathbf{S}_V y = x^\top \mathbf{S} y = y^\top \mathbf{S} x = y^\top \mathbf{S}_V x.$$

Then the operator $\mathbf{S}_V$ is symmetric and positive definite on V . If $\{e_2, \dots, e_d\}$ is an orthonormal basis of V , then $\{\psi_1, e_2, \dots, e_d\}$ is an orthonormal basis of $\mathbb{R}^d$, and hence

$$\text{tr}(\mathbf{S}_V) = \sum_{k=2}^d e_k^\top \mathbf{S}_V e_k = \sum_{k=2}^d e_k^\top \mathbf{S} e_k = \text{tr}(\mathbf{S}) - \psi_1^\top \mathbf{S} \psi_1 = \text{tr}(\mathbf{S}) - \frac{\text{tr}(\mathbf{S})}{d} = \frac{d-1}{d} \text{tr}(\mathbf{S}).$$

By the induction hypothesis, V admits an orthonormal basis $\{\psi_2, \dots, \psi_d\}$ such that

$$\psi_k^\top \mathbf{S}_V \psi_k = \frac{\text{tr}(\mathbf{S}_V)}{d-1} = \frac{\text{tr}(\mathbf{S})}{d}, \quad k = 2, \dots, d.$$

Because $\psi_k \in V$, we have $\mathbf{P}_V \psi_k = \psi_k$, and hence $\psi_k^\top \mathbf{S}_V \psi_k = \psi_k^\top \mathbf{S} \psi_k$. Together with the choice of ψ_1 , this proves that $\{\psi_1, \dots, \psi_d\}$ is an orthonormal basis of $\mathbb{R}^d$ satisfying

$$\psi_k^\top \mathbf{S} \psi_k = \frac{\text{tr}(\mathbf{S})}{d}, \quad k = 1, \dots, d.$$

We now use this basis to construct $\tilde{\mathbf{J}}^*$. We seek a matrix $\mathbf{B}^* := \mathbf{S} + \tilde{\mathbf{J}}^*$ whose eigenvalues all have real part $\frac{\text{tr}(\mathbf{S})}{d}$. Choose pairwise distinct positive numbers $q_1, \dots, q_d$, and define

$$\mathbf{Q} := \sum_{k=1}^d q_k \psi_k \psi_k^\top.$$

Then $\mathbf{Q}$ is symmetric and positive definite, and

$$\mathbf{Q} \psi_k = q_k \psi_k, \quad k = 1, \dots, d.$$

Since the eigenvalues of a real antisymmetric matrix are purely imaginary, it is sufficient to make $\mathbf{B}^*$

similar to a matrix of the form

$$\frac{\text{tr}(\mathbf{S})}{d}\mathbf{I}_d + \mathbf{K}, \quad \mathbf{K}^\top = -\mathbf{K}.$$

Consider

$$\mathbf{M} := \mathbf{Q}^{-\frac{1}{2}}\mathbf{B}^*\mathbf{Q}^{\frac{1}{2}},$$

which is similar to $\mathbf{B}^*$. The matrix

$$\mathbf{M} - \frac{\text{tr}(\mathbf{S})}{d}\mathbf{I}_d$$

is antisymmetric when

$$\mathbf{M} + \mathbf{M}^\top = 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{I}_d.$$

Then

$$\mathbf{Q}^{-\frac{1}{2}}\mathbf{B}^*\mathbf{Q}^{\frac{1}{2}} + \mathbf{Q}^{\frac{1}{2}}(\mathbf{B}^*)^\top\mathbf{Q}^{-\frac{1}{2}} = 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{I}_d.$$

Multiplying this identity on the left and right by $\mathbf{Q}^{\frac{1}{2}}$ gives

$$\mathbf{B}^*\mathbf{Q} + \mathbf{Q}(\mathbf{B}^*)^\top = 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{Q}.$$

Since $(\mathbf{B}^*)^\top = \mathbf{S} - \tilde{\mathbf{J}}^*$, it becomes

$$\tilde{\mathbf{J}}^*\mathbf{Q} - \mathbf{Q}\tilde{\mathbf{J}}^* = -\mathbf{S}\mathbf{Q} - \mathbf{Q}\mathbf{S} + 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{Q}. \quad (3.23)$$

We solve this equation entry by entry in the basis $\{\psi_1, \dots, \psi_d\}$. For $j \neq k$,

$$\psi_j^\top (\tilde{\mathbf{J}}^*\mathbf{Q} - \mathbf{Q}\tilde{\mathbf{J}}^*) \psi_k = q_k \psi_j^\top \tilde{\mathbf{J}}^* \psi_k - q_j \psi_j^\top \tilde{\mathbf{J}}^* \psi_k = (q_k - q_j) \psi_j^\top \tilde{\mathbf{J}}^* \psi_k,$$

while

$$\psi_j^\top \left(-\mathbf{S}\mathbf{Q} - \mathbf{Q}\mathbf{S} + 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{Q} \right) \psi_k = -q_k \psi_j^\top \mathbf{S} \psi_k - q_j \psi_j^\top \mathbf{S} \psi_k = -(q_j + q_k) \psi_j^\top \mathbf{S} \psi_k.$$

Therefore we define

$$\psi_j^\top \tilde{\mathbf{J}}^* \psi_k := -\frac{q_j + q_k}{q_k - q_j} \psi_j^\top \mathbf{S} \psi_k, \quad j \neq k.$$

Since $\mathbf{S}$ is symmetric,

$$\psi_k^\top \tilde{\mathbf{J}}^* \psi_j = -\frac{q_k + q_j}{q_j - q_k} \psi_k^\top \mathbf{S} \psi_j = \frac{q_j + q_k}{q_k - q_j} \psi_j^\top \mathbf{S} \psi_k = -\psi_j^\top \tilde{\mathbf{J}}^* \psi_k.$$

For $j = k$, the corresponding diagonal entry of the right-hand side is

$$\psi_k^\top \left(-\mathbf{S}\mathbf{Q} - \mathbf{Q}\mathbf{S} + 2\frac{\text{tr}(\mathbf{S})}{d}\mathbf{Q} \right) \psi_k = -2q_k \psi_k^\top \mathbf{S} \psi_k + 2q_k \frac{\text{tr}(\mathbf{S})}{d} = 0.$$

Then we set

$$\psi_k^\top \tilde{\mathbf{J}}^* \psi_k := 0, \quad k = 1, \dots, d,$$

which implies $(\tilde{\mathbf{J}}^*)^\top = -\tilde{\mathbf{J}}^*$. Moreover, since the required identity holds for every pair j, k in the orthonormal basis $\{\psi_1, \dots, \psi_d\}$, the constructed matrix satisfies (3.23). By the preceding argument, $\mathbf{M} - \frac{\text{tr}(\mathbf{S})}{d}\mathbf{I}_d$ is therefore antisymmetric. Since $\mathbf{M}$ is similar to $\mathbf{B}^*$, every eigenvalue of $\mathbf{B}^*$ has real part

$\frac{\text{tr}(\mathbf{S})}{d}$. Hence

$$\lambda^*(\mathbf{B}^*) = \frac{\text{tr}(\mathbf{S})}{d},$$

which completes the proof. $\square$

The preceding *Proposition 3.9* assumes that the symmetric part is positive definite. However, in the optimisation problem (*Theorem 3.11*) below, the diffusion matrix is allowed to be degenerate. Therefore, we need the following extension.

Corollary 3.10. *For every $\mathbf{S} \in \mathcal{S}_d^{\geq 0}(\mathbb{R})$,*

$$\max_{\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})} \lambda^*(\mathbf{S} + \tilde{\mathbf{J}}) = \frac{\text{tr}(\mathbf{S})}{d}.$$

Proof. Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\mathbf{S} + \tilde{\mathbf{J}}$, counted with algebraic multiplicity. Since an antisymmetric matrix has zero trace,

$$\sum_{k=1}^d \text{Re } \lambda_k = \text{Re } \text{tr}(\mathbf{S} + \tilde{\mathbf{J}}) = \text{tr}(\mathbf{S}),$$

and consequently

$$\lambda^*(\mathbf{S} + \tilde{\mathbf{J}}) \leq \frac{\text{tr}(\mathbf{S})}{d}$$

for every $\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})$. To prove that the upper bound is attained, fix any $\varepsilon > 0$ and set $\mathbf{S}_\varepsilon := \mathbf{S} + \varepsilon \mathbf{I}_d$. Then $\mathbf{S}_\varepsilon \in \mathcal{S}_d^{\geq 0}(\mathbb{R})$. By *Proposition 3.9*, there exists $\tilde{\mathbf{J}}_\varepsilon \in \mathcal{A}_d(\mathbb{R})$ such that

$$\lambda^*(\mathbf{S} + \varepsilon \mathbf{I}_d + \tilde{\mathbf{J}}_\varepsilon) = \frac{\text{tr}(\mathbf{S}) + d\varepsilon}{d} = \frac{\text{tr}(\mathbf{S})}{d} + \varepsilon.$$

Since

$$\lambda^*(\mathbf{S} + \varepsilon \mathbf{I}_d + \tilde{\mathbf{J}}_\varepsilon) = \lambda^*(\mathbf{S} + \tilde{\mathbf{J}}_\varepsilon) + \varepsilon,$$

cancelling ε yields

$$\lambda^*(\mathbf{S} + \tilde{\mathbf{J}}_\varepsilon) = \frac{\text{tr}(\mathbf{S})}{d},$$

proving the result. $\square$

The following theorem is a reformulation of [GM16, Theorem 2.1] for an arbitrary noise budget $\eta > 0$; the original normalisation corresponds to $\eta = d$.

Theorem 3.11. *Let $\mathbf{C} \in \mathcal{S}_d^{\geq 0}(\mathbb{R})$, and let $\eta > 0$ be the noise budget. Then*

$$\max_{\substack{\mathbf{D} \in \mathcal{S}_d^{\geq 0}(\mathbb{R}), \text{tr}(\mathbf{D}) \leq \eta \\ \mathbf{J} \in \mathcal{A}_d(\mathbb{R})}} \lambda^*((\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}) = \frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1}).$$

Moreover, if $v_{\max}$ is a unit eigenvector of $\mathbf{C}^{-1}$ corresponding to $\lambda_{\max}(\mathbf{C}^{-1})$, then the maximum is attained with $\mathbf{D}^* = \eta v_{\max} v_{\max}^\top$ and a suitable $\mathbf{J}^* \in \mathcal{A}_d(\mathbb{R})$. In particular, we can choose an optimal diffusion matrix with rank one.

Proof. Set $\mathbf{H} := \mathbf{C}^{-1}$. By *Section 2.3*, invariance implies infinitesimal invariance. Then by *Lemma 3.3*, every admissible drift matrix preserving $\mathcal{N}(0, \mathbf{C})$ is of the form

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{H},$$

where $\mathbf{D} = \sigma\sigma^\top \succeq 0$ and $\mathbf{J}^\top = -\mathbf{J}$. We first fix $\mathbf{D}$. Define

$$\tilde{\mathbf{D}} := \mathbf{H}^{\frac{1}{2}}\mathbf{D}\mathbf{H}^{\frac{1}{2}}, \quad \tilde{\mathbf{J}} := \mathbf{H}^{\frac{1}{2}}\mathbf{J}\mathbf{H}^{\frac{1}{2}}.$$

Because $\mathbf{H}^{\frac{1}{2}}$ is symmetric, $\tilde{\mathbf{D}}^\top = \tilde{\mathbf{D}}$ and $\tilde{\mathbf{J}}^\top = \mathbf{H}^{\frac{1}{2}}\mathbf{J}^\top\mathbf{H}^{\frac{1}{2}} = -\tilde{\mathbf{J}}$. Thus $\tilde{\mathbf{D}} \succeq 0$ and $\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})$. Moreover, the map $\mathbf{J} \mapsto \mathbf{H}^{\frac{1}{2}}\mathbf{J}\mathbf{H}^{\frac{1}{2}}$ is a bijection of $\mathcal{A}_d(\mathbb{R})$, with inverse $\tilde{\mathbf{J}} \mapsto \mathbf{H}^{-\frac{1}{2}}\tilde{\mathbf{J}}\mathbf{H}^{-\frac{1}{2}}$. We also have

$$\begin{aligned} \mathbf{H}^{-\frac{1}{2}}(\tilde{\mathbf{D}} + \tilde{\mathbf{J}})\mathbf{H}^{\frac{1}{2}} &= \mathbf{H}^{-\frac{1}{2}}\left[\mathbf{H}^{\frac{1}{2}}(\mathbf{D} + \mathbf{J})\mathbf{H}^{\frac{1}{2}}\right]\mathbf{H}^{\frac{1}{2}} \\ &= \left(\mathbf{H}^{-\frac{1}{2}}\mathbf{H}^{\frac{1}{2}}\right)(\mathbf{D} + \mathbf{J})\left(\mathbf{H}^{\frac{1}{2}}\mathbf{H}^{\frac{1}{2}}\right) \\ &= \mathbf{I}_d(\mathbf{D} + \mathbf{J})\mathbf{H} = (\mathbf{D} + \mathbf{J})\mathbf{H}. \end{aligned}$$

Thus $(\mathbf{D} + \mathbf{J})\mathbf{H}$ is similar to $\tilde{\mathbf{D}} + \tilde{\mathbf{J}}$, and therefore

$$\lambda^*((\mathbf{D} + \mathbf{J})\mathbf{H}) = \lambda^*(\tilde{\mathbf{D}} + \tilde{\mathbf{J}}).$$

By *Corollary 3.10*,

$$\max_{\tilde{\mathbf{J}} \in \mathcal{A}_d(\mathbb{R})} \lambda^*(\tilde{\mathbf{D}} + \tilde{\mathbf{J}}) = \frac{\text{tr}(\tilde{\mathbf{D}})}{d} = \frac{\text{tr}(\mathbf{H}^{\frac{1}{2}}\mathbf{D}\mathbf{H}^{\frac{1}{2}})}{d} = \frac{\text{tr}(\mathbf{D}\mathbf{H})}{d}.$$

Let $\lambda_1, \dots, \lambda_d$ be the eigenvalues of $\mathbf{H}$. Since $\mathbf{H}$ is real, symmetric and positive definite, the *spectral theorem* gives an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{Q}^\top \mathbf{H} \mathbf{Q} = \mathbf{\Lambda} := \text{diag}(\lambda_1, \dots, \lambda_d).$$

Equivalently, $\mathbf{H} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$. For $\mathbf{D} \succeq 0$, define $\mathbf{E} := \mathbf{Q}^\top \mathbf{D} \mathbf{Q}$. Then for every $x \in \mathbb{R}^d$,

$$x^\top \mathbf{E} x = x^\top \mathbf{Q}^\top \mathbf{D} \mathbf{Q} x = (\mathbf{Q}x)^\top \mathbf{D} (\mathbf{Q}x) \geq 0.$$

Hence $\mathbf{E} \succeq 0$. Moreover, $E_{ii} = e_i^\top \mathbf{E} e_i \geq 0$ for $i = 1, \dots, d$. The trace is invariant under orthogonal conjugation, so

$$\text{tr}(\mathbf{E}) = \text{tr}(\mathbf{Q}^\top \mathbf{D} \mathbf{Q}) = \text{tr}(\mathbf{D} \mathbf{Q} \mathbf{Q}^\top) = \text{tr}(\mathbf{D}).$$

Furthermore,

$$\begin{aligned}
\text{tr}(\mathbf{D}\mathbf{H}) &= \text{tr}(\mathbf{D}\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top) \\
&= \text{tr}(\mathbf{Q}^\top\mathbf{D}\mathbf{Q}\mathbf{\Lambda}) \\
&= \text{tr}(\mathbf{E}\mathbf{\Lambda}) \\
&= \sum_{i=1}^d \lambda_i E_{ii} \\
&\leq \lambda_{\max}(\mathbf{H}) \sum_{i=1}^d E_{ii} \\
&= \lambda_{\max}(\mathbf{H}) \text{tr}(\mathbf{E}) = \lambda_{\max}(\mathbf{H}) \text{tr}(\mathbf{D}).
\end{aligned}$$

Therefore, the noise constraint implies $\text{tr}(\mathbf{D}\mathbf{H}) \leq \eta \lambda_{\max}(\mathbf{H})$. Consequently, by the similarity relation and the bijection between $\mathbf{J}$ and $\tilde{\mathbf{J}}$ established above,

$$\lambda^*((\mathbf{D} + \mathbf{J})\mathbf{H}) \leq \max_{\mathbf{K} \in \mathcal{A}_d(\mathbb{R})} \lambda^*((\mathbf{D} + \mathbf{K})\mathbf{H}) = \max_{\tilde{\mathbf{K}} \in \mathcal{A}_d(\mathbb{R})} \lambda^*(\tilde{\mathbf{D}} + \tilde{\mathbf{K}}) = \frac{\text{tr}(\mathbf{D}\mathbf{H})}{d} \leq \frac{\eta}{d} \lambda_{\max}(\mathbf{H}).$$

It remains to prove that this upper bound is attained. Let $v_{\max}$ be a unit eigenvector satisfying

$$\mathbf{H}v_{\max} = \lambda_{\max}(\mathbf{H})v_{\max}, \quad \|v_{\max}\| = 1.$$

Define the rank-one diffusion matrix $\mathbf{D}^* := \eta v_{\max} v_{\max}^\top$. Since

$$x^\top \mathbf{D}^* x = \eta (v_{\max}^\top x)^2 \geq 0,$$

it is positive semidefinite. Moreover,

$$\text{tr}(\mathbf{D}^*) = \eta \text{tr}(v_{\max} v_{\max}^\top) = \eta v_{\max}^\top v_{\max} = \eta.$$

Therefore,

$$\text{tr}(\mathbf{D}^* \mathbf{H}) = \eta \text{tr}(v_{\max} v_{\max}^\top \mathbf{H}) = \eta v_{\max}^\top \mathbf{H} v_{\max} = \eta \lambda_{\max}(\mathbf{H}) v_{\max}^\top v_{\max} = \eta \lambda_{\max}(\mathbf{H}).$$

Let $\tilde{\mathbf{D}}^* := \mathbf{H}^{\frac{1}{2}} \mathbf{D}^* \mathbf{H}^{\frac{1}{2}}$. By *Corollary 3.10*, there exists $\tilde{\mathbf{J}}^* \in \mathcal{A}_d(\mathbb{R})$ such that

$$\lambda^*(\tilde{\mathbf{D}}^* + \tilde{\mathbf{J}}^*) = \frac{\text{tr}(\tilde{\mathbf{D}}^*)}{d}.$$

Define the corresponding antisymmetric matrix in the original coordinates by $\mathbf{J}^* := \mathbf{H}^{-\frac{1}{2}} \tilde{\mathbf{J}}^* \mathbf{H}^{-\frac{1}{2}}$. Since $\mathbf{H}^{-\frac{1}{2}}$ is symmetric,

$$(\mathbf{J}^*)^\top = \mathbf{H}^{-\frac{1}{2}} (\tilde{\mathbf{J}}^*)^\top \mathbf{H}^{-\frac{1}{2}} = -\mathbf{H}^{-\frac{1}{2}} \tilde{\mathbf{J}}^* \mathbf{H}^{-\frac{1}{2}} = -\mathbf{J}^*.$$

Thus $\mathbf{J}^* \in \mathcal{A}_d(\mathbb{R})$. Moreover,

$$\mathbf{H}^{-\frac{1}{2}} (\tilde{\mathbf{D}}^* + \tilde{\mathbf{J}}^*) \mathbf{H}^{\frac{1}{2}} = \mathbf{D}^* \mathbf{H} + \mathbf{J}^* \mathbf{H} = (\mathbf{D}^* + \mathbf{J}^*) \mathbf{H}.$$

Therefore, $(\mathbf{D}^* + \mathbf{J}^*)\mathbf{H}$ is similar to $\tilde{\mathbf{D}}^* + \tilde{\mathbf{J}}^*$, and hence

$$\lambda^*((\mathbf{D}^* + \mathbf{J}^*)\mathbf{H}) = \lambda^*(\tilde{\mathbf{D}}^* + \tilde{\mathbf{J}}^*) = \frac{\text{tr}(\tilde{\mathbf{D}}^*)}{d} = \frac{\text{tr}(\mathbf{D}^*\mathbf{H})}{d} = \frac{\eta}{d}\lambda_{\max}(\mathbf{H}).$$

Since $\mathbf{H} = \mathbf{C}^{-1}$, this proves the theorem. $\square$

Remark. The two-dimensional noise-constrained result is a special case of the theorem. If $d = 2$, $\eta = 1$, and $\mathbf{C} = \text{diag}(u, v)$,

$$\frac{\eta}{d}\lambda_{\max}(\mathbf{C}^{-1}) = \frac{1}{2} \max\left\{\frac{1}{u}, \frac{1}{v}\right\}.$$

When $u \geq v$, the unit eigenvector corresponding to $\lambda_{\max}(\mathbf{C}^{-1}) = \frac{1}{v}$ is e_2 , and the rank-one optimiser is

$$\mathbf{D}^* = e_2 e_2^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

In the notation of the two-dimensional case, this is exactly the choice $\varepsilon = 0$ and $\gamma = 1$ appearing in Theorem 3.7.

The following Example 3.12 is an explicit three-dimensional construction that attains the upper bound in the theorem. The antisymmetric drift redistributes the real parts of the eigenvalues until they all coincide with their trace-constrained average.

Example 3.12 (Three-Dimensional Case). Let $\mathbf{C} = \text{diag}(u, v, w)$ for $u, v, w > 0$. Although the cubic formula is available, using it to optimise the smallest real part obscures the mechanism and quickly becomes unwieldy. Nevertheless, we can still obtain an informative check by the preceding theorem. If $u \geq v \geq w$, then $\lambda_{\max}(\mathbf{C}^{-1}) = \frac{1}{w}$. Under the noise constraint $\text{tr}(\mathbf{D}) \leq \eta$, the theorem predicts the optimal rate $\frac{\eta}{3w}$. We place all the noise in the direction corresponding to the smallest variance w , and choose

$$\mathbf{D}^* = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \eta \end{pmatrix}, \quad \mathbf{J}^* = \begin{pmatrix} 0 & \beta_{12} & \beta_{13} \\ -\beta_{12} & 0 & 0 \\ -\beta_{13} & 0 & 0 \end{pmatrix},$$

where

$$\beta_{12}^2 = \frac{2uv\eta^2}{27w^2}, \quad \beta_{13}^2 = \frac{10u\eta^2}{27w}.$$

The corresponding drift matrix is

$$\mathbf{A}^* = (\mathbf{D}^* + \mathbf{J}^*)\mathbf{C}^{-1} = \begin{pmatrix} 0 & \frac{\beta_{12}}{v} & \frac{\beta_{13}}{w} \\ -\frac{\beta_{12}}{u} & 0 & 0 \\ -\frac{\beta_{13}}{u} & 0 & \frac{\eta}{w} \end{pmatrix}.$$

Its characteristic polynomial is

$$\det(\lambda\mathbf{I} - \mathbf{A}^*) = \lambda^3 - \frac{\eta}{w}\lambda^2 + \left(\frac{\beta_{12}^2}{uv} + \frac{\beta_{13}^2}{uw}\right)\lambda - \frac{\eta\beta_{12}^2}{uvw} = \left(\lambda - \frac{\eta}{3w}\right)\left[\left(\lambda - \frac{\eta}{3w}\right)^2 + \frac{\eta^2}{9w^2}\right].$$

Hence

$$\text{spec}(\mathbf{A}^*) = \left\{\frac{\eta}{3w}, \frac{\eta}{3w} + i\frac{\eta}{3w}, \frac{\eta}{3w} - i\frac{\eta}{3w}\right\}.$$

All three eigenvalues therefore have the same real part, and

$$\lambda^*(\mathbf{A}^*) = \frac{\eta}{3w} = \frac{\eta}{3} \lambda_{\max}(\mathbf{C}^{-1}).$$

3.3 Application to Langevin Dynamics

Degenerate noise does not leave the unforced variables untouched. Langevin dynamics provides a simple example: Brownian noise is injected only into the *velocity*, but the deterministic coupling continually transfers its effect to the *position*. This makes Langevin dynamics a natural application of the preceding Gaussian theory.

Let $U \in C^2(\mathbb{R}^d)$ satisfy

$$Z_U := \int_{\mathbb{R}^d} e^{-U(x)} dx < \infty.$$

We additionally assume that ∇U is globally Lipschitz. Then the drift $(x, v) \mapsto (v, -\nabla U(x) - \gamma v)$ is globally Lipschitz and has linear growth. Hence, for every square-integrable initial condition (X_0, V_0) independent of the driving BM, applying *Theorem 2.16* on each finite time interval yields a unique global strong solution. This assumption is automatic for the Gaussian potential considered later. For a friction coefficient $\gamma \geq 0$, the *kinetic Langevin dynamics* is

$$dX_t = V_t dt, \tag{3.24}$$

$$dV_t = -\nabla U(X_t) dt - \gamma V_t dt + \sqrt{2\gamma} dB_t, \tag{3.25}$$

where $(B_t)_{t \geq 0}$ is a standard BM on $\mathbb{R}^d$ [BE26, (4.68)]. Define the Hamiltonian

$$H(x, v) := U(x) + \frac{1}{2}|v|^2.$$

The generator decomposes as

$$\mathcal{L}f = \underbrace{v \cdot \nabla_x f - \nabla U(x) \cdot \nabla_v f}_{:= \mathcal{L}_H f} + \underbrace{\gamma(\Delta_v f - v \cdot \nabla_v f)}_{:= \mathcal{L}_{OU} f}.$$

The Hamiltonian vector field

$$b_H(x, v) = \begin{pmatrix} v \\ -\nabla U(x) \end{pmatrix}$$

is divergence-free and preserves H :

$$\nabla_{x,v} \cdot b_H = \nabla_x \cdot v + \nabla_v \cdot (-\nabla U(x)) = 0, \quad b_H \cdot \nabla H = v \cdot \nabla U(x) - \nabla U(x) \cdot v = 0.$$

Consequently,

$$\mathcal{L}_H^* e^{-H} = -\nabla_{x,v} \cdot (b_H e^{-H}) = -(\nabla_{x,v} \cdot b_H - b_H \cdot \nabla H) e^{-H} = 0.$$

For the Ornstein–Uhlenbeck part,

$$\mathcal{L}_{OU}^* g = \gamma \nabla_v \cdot (vg + \nabla_v g).$$

Because $\nabla_v e^{-H} = -v e^{-H}$, we also have $\mathcal{L}_{OU}^* e^{-H} = 0$. Thus $\mathcal{L}^* e^{-H} = 0$, so the Boltzmann–Gibbs

measure

$$\mu(\mathrm{d}x\mathrm{d}v) = \frac{1}{Z_U(2\pi)^{\frac{d}{2}}} \exp\left(-U(x) - \frac{1}{2}|v|^2\right) \mathrm{d}x\mathrm{d}v \quad (3.26)$$

is infinitesimally invariant [BE26, (4.69)]. In the Gaussian case considered below, its invariance will follow directly from *Theorem 3.4*. When $\gamma = 0$, both the noise and the friction disappear. The resulting Hamiltonian flow still preserves μ , but it does not generally converge to it.

We assume $\gamma > 0$ from now on, and consider the Gaussian case. Let

$$U(x) = \frac{1}{2}x^\top \mathbf{C}^{-1}x,$$

where $\mathbf{C} \in \mathbb{R}^{d \times d}$ is symmetric and positive definite. The invariant measure is then

$$\mu = \mathcal{N}(0, \mathbf{C}) \otimes \mathcal{N}(0, \mathbf{I}_d), \quad \bar{\mathbf{C}} = \begin{pmatrix} \mathbf{C} & 0 \\ 0 & \mathbf{I}_d \end{pmatrix}.$$

The Langevin equation becomes

$$\mathrm{d} \begin{pmatrix} X_t \\ V_t \end{pmatrix} = - \underbrace{\begin{pmatrix} 0 & -\mathbf{I}_d \\ \mathbf{C}^{-1} & \gamma \mathbf{I}_d \end{pmatrix}}_{:=\mathcal{A}_\gamma} \begin{pmatrix} X_t \\ V_t \end{pmatrix} \mathrm{d}t + \begin{pmatrix} 0 \\ \sqrt{2\gamma} \mathrm{d}B_t \end{pmatrix}.$$

Let

$$\mathbf{D}_\gamma = \begin{pmatrix} 0 & 0 \\ 0 & \gamma \mathbf{I}_d \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & -\mathbf{I}_d \\ \mathbf{I}_d & 0 \end{pmatrix}.$$

Then

$$\mathcal{A}_\gamma = (\mathbf{D}_\gamma + \mathbf{J})\bar{\mathbf{C}}^{-1}, \quad \mathcal{A}_\gamma \bar{\mathbf{C}} + \bar{\mathbf{C}} \mathcal{A}_\gamma^\top = 2\mathbf{D}_\gamma.$$

Although the Gaussian results above were stated with a square diffusion matrix, the same proofs apply when the BM has lower dimension than the state space. *Theorem 3.4* confirms that $\mathcal{N}(0, \bar{\mathbf{C}})$ is invariant. As in *Section 3.2*, choose an orthogonal matrix $\mathbf{Q}$ such that $\mathbf{Q}^\top \mathbf{C} \mathbf{Q} = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$, where $\sigma_i > 0$ for $1 \leq i \leq d$. Define

$$\hat{X}_t := \mathbf{Q}^\top X_t, \quad \hat{V}_t := \mathbf{Q}^\top V_t, \quad \hat{B}_t := \mathbf{Q}^\top B_t.$$

Since $\mathbf{Q}$ is orthogonal, $(\hat{B}_t)$ is again a standard BM. In these coordinates, the dynamics splits into d two-dimensional systems:

$$\begin{aligned} \mathrm{d}\hat{X}_t^i &= \hat{V}_t^i \mathrm{d}t, \\ \mathrm{d}\hat{V}_t^i &= -\frac{1}{\sigma_i^2} \hat{X}_t^i \mathrm{d}t - \gamma \hat{V}_t^i \mathrm{d}t + \sqrt{2\gamma} \mathrm{d}\hat{B}_t^i. \end{aligned}$$

These component processes are independent whenever their initial coordinate pairs are independent. After rearranging the coordinates as $(\hat{X}^1, \hat{V}^1, \dots, \hat{X}^d, \hat{V}^d)$, the drift matrix becomes block diagonal. Since this rearrangement does not change the spectrum,

$$\text{spec}(\mathcal{A}_\gamma) = \bigcup_{i=1}^d \text{spec}(\mathcal{A}_{\gamma,i}), \quad \mathcal{A}_{\gamma,i} = \begin{pmatrix} 0 & -1 \\ \sigma_i^{-2} & \gamma \end{pmatrix}.$$

Theorem 3.13 ([BE26, Theorem 4.35]). *The friction coefficient that maximises the asymptotic Wasserstein contraction rate is*

$$\gamma^* = \frac{2}{\max_{1 \leq i \leq d} \sigma_i} = \frac{2}{\sqrt{\varrho(\mathbf{C})}}.$$

The corresponding optimal rate is

$$\lambda^* = \frac{1}{\max_{1 \leq i \leq d} \sigma_i} = \frac{1}{\sqrt{\varrho(\mathbf{C})}}.$$

Proof. For the i -th component,

$$\det(\lambda \mathbf{I}_2 - \mathcal{A}_{\gamma,i}) = \lambda^2 - \gamma\lambda + \frac{1}{\sigma_i^2},$$

so its eigenvalues are

$$\lambda_{i,\pm} = \frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \frac{1}{\sigma_i^2}}.$$

Its asymptotic rate is therefore

$$r_i(\gamma) = \begin{cases} \frac{\gamma}{2}, & 0 < \gamma \leq \frac{2}{\sigma_i} \\ \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \frac{1}{\sigma_i^2}}, & \gamma > \frac{2}{\sigma_i} \end{cases}.$$

For fixed γ , the rate equals $\frac{\gamma}{2}$ when $\sigma_i \leq \frac{2}{\gamma}$, whereas for $\sigma_i > \frac{2}{\gamma}$,

$$\frac{\partial r_i}{\partial \sigma_i} = -\frac{\sigma_i^{-3}}{\sqrt{\frac{\gamma^2}{4} - \sigma_i^{-2}}} < 0.$$

Hence $r_i(\gamma)$ is non-increasing in σ_i . Thus the minimum is attained at $\sigma_{\max} := \max_{1 \leq i \leq d} \sigma_i$, and the rate of the full dynamics is

$$r(\gamma) = \min_{1 \leq i \leq d} r_i(\gamma) = \begin{cases} \frac{\gamma}{2}, & 0 < \gamma \leq \frac{2}{\sigma_{\max}} \\ \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \frac{1}{\sigma_{\max}^2}}, & \gamma > \frac{2}{\sigma_{\max}} \end{cases}.$$

The first expression is strictly increasing. For $\gamma > \frac{2}{\sigma_{\max}}$

$$r'(\gamma) = \frac{1}{2} - \frac{\gamma}{4\sqrt{\frac{\gamma^2}{4} - \sigma_{\max}^{-2}}} < 0,$$

because

$$\sqrt{\frac{\gamma^2}{4} - \frac{1}{\sigma_{\max}^2}} < \frac{\gamma}{2}.$$

Thus $r(\gamma)$ increases up to $\gamma = \frac{2}{\sigma_{\max}}$ and decreases afterwards. Its unique maximiser is $\gamma^* = \frac{2}{\sigma_{\max}}$, and substitution gives $\lambda^* = r(\gamma^*) = \frac{1}{\sigma_{\max}}$, which proves the result. $\square$

Remark. *At the optimal friction, every block with $\sigma_i = \sigma_{\max}$ is critically damped and has the defective repeated eigenvalue $\lambda^* = \frac{1}{\sigma_{\max}}$. Consequently, although λ^* is the asymptotic logarithmic-slope rate, the corresponding non-trivial Jordan block prevents this rate from being attained in a uniform contraction*

estimate with the same exponent.

We now compare the kinetic dynamics with related systems having one fewer or one additional state variable. Throughout this comparison, $d = 1$, and “one-, two-, and three-level” refer only to the number of state variables. Let

$$U(x) = \frac{k}{2}x^2, \quad k = \frac{1}{\sigma^2}. \quad (3.27)$$

For the two-level kinetic dynamics,

$$\gamma_{kl}^* = 2\sqrt{k}, \quad \lambda_{kl}^* = \sqrt{k} = \frac{1}{\sigma}.$$

Overdamped Langevin Dynamics

The one-level, or *overdamped*, Langevin dynamics is the reversible Ornstein–Uhlenbeck process obtained from [GM16, (1.1)] by taking $U(x) = \frac{k}{2}x^2$:

$$dX_t = -kX_t dt + \sqrt{2}dB_t.$$

Here $\mathbf{A} = k$, $\mathbf{D} = 1$, and the invariant covariance is $\mathbf{C} = k^{-1}$. Indeed,

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = k \cdot k^{-1} + k^{-1} \cdot k = 2 = 2\mathbf{D}.$$

By *Theorem 3.4*, its invariant measure is $\mathcal{N}(0, k^{-1}) = \mathcal{N}(0, \sigma^2)$, while *Theorem 3.5* gives the asymptotic Wasserstein contraction rate

$$\lambda_{\text{od}} = \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda = k = \frac{1}{\sigma^2}.$$

Consequently,

$$\lambda_{kl}^* > \lambda_{\text{od}} \iff \frac{1}{\sigma} > \frac{1}{\sigma^2} \iff \sigma > 1.$$

Thus, under this normalisation, even the optimally tuned kinetic dynamics is faster than the overdamped dynamics only when $\sigma > 1$.

A Three-Level Langevin Lift

We next insert one further relay. Noise first acts on an auxiliary variable Y_t , is then transmitted to V_t , and finally reaches X_t . For $\alpha, \gamma > 0$, set

$$\begin{aligned} dX_t &= V_t dt, \\ dV_t &= -\nabla U(X_t) dt + \alpha Y_t dt, \\ dY_t &= -\alpha V_t dt - \gamma Y_t dt + \sqrt{2\gamma} dB_t. \end{aligned}$$

More precisely, this is the $m = 1$, $\beta = 1$, $\lambda_1 = \alpha$, and $\alpha_1 = \gamma$ instance of the Markovian extension studied in [OP11, (7)]. Define

$$H_3(x, v, y) = U(x) + \frac{1}{2}|v|^2 + \frac{1}{2}|y|^2, \quad b_3(x, v, y) = \begin{pmatrix} v \\ -\nabla U(x) + \alpha y \\ -\alpha v \end{pmatrix}.$$

The vector field b_3 is divergence-free:

$$\nabla_{x,v,y} \cdot b_3 = \nabla_x \cdot v + \nabla_v \cdot (-\nabla U(x) + \alpha y) + \nabla_y \cdot (-\alpha v) = 0.$$

It also preserves H_3 :

$$b_3 \cdot \nabla H_3 = v \cdot \nabla U(x) + (-\nabla U(x) + \alpha y) \cdot v - \alpha v \cdot y = 0.$$

The generator of the three-level process is

$$\mathcal{L}_3 f = b_3 \cdot \nabla f + \gamma(\Delta_y f - y \cdot \nabla_y f).$$

Let

$$\rho_3(x, v, y) := \frac{1}{2\pi Z_U} \exp(-H_3(x, v, y)).$$

Its formal adjoint with respect to Lebesgue measure is

$$\mathcal{L}_3^* \rho_3 = -\nabla_{x,v,y} \cdot (b_3 \rho_3) + \gamma \nabla_y \cdot (y \rho_3 + \nabla_y \rho_3).$$

For the deterministic part, the product rule gives

$$-\nabla \cdot (b_3 \rho_3) = -((\nabla \cdot b_3) \rho_3 + b_3 \cdot \nabla \rho_3) = -(\nabla \cdot b_3 - b_3 \cdot \nabla H_3) \rho_3 = 0.$$

Moreover, $\nabla_y \rho_3 = -y \rho_3$, and hence $\gamma \nabla_y \cdot (y \rho_3 + \nabla_y \rho_3) = 0$. Consequently, $\mathcal{L}_3^* \rho_3 = 0$, so

$$\mu_3(dx dv dy) = \rho_3(x, v, y) dx dv dy = \frac{1}{2\pi Z_U} \exp\left(-U(x) - \frac{1}{2}|v|^2 - \frac{1}{2}|y|^2\right) dx dv dy$$

is infinitesimally invariant [OP11, (10)].

For the one-dimensional Gaussian in (3.27), set

$$\bar{\mathbf{C}}_3 = \text{diag}(k^{-1}, 1, 1), \quad \mathbf{D}_3 = \text{diag}(0, 0, \gamma), \quad \mathbf{J}_\alpha = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\alpha \\ 0 & \alpha & 0 \end{pmatrix},$$

where $\mathbf{J}_\alpha$ is antisymmetric. Writing $\mathbf{Z}_t = (X_t, V_t, Y_t)^\top$, the dynamics becomes

$$d\mathbf{Z}_t = -\mathcal{M}_{\alpha,\gamma} \mathbf{Z}_t dt + \sqrt{2\gamma} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} dB_t,$$

where

$$\mathcal{M}_{\alpha,\gamma} = (\mathbf{D}_3 + \mathbf{J}_\alpha) \bar{\mathbf{C}}_3^{-1} = \begin{pmatrix} 0 & -1 & 0 \\ k & 0 & -\alpha \\ 0 & \alpha & \gamma \end{pmatrix}.$$

Because

$$\mathcal{M}_{\alpha,\gamma} \bar{\mathbf{C}}_3 + \bar{\mathbf{C}}_3 \mathcal{M}_{\alpha,\gamma}^\top = 2\mathbf{D}_3,$$

Theorem 3.4 shows that

$$\mathcal{N}(0, \bar{\mathbf{C}}_3) = \mathcal{N}(0, k^{-1}) \otimes \mathcal{N}(0, 1) \otimes \mathcal{N}(0, 1)$$

is invariant. Moreover, *Theorem 3.5* identifies the spectral quantity governing the asymptotic logarithmic slope as

$$r(\alpha, \gamma) := \min_{\lambda \in \text{spec}(\mathcal{M}_{\alpha, \gamma})} \text{Re } \lambda.$$

The characteristic polynomial is

$$p_{\alpha, \gamma}(\lambda) = \det(\lambda \mathbf{I}_3 - \mathcal{M}_{\alpha, \gamma}) = \det \begin{pmatrix} \lambda & 1 & 0 \\ -k & \lambda & \alpha \\ 0 & -\alpha & \lambda - \gamma \end{pmatrix} = \lambda^3 - \gamma \lambda^2 + (k + \alpha^2) \lambda - k \gamma.$$

For $k = 1$ and three real eigenvalues, the coefficient comparison used below also appears in [SB25, Lemma III.1]. We extend it to arbitrary $k > 0$ and include the case of a complex-conjugate pair.

Proposition 3.14. *Within this three-level family,*

$$\max_{\alpha, \gamma > 0} \min_{\lambda \in \text{spec}(\mathcal{M}_{\alpha, \gamma})} \text{Re } \lambda = \sqrt{3k}.$$

The maximum is uniquely attained at

$$\alpha^* = \sqrt{8k}, \quad \gamma^* = 3\sqrt{3k}.$$

Proof. Let $\lambda_1, \lambda_2, \lambda_3$ be the roots of $p_{\alpha, \gamma}$, counted with algebraic multiplicity, and write $r := \min_{1 \leq j \leq 3} \text{Re } \lambda_j$.

If $r \leq 0$, then $r \leq \sqrt{3k}$ is immediate. We may therefore assume that $r > 0$. Suppose first that all three roots are real. Since $r > 0$, we have $\lambda_j \geq r$ for every j . Comparing coefficients in

$$p_{\alpha, \gamma}(\lambda) = \prod_{j=1}^3 (\lambda - \lambda_j)$$

gives

$$\lambda_1 + \lambda_2 + \lambda_3 = \gamma, \quad \lambda_1 \lambda_2 \lambda_3 = k \gamma.$$

It follows that

$$\lambda_1 \lambda_2 \lambda_3 = k(\lambda_1 + \lambda_2 + \lambda_3).$$

Dividing by $k \lambda_1 \lambda_2 \lambda_3 > 0$, we obtain

$$\frac{1}{k} = \frac{1}{\lambda_1 \lambda_2} + \frac{1}{\lambda_1 \lambda_3} + \frac{1}{\lambda_2 \lambda_3} \leq \frac{3}{r^2}.$$

Therefore, $r \leq \sqrt{3k}$.

Now suppose that the roots are a , $b + ic$, and $b - ic$. Since $r > 0$, we have $a, b \geq r > 0$. The trace and determinant identities become

$$a + 2b = \gamma, \quad a(b^2 + c^2) = k\gamma = k(a + 2b).$$

Dividing the second identity by $ab^2 > 0$ gives

$$1 + \frac{c^2}{b^2} = \frac{k}{b^2} + \frac{2k}{ab}.$$

Since $a, b \geq r$,

$$1 \leq 1 + \frac{c^2}{b^2} \leq \frac{k}{r^2} + \frac{2k}{r^2} = \frac{3k}{r^2}.$$

Hence $r \leq \sqrt{3k}$ in this case as well.

We now determine when equality can occur. Let $s = \sqrt{3k}$ and suppose that $r = s$. In the real-root case, equality in

$$\frac{1}{k} = \frac{1}{\lambda_1 \lambda_2} + \frac{1}{\lambda_1 \lambda_3} + \frac{1}{\lambda_2 \lambda_3} \leq \frac{3}{r^2}$$

requires

$$\lambda_1 \lambda_2 = \lambda_1 \lambda_3 = \lambda_2 \lambda_3 = r^2.$$

Since all three roots are positive, this implies $\lambda_1 = \lambda_2 = \lambda_3 = r = s$. In the complex-conjugate case, $r = s$ gives

$$1 \leq 1 + \frac{c^2}{b^2} \leq \frac{3k}{r^2} = 1.$$

Hence $c = 0$. Equality in the upper bound also forces $b = r$ and $ab = r^2$, and therefore $a = b = r = s$. Thus, in either case, equality can occur only when all three roots are equal to s . To attain equality, we choose the parameters such that

$$p_{\alpha, \gamma}(\lambda) = (\lambda - s)^3 = \lambda^3 - 3s\lambda^2 + 3s^2\lambda - s^3.$$

Comparing coefficients with

$$p_{\alpha, \gamma}(\lambda) = \lambda^3 - \gamma\lambda^2 + (k + \alpha^2)\lambda - k\gamma$$

gives

$$\gamma = 3s = 3\sqrt{3k}, \quad k + \alpha^2 = 3s^2 = 9k, \quad k\gamma = 3ks = s^3.$$

Thus $\alpha^2 = 8k$. Because $\alpha > 0$, the only possible choice is $\alpha^* = \sqrt{8k}$. For these parameters,

$$p_{\alpha^*, \gamma^*}(\lambda) = \left(\lambda - \sqrt{3k}\right)^3.$$

Therefore the maximal asymptotic spectral rate is $\lambda_{3L}^* = \sqrt{3k}$. □

Remark. *At the optimal parameters, all three eigenvalues coalesce at $s = \sqrt{3k}$. Since $\mathcal{M}_{\alpha^*, \gamma^*} \neq s\mathbf{I}_3$, this repeated eigenvalue is defective. As in the critically damped kinetic case above, s is the asymptotic logarithmic-slope rate, but it cannot be attained in a uniform contraction estimate with the same exponent.*

Under the normalisation (3.27), the three asymptotic spectral rates are

$$\lambda_{\text{od}} = \frac{1}{\sigma^2}, \quad \lambda_{\text{kl}}^* = \frac{1}{\sigma}, \quad \lambda_{3L}^* = \frac{\sqrt{3}}{\sigma}.$$

Thus, under these parameter choices, the optimal three-level spectral slope is $\sqrt{3}$ times the optimal

kinetic spectral slope: $\lambda_{3L}^* = \sqrt{3}\lambda_{kl}^*$. Its comparison with overdamped Langevin depends on the variance:

$$\lambda_{3L}^* > \lambda_{od} \iff \frac{\sqrt{3}}{\sigma} > \frac{1}{\sigma^2} \iff \sigma > \frac{1}{\sqrt{3}}.$$

Consequently,

$$\begin{cases} \lambda_{od} > \lambda_{3L}^* > \lambda_{kl}^*, & 0 < \sigma < \frac{1}{\sqrt{3}} \\ \lambda_{od} = \lambda_{3L}^* > \lambda_{kl}^*, & \sigma = \frac{1}{\sqrt{3}} \\ \lambda_{3L}^* > \max\{\lambda_{od}, \lambda_{kl}^*\}, & \sigma > \frac{1}{\sqrt{3}} \end{cases}.$$

However, the three systems have different state-space dimensions and do not use the same noise budget. If $\eta = \text{tr } \mathbf{D}$, then, for the parameter choices considered above,

$$\eta_{od} = 1, \quad \eta_{kl} = 2\sqrt{k}, \quad \eta_{3L} = 3\sqrt{3k}.$$

Therefore, the comparison above concerns only the large-time spectral rates of the continuous-time models under the chosen normalisation. It does not determine whether one discretised sampling algorithm is uniformly more efficient than the others.

Chapter 4

Beyond the Gaussian Case

The preceding chapter exploited the constant Hessian of a quadratic potential to reduce synchronous coupling to a linear equation with a fixed drift matrix. We now retain the kinetic Langevin dynamics but allow the potential to be non-quadratic. The noise still cancels under synchronous coupling, but the resulting difference equation has time-dependent coefficients. Under uniform curvature bounds, however, a single twisted norm controls all these coefficients simultaneously.

The argument makes explicit the extension outlined in [BE26, page 133]. It uses the twisted-metric calculation from [DR20, Theorem 1 and Proposition 1], adapted here to the full position–velocity process and to every Wasserstein order $q \in [1, \infty)$.

Fix $\gamma > 0$, and consider the kinetic Langevin equation (3.24)–(3.25). Suppose that $U \in C^2(\mathbb{R}^d)$ and that there are constants $0 < K \leq L$ such that

$$K\mathbf{I}_d \preceq \nabla^2 U(x) \preceq L\mathbf{I}_d, \quad x \in \mathbb{R}^d.$$

Equivalently,

$$K\|z\|^2 \leq z^\top \nabla^2 U(x) z \leq L\|z\|^2, \quad x, z \in \mathbb{R}^d.$$

The *fundamental theorem of calculus* gives

$$\nabla U(x) - \nabla U(y) = \int_0^1 \nabla^2 U(y + s(x - y))(x - y) ds.$$

Since $\|\nabla^2 U(x)\| \leq L$,

$$\|\nabla U(x) - \nabla U(y)\| \leq \int_0^1 \|\nabla^2 U(y + s(x - y))\| \|x - y\| ds \leq L\|x - y\|.$$

Thus ∇U is globally L -Lipschitz. Consequently, the drift

$$b(x, v) = \begin{pmatrix} v \\ -\nabla U(x) - \gamma v \end{pmatrix},$$

is globally Lipschitz on $\mathbb{R}^{2d}$, and therefore satisfies a linear-growth bound. Since the diffusion coefficient is constant, applying *Theorem 2.16* on each finite time interval yields a unique global strong solution for every square-integrable initial condition that is independent of the driving BM. *Taylor's theorem*

with integral remainder gives

$$U(x) - U(y) - \langle \nabla U(y), x - y \rangle = \int_0^1 (1-s)(x-y)^\top \nabla^2 U(y + s(x-y))(x-y) ds \geq \frac{K}{2} \|x - y\|^2.$$

Thus U is K -strongly convex. Taking $y = 0$ and completing the square,

$$U(x) \geq U(0) + \langle \nabla U(0), x \rangle + \frac{K}{2} \|x\|^2 = U(0) - \frac{\|\nabla U(0)\|^2}{2K} + \frac{K}{2} \left\| x + \frac{\nabla U(0)}{K} \right\|^2.$$

It follows that

$$\begin{aligned} Z_U &= \int_{\mathbb{R}^d} e^{-U(x)} dx \\ &\leq \exp \left(-U(0) + \frac{\|\nabla U(0)\|^2}{2K} \right) \int_{\mathbb{R}^d} \exp \left(-\frac{K}{2} \left\| x + \frac{\nabla U(0)}{K} \right\|^2 \right) dx \\ &= \exp \left(-U(0) + \frac{\|\nabla U(0)\|^2}{2K} \right) \left(\frac{2\pi}{K} \right)^{\frac{d}{2}} < \infty. \end{aligned}$$

The calculation in *Section 3.3* shows that the Boltzmann–Gibbs measure (3.26) is infinitesimally invariant, while the estimate above shows that it is well-defined. Standard results for Langevin diffusions with globally Lipschitz gradients show that the solution family defines a time-homogeneous Markov process and that μ is invariant for its transition semigroup; see [EGZ19, Section 1]. We denote its transition kernel by p_t .

Consider two solutions $(X_t, V_t)_{t \geq 0}$ and $(\tilde{X}_t, \tilde{V}_t)_{t \geq 0}$ driven by the same BM:

$$\begin{aligned} dX_t &= V_t dt, & d\tilde{X}_t &= \tilde{V}_t dt, \\ dV_t &= -\nabla U(X_t) dt - \gamma V_t dt + \sqrt{2\gamma} dB_t, & d\tilde{V}_t &= -\nabla U(\tilde{X}_t) dt - \gamma \tilde{V}_t dt + \sqrt{2\gamma} dB_t, \end{aligned}$$

and set

$$Z_t := X_t - \tilde{X}_t, \quad W_t := V_t - \tilde{V}_t.$$

Their stochastic terms cancel, so

$$\frac{d}{dt} \begin{pmatrix} Z_t \\ W_t \end{pmatrix} = \begin{pmatrix} V_t - \tilde{V}_t \\ -\nabla U(X_t) + \nabla U(\tilde{X}_t) - \gamma(V_t - \tilde{V}_t) \end{pmatrix} = \begin{pmatrix} W_t \\ -\nabla U(X_t) + \nabla U(\tilde{X}_t) - \gamma W_t \end{pmatrix}.$$

Applying the same integral identity gives

$$\nabla U(X_t) - \nabla U(\tilde{X}_t) = \mathbf{H}_t (X_t - \tilde{X}_t) = \mathbf{H}_t Z_t,$$

where the symmetric matrix $\mathbf{H}_t$ is defined by

$$\mathbf{H}_t := \int_0^1 \nabla^2 U(\tilde{X}_t + s(X_t - \tilde{X}_t)) ds.$$

Moreover, for every $z \in \mathbb{R}^d$,

$$z^\top \mathbf{H}_t z = \int_0^1 z^\top \nabla^2 U(\tilde{X}_t + s(X_t - \tilde{X}_t)) z ds,$$

and hence

$$K\|z\|^2 \leq z^\top \mathbf{H}_t z \leq L\|z\|^2.$$

Thus $K\mathbf{I}_d \preceq \mathbf{H}_t \preceq L\mathbf{I}_d$. The difference process satisfies the ODE:

$$\frac{d}{dt} \begin{pmatrix} Z_t \\ W_t \end{pmatrix} = \begin{pmatrix} W_t \\ -\mathbf{H}_t Z_t - \gamma W_t \end{pmatrix} = - \underbrace{\begin{pmatrix} 0 & -\mathbf{I}_d \\ \mathbf{H}_t & \gamma \mathbf{I}_d \end{pmatrix}}_{:= \mathbf{A}_t} \begin{pmatrix} Z_t \\ W_t \end{pmatrix}. \quad (4.1)$$

Unlike the Gaussian case, $\mathbf{A}_t$ is not constant, so its spectrum at a single time does not determine the contraction rate. We instead seek one quadratic form that contracts for every symmetric matrix whose eigenvalues lie in $[K, L]$.

From now on, suppose that $\gamma > \sqrt{L}$. Define

$$\mathbf{G}_\gamma := \begin{pmatrix} \gamma^2 \mathbf{I}_d & \gamma \mathbf{I}_d \\ \gamma \mathbf{I}_d & 2\mathbf{I}_d \end{pmatrix}.$$

For $\xi = (z, w)^\top \in \mathbb{R}^{2d}$,

$$\|\xi\|_{\mathbf{G}_\gamma}^2 := \xi^\top \mathbf{G}_\gamma \xi = \|\gamma z + w\|^2 + \|w\|^2.$$

The matrix $\mathbf{G}_\gamma$ is symmetric, and the displayed quadratic form vanishes only when $w = 0$ and $z = 0$. Hence, $\mathbf{G}_\gamma$ is positive definite. Set

$$a := \min \{K, \gamma^2 - L\}, \quad c(\gamma) := \frac{a}{\gamma}. \quad (4.2)$$

Because $\gamma > \sqrt{L}$, we have $a > 0$.

Theorem 4.1 ([BE26, page 133] and [DR20, Theorem 1 and Proposition 1]). *For every $q \in [1, \infty)$, p_t maps $\mathcal{P}_q(\mathbb{R}^{2d})$ into itself and*

$$\alpha_q(p_t) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t}, \quad t \geq 0.$$

Consequently, for every $\nu \in \mathcal{P}_q(\mathbb{R}^{2d})$,

$$\mathcal{W}_q(\nu p_t, \mu) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} \mathcal{W}_q(\nu, \mu).$$

Among all $\gamma > \sqrt{L}$, the rate bound $c(\gamma)$ is maximised at

$$\gamma^* = \sqrt{K + L}, \quad c(\gamma^*) = \frac{K}{\sqrt{K + L}}.$$

Proof. Direct multiplication gives

$$\mathbf{A}_t^\top \mathbf{G}_\gamma + \mathbf{G}_\gamma \mathbf{A}_t - 2c(\gamma) \mathbf{G}_\gamma = 2 \begin{pmatrix} \gamma(\mathbf{H}_t - a\mathbf{I}_d) & \mathbf{H}_t - a\mathbf{I}_d \\ \mathbf{H}_t - a\mathbf{I}_d & \frac{\gamma^2 - 2a}{\gamma} \mathbf{I}_d \end{pmatrix}.$$

For each fixed t , choose an orthogonal matrix $\mathbf{Q}$ such that $\mathbf{Q}^\top \mathbf{H}_t \mathbf{Q}$ is diagonal. Conjugating by $\text{diag}(\mathbf{Q}, \mathbf{Q})$, and then grouping the corresponding coordinates, reduces the matrix above to d symmetric 2×2 blocks. The block associated with an eigenvalue $h \in [K, L]$ is

$$\mathbf{B}(h) = \begin{pmatrix} \gamma(h-a) & h-a \\ h-a & \frac{\gamma^2-2a}{\gamma} \end{pmatrix}.$$

Because $a \leq K \leq h$, $\gamma(h-a) \geq 0$. Furthermore, $a \leq K \leq L$ and $a \leq \gamma^2 - L$, so $2a \leq L + (\gamma^2 - L) = \gamma^2$. Therefore, $\frac{\gamma^2-2a}{\gamma} \geq 0$. Finally,

$$\det \mathbf{B}(h) = \gamma(h-a) \frac{\gamma^2-2a}{\gamma} - (h-a)^2 = (h-a)(\gamma^2 - a - h).$$

Since $h \leq L$ and $a \leq \gamma^2 - L$, $\gamma^2 - a - h \geq \gamma^2 - a - L \geq 0$. Thus $\det \mathbf{B}(h) \geq 0$, so every block $\mathbf{B}(h)$ is positive semidefinite. It follows that

$$\mathbf{A}_t^\top \mathbf{G}_\gamma + \mathbf{G}_\gamma \mathbf{A}_t \succeq 2c(\gamma) \mathbf{G}_\gamma.$$

Let $\mathbf{Y}_t := (Z_t, W_t)^\top$. By (4.1), $\dot{\mathbf{Y}}_t = -\mathbf{A}_t \mathbf{Y}_t$. Then

$$\frac{d}{dt} \|\mathbf{Y}_t\|_{\mathbf{G}_\gamma}^2 = \frac{d}{dt} (\mathbf{Y}_t^\top \mathbf{G}_\gamma \mathbf{Y}_t) = -\mathbf{Y}_t^\top (\mathbf{A}_t^\top \mathbf{G}_\gamma + \mathbf{G}_\gamma \mathbf{A}_t) \mathbf{Y}_t \leq -2c(\gamma) \|\mathbf{Y}_t\|_{\mathbf{G}_\gamma}^2.$$

Equivalently,

$$\frac{d}{dt} \left(e^{2c(\gamma)t} \|\mathbf{Y}_t\|_{\mathbf{G}_\gamma}^2 \right) \leq 0.$$

Hence,

$$\|\mathbf{Y}_t\|_{\mathbf{G}_\gamma} \leq e^{-c(\gamma)t} \|\mathbf{Y}_0\|_{\mathbf{G}_\gamma}.$$

Using the norm equivalence established in *Chapter 2*, we obtain the pathwise estimate

$$\|\mathbf{Y}_t\| \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} \|\mathbf{Y}_0\|. \quad (4.3)$$

The quadratic lower bound on U , together with the Gaussian velocity factor, implies that $\mu \in \mathcal{P}_r(\mathbb{R}^{2d})$ for every $r \in [1, \infty)$. In particular, $\mu \in \mathcal{P}_2(\mathbb{R}^{2d})$. Fix $q \in [1, \infty)$, and set

$$m_q(\mu) := \left(\int_{\mathbb{R}^{2d}} \|z\|^q \mu(dz) \right)^{\frac{1}{q}} < \infty.$$

For a deterministic initial point y , synchronously couple the process started from y with a second process whose initial distribution is μ . By invariance, the second process has distribution μ at every time. Hence, by (4.3) and *Minkowski's inequality*,

$$m_q(p_t(y, \cdot)) \leq m_q(\mu) + \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} (\|y\| + m_q(\mu)).$$

Therefore, for every $\nu \in \mathcal{P}_q(\mathbb{R}^{2d})$, integrating over $y \sim \nu$ and applying *Minkowski's inequality* once

more gives

$$\begin{aligned}
m_q(\nu p_t) &= \left(\int_{\mathbb{R}^{2d}} m_q(p_t(y, \cdot))^q \nu(dy) \right)^{\frac{1}{q}} \\
&\leq \left(\int_{\mathbb{R}^{2d}} \left[m_q(\mu) + \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} (\|y\| + m_q(\mu)) \right]^q \nu(dy) \right)^{\frac{1}{q}} \\
&\leq m_q(\mu) + \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} (m_q(\nu) + m_q(\mu)) < \infty.
\end{aligned}$$

Thus p_t maps $\mathcal{P}_q(\mathbb{R}^{2d})$ into itself. Now start the two synchronously coupled processes from deterministic points y and $\tilde{y}$. Their joint law at time t is a coupling of $p_t(y, \cdot)$ and $p_t(\tilde{y}, \cdot)$. Hence, for every $q \in [1, \infty)$,

$$\mathcal{W}_q(p_t(y, \cdot), p_t(\tilde{y}, \cdot)) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} \|y - \tilde{y}\|.$$

The pointwise characterisation from *Theorem 2.29* therefore yields

$$\alpha_q(p_t) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t}, \quad q \in [1, \infty), \quad t \geq 0.$$

Since μ is invariant, it follows that, for every $\nu \in \mathcal{P}_q(\mathbb{R}^{2d})$,

$$\mathcal{W}_q(\nu p_t, \mu) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t} \mathcal{W}_q(\nu, \mu).$$

By (4.2),

$$c(\gamma) = \begin{cases} \gamma - \frac{L}{\gamma}, & \sqrt{L} < \gamma \leq \sqrt{K+L} \\ \frac{K}{\gamma}, & \gamma > \sqrt{K+L} \end{cases}.$$

The first branch is strictly increasing because

$$c'(\gamma) = 1 + \frac{L}{\gamma^2} > 0, \quad \sqrt{L} < \gamma \leq \sqrt{K+L},$$

whereas the second branch is strictly decreasing because

$$c'(\gamma) = -\frac{K}{\gamma^2} < 0, \quad \gamma > \sqrt{K+L}.$$

Since the two branches agree at $\sqrt{K+L}$, the rate bound is maximised at $\gamma^* = \sqrt{K+L}$. For this choice,

$$c^* = c(\gamma^*) = \frac{K}{\sqrt{K+L}} = \frac{\sqrt{K}}{\sqrt{1 + \frac{L}{K}}}.$$

□

Remark. *The uniform lower and upper Hessian bounds are essential to the argument above. Without a suitable confining condition, an invariant probability measure need not exist. Beyond strong convexity, more elaborate couplings and distance functions are generally required; see [EGZ19]. We do not pursue this direction here.*

Chapter 5

A Gaussian Illustration

After the non-Gaussian extension of *Chapter 4*, we return to the Gaussian setting of *Chapter 3* and illustrate the effect of non-reversibility in a concrete two-dimensional example. We compare the optimal reversible diffusion with an optimal non-reversible diffusion under the same unit noise budget. The construction is the case $u = 9$ and $v = 1$ of the two-dimensional optimisation; see *Lemma 3.6* and *Theorem 3.7*. Let

$$\mu = \mathcal{N}(0, \mathbf{C}), \quad \mathbf{C} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}.$$

The level sets of the density of μ are the ellipses

$$x^\top \mathbf{C}^{-1} x = c \iff \frac{x_1^2}{9c} + \frac{x_2^2}{c} = 1, \quad c > 0.$$

Thus their principal semiaxes have lengths $3\sqrt{c}$ and $\sqrt{c}$. In particular, $\kappa(\mathbf{C}) = 9$, so the target is sufficiently anisotropic for the acceleration to be clearly visible. We impose the total noise constraint

$$\text{tr}(\mathbf{D}) \leq 1, \quad \mathbf{D} = \sigma \sigma^\top.$$

For the reversible dynamics, the calculation following *Theorem 3.7* yields

$$\mathbf{D}_{\text{rev}} = \begin{pmatrix} \frac{9}{10} & 0 \\ 0 & \frac{1}{10} \end{pmatrix}, \quad \mathbf{J}_{\text{rev}} = 0.$$

Consequently,

$$\mathbf{A}_{\text{rev}} = \mathbf{D}_{\text{rev}} \mathbf{C}^{-1} = \begin{pmatrix} \frac{9}{10} & 0 \\ 0 & \frac{1}{10} \end{pmatrix} \begin{pmatrix} \frac{1}{9} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{10} & 0 \\ 0 & \frac{1}{10} \end{pmatrix} = \frac{1}{10} \mathbf{I}_2.$$

It follows that

$$\text{spec}(\mathbf{A}_{\text{rev}}) = \left\{ \frac{1}{10} \right\}, \quad \lambda_{\text{rev}}^* = \frac{1}{10},$$

where the eigenvalue has algebraic multiplicity two. The mean equation gives

$$m_t^{\text{rev}} = e^{-t\mathbf{A}_{\text{rev}}} m_0 = e^{-\frac{t}{10}} m_0.$$

Every mean trajectory therefore approaches the origin along a straight line.

For the non-reversible comparison, we instead place the entire noise budget in the second coordinate

and couple the two directions through an antisymmetric matrix:

$$\mathbf{D}_{\text{nr}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{J}_{\text{nr}} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}.$$

The optimality condition from *Theorem 3.7* is

$$|\beta| \geq \frac{1}{2} \sqrt{\frac{u}{v}} = \frac{3}{2}.$$

Thus, $\beta = 2$ attains the optimal rate. Its drift matrix is

$$\mathbf{A}_{\text{nr}} = (\mathbf{D}_{\text{nr}} + \mathbf{J}_{\text{nr}}) \mathbf{C}^{-1} = \begin{pmatrix} 0 & 2 \\ -\frac{2}{9} & 1 \end{pmatrix}.$$

Both constructions preserve the same target measure. Indeed, in either case,

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = (\mathbf{D} + \mathbf{J}) + (\mathbf{D} - \mathbf{J}) = 2\mathbf{D}.$$

It follows from *Theorem 3.4* that μ is invariant for both processes. Moreover, $\text{tr}(\mathbf{D}_{\text{rev}}) = \text{tr}(\mathbf{D}_{\text{nr}}) = 1$, so the two dynamics use the same total noise budget. For the non-reversible matrix,

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}_{\text{nr}}) = \lambda^2 - \lambda + \frac{4}{9}.$$

Its eigenvalues are

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{\sqrt{7}}{6}i.$$

Therefore,

$$\lambda_{\text{nr}}^* = \frac{1}{2}, \quad \frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} = 5.$$

Thus the optimal non-reversible asymptotic rate is five times the optimal reversible rate under the same noise constraint.

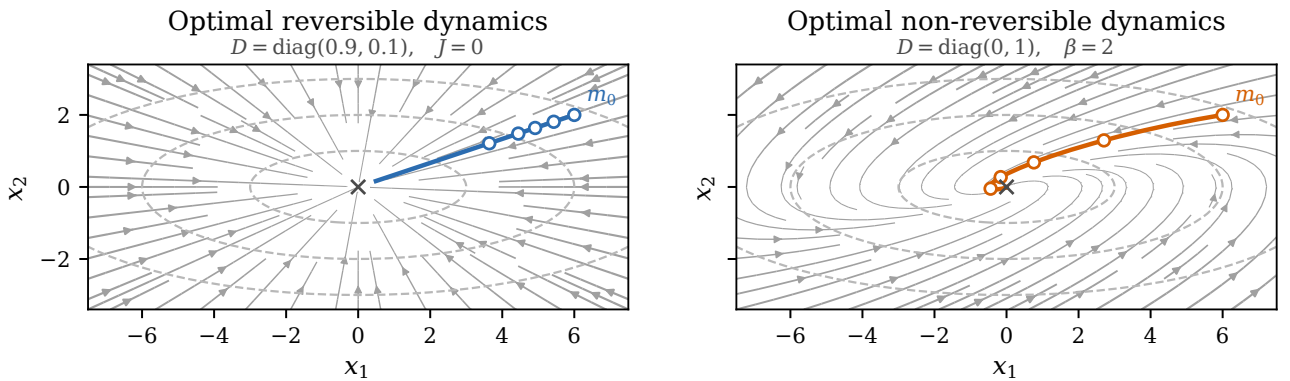


Figure 5.1: Drift geometry of the reversible and non-reversible Gaussian diffusions. The grey curves are trajectories of the deterministic equation $\dot{x} = -\mathbf{A}x$ from different initial points, with arrows indicating increasing time. The dashed ellipses are level sets of the invariant density. The coloured curves show the mean trajectory $m_t = e^{-t\mathbf{A}}m_0$ from $m_0 = (6, 2)^\top$, and the open circles mark the times $t = 0, 1, 2, 3, 5$.

To connect the plotted mean trajectories directly to convergence in Wasserstein distance, set

$$\nu_0 := \mathcal{N}(m_0, \mathbf{C}), \quad m_0 = (6, 2)^\top.$$

Let $X_0 \sim \nu_0$ be independent of the driving BM, and denote the distribution of X_t by ν_t . Since the equation is linear, the explicit solution (3.2) shows that X_t is Gaussian. Its mean is $m_t = e^{-t\mathbf{A}}m_0$. Moreover, the identity $\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = 2\mathbf{D}$ shows that $\Sigma_t \equiv \mathbf{C}$ solves the covariance equation with initial condition $\Sigma_0 = \mathbf{C}$. By uniqueness of solutions,

$$\nu_t = \mathcal{N}(e^{-t\mathbf{A}}m_0, \mathbf{C}).$$

Since ν_t and μ have the same covariance, the translation argument used in *Theorem 3.5* yields the exact identity

$$\mathcal{W}_2(\nu_t, \mu) = \|e^{-t\mathbf{A}}m_0\|.$$

Thus, the Euclidean norm of the plotted mean at time t is exactly the Wasserstein distance from ν_t to equilibrium. The coloured curves are deterministic mean trajectories, not simulated sample paths.

In the reversible system, the drift points directly towards the origin, and the mean approaches it slowly along a straight line. In the non-reversible system, the antisymmetric coupling continuously transfers displacement between the two coordinates, producing a damped spiral. Although $\|m_t^{\text{nr}}\|$ need not decrease monotonically, its asymptotic envelope decays at rate $\frac{1}{2}$, rather than $\frac{1}{10}$. For example, at $t = 3$,

$$\|m_3^{\text{rev}}\| \approx 4.69, \quad \|m_3^{\text{nr}}\| \approx 0.32.$$

The figure therefore makes the spectral acceleration visible while keeping both the target distribution and the total noise budget unchanged.

Chapter 6

Conclusion

This thesis studied how the drift and diffusion coefficients can be arranged to accelerate convergence to a prescribed Gaussian equilibrium. For a linear Ornstein–Uhlenbeck process, invariance of $\mathcal{N}(0, \mathbf{C})$ is equivalent to writing the drift matrix as $\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}$, where $\mathbf{D} = \sigma\sigma^\top$ is positive semidefinite and $\mathbf{J}$ is antisymmetric. This decomposition separates the directions in which noise is injected from the non-reversible coupling that transports its effect through the system.

Synchronous coupling shows that the L^2 -Wasserstein contraction coefficient of the Gaussian transition semigroup is exactly $\|e^{-t\mathbf{A}}\|$. The asymptotic convergence rate is therefore determined by the smallest real part of the eigenvalues of $\mathbf{A}$. Under the noise constraint $\text{tr}(\mathbf{D}) \leq \eta$, the optimal rate in dimension d is $\frac{\eta}{d}\lambda_{\max}(\mathbf{C}^{-1})$. It can be attained using a rank-one noise covariance matrix and a suitable antisymmetric drift. In dimension two, under the same total noise constraint, the acceleration factor relative to the optimal reversible dynamics is $\frac{1+\kappa(\mathbf{C})}{2}$, so non-reversibility is especially effective for strongly anisotropic targets.

Langevin dynamics illustrates how degenerate noise can nevertheless affect the entire system through deterministic coupling. For a Gaussian target, the optimal kinetic friction is $\gamma^* = \frac{2}{\sqrt{\varrho(\mathbf{C})}}$, with rate $\frac{1}{\sqrt{\varrho(\mathbf{C})}}$. In the one-dimensional quadratic comparison, the optimal three-level rate is $\sqrt{3}$ times the optimal kinetic rate under the chosen normalisation, but the noise budgets differ. Beyond the Gaussian case, synchronous coupling in a twisted metric yields exponential contraction when $K\mathbf{I}_d \preceq \nabla^2 U \preceq L\mathbf{I}_d$. The resulting contraction-rate bound is maximised at $\gamma = \sqrt{K+L}$, giving the rate $\frac{K}{\sqrt{K+L}}$.

These results show that convergence depends not only on how much noise is used, but also on where it is injected and how the drift transports its effect. They also expose an important limitation: the optimal spectral rate is an asymptotic continuous-time quantity. When an eigenvalue whose real part equals $\lambda^*(\mathbf{A})$ is defective, the asymptotic rate need not be attained by a uniform contraction estimate. Moreover, a large non-reversible drift may make the dynamics numerically stiff and require smaller discretisation steps.

Future work could determine whether the asymptotic continuous-time acceleration found here yields a finite-time advantage after stable discretisation and under a fair computational budget. A further direction is to extend the analysis beyond uniformly strongly convex targets.

Popular Summary

Suppose we ask a computer to generate a collection of points that form a bell-shaped cloud. The computer, being a rather literal creature, may reasonably ask: *How?*

One answer is to release a restless particle and let it wander randomly, recording its position at regular intervals. If the motion is designed correctly, then after an initial waiting period the recorded positions will occupy the cloud in approximately the right proportions. This long-run pattern is described by the process's *equilibrium distribution*. In this thesis, the desired distribution is usually Gaussian.

Reaching the correct equilibrium, however, is not the whole story. A particle may retain a noticeable memory of its starting point for a long time. For sampling, we would like that memory to *disappear* quickly. The question is therefore not merely “*Does the particle reach the correct equilibrium?*” but rather “*How quickly does it forget where it came from?*”

Many motions can have the same equilibrium. One particle may move fairly directly towards the centre; another may follow curved or spiralling paths. Some motions are *reversible*: once equilibrium has been reached, a film of the particle played backwards would look statistically ordinary. Others contain a preferred circulation and are called *non-reversible*. This circulation leaves the final cloud unchanged, but it can alter how quickly the particle explores it.

To measure this loss of memory, start two particles from different points and give them the same random pushes. The shared randomness then cancels when we subtract their positions. If the distance between them shrinks rapidly, the process rapidly forgets its starting point. In the Gaussian case, linear algebra determines exactly how fast this forgetting occurs over long times.

The most revealing example is a Gaussian density whose level curves are long, thin ellipses. When the two coordinate directions evolve independently, one direction may approach equilibrium much more slowly than the other. A non-reversible motion transfers movement between the two directions, preventing either from becoming a bottleneck.

The same idea appears in *Langevin dynamics*, where a particle has both position and velocity. Random pushes act on velocity, which then moves position. We determine the friction giving the fastest Gaussian convergence and study a three-level model in which the noise passes through an additional variable before reaching position.

There is, of course, no free lunch. A faster mathematical rate does not automatically produce a faster computer program. Strong circulation may require smaller simulation steps, while additional variables require additional work. Nevertheless, the Gaussian case reveals an important principle: faster convergence does not always require more noise. It can also be achieved by arranging the motion so that its different directions help one another and no single direction holds back the entire system.

Bibliography

- [AAS15] Franz Achleitner, Anton Arnold, and Dominik Stürzer. “Large-Time Behavior in Non-Symmetric Fokker–Planck Equations”. In: *Rivista di Matematica della Università di Parma* 6.1 (2015), pages 1–68. <https://www.rivmat.unipr.it/vols/2015-6-1/01.html> (cited on pages 4, 9).
- [BE26] Nawaf Bou-Rabee and Andreas Eberle. *Markov Chain Monte Carlo Methods*. Lecture notes, University of Bonn. 2026. <https://uni-bonn.sciebo.de/s/kzTUFff5FrWGAay> (visited on 04/13/2026) (cited on pages 2, 4, 16–22, 27–29, 34, 35, 37, 48–50, 56, 58).
- [Bil95] Patrick Billingsley. *Probability and Measure*. 3rd edition. New York: John Wiley & Sons, 1995. ISBN: 978-0-471-00710-4 (cited on pages 4, 11).
- [DR20] Arnak S. Dalalyan and Lionel Riou-Durand. “On Sampling from a Log-Concave Density Using Kinetic Langevin Diffusions”. In: *Bernoulli* 26.3 (2020), pages 1956–1988. <https://doi.org/10.3150/19-BEJ1178>. <https://arxiv.org/abs/1807.09382> (cited on pages 4, 56, 58).
- [EGZ19] Andreas Eberle, Arnaud Guillin, and Raphael Zimmer. “Couplings and Quantitative Contraction Rates for Langevin Dynamics”. In: *The Annals of Probability* 47.4 (2019), pages 1982–2010. <https://doi.org/10.1214/18-AOP1299>. <https://arxiv.org/abs/1703.01617> (cited on pages 4, 57, 60).
- [Fol99] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. 2nd edition. New York: John Wiley & Sons, 1999. ISBN: 978-0-471-31716-6 (cited on page 26).
- [GM16] Arnaud Guillin and Pierre Monmarché. “Optimal Linear Drift for the Speed of Convergence of an Hypocoelliptic Diffusion”. In: *Electronic Communications in Probability* 21 (2016). Paper No. 74, pages 1–14. <https://doi.org/10.1214/16-ECP25> (cited on pages 2–5, 24, 27, 44, 51).
- [LNP13] Tony Lelièvre, Francis Nier, and Grigorios A. Pavliotis. “Optimal Non-reversible Linear Drift for the Convergence to Equilibrium of a Diffusion”. In: *Journal of Statistical Physics* 152.2 (2013), pages 237–274. <https://doi.org/10.1007/s10955-013-0769-x> (cited on pages 2, 4, 41).
- [LP17] David A. Levin and Yuval Peres. *Markov Chains and Mixing Times*. 2nd edition. Volume 107. AMS Non-Series Monographs. Providence, RI: American Mathematical Society, 2017. ISBN: 978-1-4704-2962-1. <https://doi.org/10.1090/mbk/107> (cited on page 2).
- [Øks98] Bernt Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. 5th edition. Universitext. Berlin: Springer-Verlag, 1998. ISBN: 978-3-540-63720-2. <https://doi.org/10.1007/978-3-662-03620-4> (cited on pages 4, 11–15, 17).

- [OP11] Michela Ottobre and Grigorios A. Pavliotis. “Asymptotic Analysis for the Generalized Langevin Equation”. In: *Nonlinearity* 24.5 (2011), pages 1629–1653. <https://doi.org/10.1088/0951-7715/24/5/013> (cited on pages 4, 51, 52).
- [SB25] Benjamin Sterling and Mónica F. Bugallo. “Critically-Damped Third-Order Langevin Dynamics”. In: *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pages 1–5. <https://doi.org/10.1109/ICASSP49660.2025.10889657>. <https://arxiv.org/abs/2409.07697> (cited on pages 4, 53).
- [Sep14] Timo Seppäläinen. *Basics of Stochastic Analysis*. Lecture notes, University of Wisconsin–Madison. 2014. <https://people.math.wisc.edu/~seppalai/courses/735/notes2014.pdf> (visited on 04/01/2025) (cited on pages 4, 11–13, 15, 23).
- [SS90] G. W. Stewart and Ji-Guang Sun. *Matrix Perturbation Theory*. Computer Science and Scientific Computing. Boston: Academic Press, 1990. ISBN: 978-0-12-670230-9 (cited on pages 4–9).
- [Tro22] Joel A. Tropp. *ACM 204: Matrix Analysis*. Accepted version. 2022. <https://doi.org/10.7907/m421-yb89> (cited on pages 4, 6).